

Mathematical Operators

Different types of questions covered in this chapter are as follows

- Symbol Substitution
- Balancing the Equation
- Interchange of Signs and Numbers
- Trick Based Mathematical Operations

Note:- While solving a mathematical expression, proceed according to the rule **BODMAS** — i.e., Brackets, Of, Division, Multiplication, Addition, Subtraction.

Example:

$$(36 - 12) \div 4 + 6 + 2 \times 3 = 24 \div 4 + 6 + 2 \times 3 \text{ (Solving Bracket)}$$

$$= 6 + 6 + 2 \times 3 \text{ (Solving Division)}$$

$$= 6 + 6 + 6 \text{ (Solving Multiplication)}$$

$$= 18 \text{ (Solving Addition)}$$

Type 1: Symbol Substitution

In this type of question, a candidate is provided with the substitutes for various mathematical symbols followed by a question involving calculation of an expression or choosing the correct/ incorrect equation. The candidate is required to put in the real signs in the given equation and then solve the questions as required.

Ex 1: if ' $\times$ ' means ' $-$ ', ' $\div$ ' means ' $+$ ', $+$ means ' $\times$ ', then $18 \times 5 \div 5 + 6$ is equal to

1. 58
2. 49
3. 43
4. 37

Solution: (c)

Change of symbols according to the question,

$$? = 18 \times 5 \div 5 + 6 = 18 - 5 + 5 \times 6$$

$$= 18 - 5 + 30 = (13 + 30) = 43$$

Type 2: Balancing the Equation

In this type of questions, the signs in one of the alternatives are required to fill up the blank spaces in order to balance the given equation

Ex: Choose the correct option in order to balance the following equation.

$$24 \quad 6 \quad 12 \quad 16 = 0$$

1. -, + and +
2. ÷, + and ÷
3. -, - and -
4. ÷, + and -

Solution: (d)

From Option (d)

$$24 \div 6 + 12 - 16 = 0$$

$$\frac{24}{6} + 12 - 16 = 0$$

$$4 + 12 - 16 = 0$$

$$16 - 16 = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence, option (d) is correct.

Type 3: Interchange of Signs and Numbers

In this type of questions, the given equation becomes correct and fully balanced when either two signs of the equation or both the numbers and the signs of the equation are interchanged. The candidate is required to find the correct pair of signs and numbers from the given options.

Ex: Choose the correct interchange option in order to make the given equation correct

$$10 - 2 + 9 \times 2 \div 4 = 19$$

1. ÷, +, × and -
2. -, +, × and +

3. $\div, \div, \div$ and $\times$
4. $\times, \div, \times$ and $\div$

Solution: (a)

Let us check the options one by one

From option (a),

$$10 - 2 + 9 \times 2 \div 4 = 19$$

$\downarrow$
 $\div$

$\downarrow$
 $-$

$$10 - 2 + 9 \times 2 - 4 = 19$$

$$5 + 9 \times 2 - 4 = 19$$

$$5 + 18 - 4 = 19 \quad \text{D} \quad 23 - 4 = 19$$

$$\setminus 19 = 19$$

As options (a) gives us the correct answer. Hence, there is no need to check other options

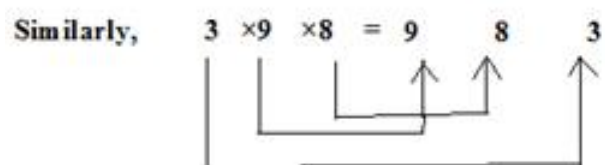
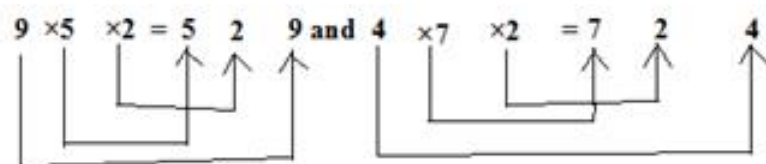
Type 4: Trick Based Mathematical Operations

The questions are based on simple mathematical operations that do not follow a universal rule. These questions can be based on several different patterns.

Ex: If $9 \times 5 \times 2 = 529$ and $4 \times 7 \times 2 = 724$, then $3 \times 9 \times 8 = ?$

1. 983
2. 839
3. 938
4. 893

Solution: (a)



= 983

Mathematical Opeartions

Instructions

For the following questions answer them individually

Question 1

If '+' means '-', '-' means '*', '*' means '/' and '/' means '+', then what is the value of the following expression?

$$13 - 13 / 3 - 63 * 9 + 2 - 5$$

- A 170
- B 180
- C 190
- D 200

Answer: B

Explanation:

Let us replace the symbols with the actual operators.

$$13 - 13 / 3 - 63 * 9 + 2 - 5 = 13 * 13 + 3 * 63 / 9 - 2 * 5$$

Applying BODMAS rule, we get, $169 + 3 * 7 - 10 = 180$

Therefore, option B is the right answer.

Question 2

If '+' means '*', '-' means '+', '*' means '/' and '/' means '-', then what is the value of the following expression?

$$119 * 17 + 6 / 16 * 2 + 5$$

- A 1
- B 3
- C 2
- D 4

Answer: C

Explanation:

Let us replace the symbols with the actual operators.

$$119 * 17 + 6 / 16 * 2 + 5 = 119 / 17 * 6 - 16 / 2 * 5$$

Applying BODMAS rule, we get,

$$\begin{aligned} 119 / 17 * 6 - 16 / 2 * 5 &= 7 * 6 - 8 * 5 \\ &= 42 - 40 \\ &= 2 \end{aligned}$$

Therefore, option C is the right answer.

Question 3

$$\sqrt{0.07 + \sqrt{0.0967 - 0.0643}} = ?$$

- A 0.8
- B 0.7
- C 0.6
- D 0.5

Answer: D

Explanation:

$$\sqrt{0.07 + \sqrt{0.0967 - 0.0643}} = \sqrt{0.07 + \sqrt{0.0324}} = \sqrt{0.07 + 0.18} = \sqrt{0.25} = 0.5$$

So the answer is option D.

Question 4

Find x^{2018} , if $3x - 4.5 + 2.4 = -2.1$

A 0

B 1

C 7^{2018}

D 0.7^{2018}

Answer: A

Explanation:

$$3x - 4.5 + 2.4 = -2.1$$

$$3x - 2.1 = -2.1$$

$$3x = 0$$

$$x = 0$$

$$x^{2018} = 0$$

So the answer is option A.

Question 5

2.1221-3.1331+4.1441 = ?

A 9.3993

B 3.1331

C 6.2662

D none of these

Answer: B

Explanation:

$$2.1221-3.1331+4.1441 = -1.0110+4.1441 = 3.1331$$

So the answer is option B.

Question 6

$$\frac{5}{10} \times \frac{100}{125} \times \frac{25}{x} = 1, \text{ find } x ?$$

A 20

B 15

C 5

D 10

Answer: D

Explanation:

$$\frac{5}{10} \times \frac{100}{125} \times \frac{25}{x} = 1$$

$$\frac{10}{25} \times \frac{25}{x} = 1$$

$$\frac{10}{x} = 1$$

$$x = 10$$

So the answer is option D.

Question 7

$$\sqrt{0.9 - \sqrt{0.0234 - 0.0153}} = ?$$

A 0.9

B 0.99

C 0.999

D 9.0

Answer: A

Explanation:

$$\sqrt{0.9 - \sqrt{0.0234 - 0.0153}} = \sqrt{0.9 - \sqrt{0.0081}} = \sqrt{0.9 - 0.09} = \sqrt{0.81} = 0.9$$

So the answer is option A.

Question 8

$$\frac{3}{4} \times \frac{16}{27} \div \frac{2}{3} = ?$$

A 2/3

B 3/2

C 9/4

D 4/9

Answer: A

Explanation:

$$\frac{3}{4} \times \frac{16}{27} \div \frac{2}{3} = \frac{3}{4} \times \frac{16}{27} \times \frac{3}{2} = \frac{4}{9} \times \frac{3}{2} = \frac{2}{3}$$

So the answer is option A.

Question 9

$$0.1123 - 0.0012 + 1.0032 - 0.101 = ?$$

A 1.0033

B 1.1333

C 1.0133

D 1.1033

Answer: C

Explanation:

$$0.1123 - 0.0012 + 1.0032 - 0.101 = 0.1111 + 0.9022 = 1.0133$$

So the answer is option C.

Question 10

Find x , if $4x - 0.3 + 0.22 = 2$?

A 5.002

B 5.200

C 5.020

D 0.520

Answer: D

Explanation:

$$4x - 0.3 + 0.22 = 2$$

$$4x - 0.08 = 2$$

$$4x = 2.08$$

$$x = 0.52$$

So the answer is option D.

Question 11

$15 \div 3 + 2 \times 4 - 2 = x + 1$, find x ?

A 9

B 10

C 11

D 12

Answer: B

Explanation:

$$15 \div 3 + 2 \times 4 - 2 = x + 1$$

$$5 + 8 - 2 = x + 1$$

$$11 = x + 1$$

$$x = 10$$

So the answer is option B.

Question 12

$\frac{6}{8} \times \frac{16}{18} \times \frac{3}{4} \times X = \frac{4}{7}$, find $1/X$?

A $4/7$

B $8/7$

C $7/8$

D $7/4$

Answer: B

Explanation:

$$\frac{6}{8} \times \frac{16}{18} \times \frac{3}{4} \times X = \frac{4}{7}$$

$$\frac{3}{4} \times \frac{8}{9} \times \frac{3}{4} \times X = \frac{4}{7}$$

$$\frac{1}{2} \times X = \frac{4}{7}$$

$$X = 8/7$$

$$1/X = 7/8$$

So the answer is option B.

Question 13

576.78+456.87-x = 658.65, Find x ?

A 375

B 376

C 373

D 374

Answer: A

Explanation:

$$576.78+456.87-x = 658.65$$

$$1033.65 - x = 658.65$$

$$x = 1033.65 - 658.65$$

$$x = 375$$

So the answer is option A.

Question 14

Find $((4096)^{1/3})^{1/2} = ?$

A 2

B 3

C 4

D 5

Answer: C

Explanation:

$$((4096)^{1/3})^{1/2} = 16^{1/2} = 4$$

So the answer is option C.

Question 15

3234 + 5678 - 11223 + 4456 = ?

A 4585

B 3545

C 5645

D 2145

Answer: D

Explanation:

$$3234 + 5678 - 11223 + 4456 = 2145.$$

So the answer is option D.

Question 16

Find $[(16)^4]^3]^2 = ?$

A 256

B 64

C 16

D 4

Answer: D

Explanation:

$$[(16)^4]^3]^2 = [16]^{4 \times 3 \times 2} = [16]^6 = 4$$

So the answer is option D.

Question 17

$123.45+45.678-3.1345+13.234 = ?$

A 146.3455

B 169.2275

C 156.3455

D 179.2275

Answer: D

Explanation:

$$123.45+45.678-3.1345+13.234 = 179.2275$$

So the answer is option D.

Question 18

$45.23+23.17-12.46+34.67 = ?$

A 47.86

B 56.64

C 67.57

D 90.61

Answer: D

Explanation:

$$45.23+23.17-12.46+34.67 = 90.61$$

So the answer is option D.

Question 19

$95^2 - 5^2 = ?$

A 8100

B 9000

C 9025

D 9205

Answer: B

Explanation:

$$95^2 - 5^2 = (95 + 5)(95 - 5) = 100 * 90 = 9000$$

So the answer is option B.

Question 20

Which of the following is divisible by 3, 7, 9 and 11 ?

A 2645

B 4158

C 3791

D 1188

Answer: B

Explanation:

$$3*7*9*11 = 2079$$

$$2079*2 = 4158 \text{ is divisible by 3, 7, 9, 11}$$

So the answer is option B.

Question 21

3245 - 123 - 456 - 347 + x = 3650 ?

A 2134

B 1331

C 1567

D 1467

Answer: B

Explanation:

$$3245 - 123 - 456 - 347 + x = 3650$$

$$2319 + x = 3650$$

$$x = 1331$$

So the answer is option B.

Question 22

$$600 \div 12 \times \frac{1}{2} \div 5 = ?$$

A 4

B 5

C 6

D 8

Answer: B

Explanation:

$$600 \div 12 \times \frac{1}{2} \div 5 = 50 \times 1/10 = 5$$

So the answer is option B.

Question 23

$$123.45 + 6789.01 - 2345.67 = ?$$

A 4565.79

B 4655.79

C 4566.79

D 4665.79

Answer: C

Explanation:

$$123.45 + 6789.01 - 2345.67 = 6912.46 - 2345.67 = 4566.79$$

So the answer is option C.

Question 24

$$4591.15 - 528.116 = x + 456.123. \text{ Find } x ?$$

A 3660.911

B 3666.911

C 3006.911

D 3606.911

Answer: D

Explanation:

$$4591.15 - 528.116 = x + 456.123$$

$$4063.034 = x + 456.123$$

$$4063.034 - 456.123 = x$$

$$3606.911 = x$$

So the answer is option D.

Question 25

$$987.65 - 456.78 - 43.321 = ?$$

A 478.549

B 487.549

C 487.459

D 478.459

Answer: B

Explanation:

$$987.65 - 456.78 - 43.321 = 530.87 - 43.321 = 487.549$$

So the answer is option B.

Mathematical Operator

TYPE-I (i)

Directions: In the following question you have to identify the correct response from the given premises stated according to following symbols :

1. If '+' stands for division, '÷' stands for multiplication, 'x' stands for subtraction and '-' stands for addition, which one of the following is correct?

- (1) $18 \div 6 - 7 + 5 \times 2 = 20$
 (2) $18 + 6 \div 7 \times 5 - 2 = 18$
 (3) $18 \times 6 + 7 \div 5 - 2 = 16$
 (4) $18 \div 6 \times 7 + 5 - 2 = 22$

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Ist Sitting))

2. If '-' stands for division, '+' for multiplication, '÷' for subtraction and 'x' for addition, which one of the following equations is correct?

- (1) $18 \div 3 \times 2 + 8 - 6 = 10$
 (2) $18 - 3 + 2 \times 8 \div 6 = 14$
 (3) $18 - 3 \div 2 \times 8 + 6 = 17$
 (4) $18 \times 3 + 2 \div 8 - 6 = 15$

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Ist Sitting))

3. In an imaginary mathematical operation '+' means multiplication, 'x' means subtraction, '÷' means addition and '-' means division. All other rules in mathematical operation are the same as in the existing system.

Which one of the following gives the result of

$$175 - 25 \div 5 + 20 \times 3 + 10 ?$$

- (1) 160 (2) 2370
 (3) 77 (4) 240

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (IInd Sitting))

4. If '-' stands for division, '+' for multiplication, '÷' for subtraction and 'x' for addition, which one of the following equations is correct?

- (1) $6 \div 20 \times 12 + 7 - 1 = 70$
 (2) $6 + 20 - 12 \div 7 \times 1 = 62$
 (3) $6 - 20 \div 12 \times 7 + 1 = 57$
 (4) $6 + 20 - 12 \div 7 - 1 = 38$

(SSC Combined Graduate Level Prelim Exam. 27.02.2000 (IInd Sitting))

5. If '+' means '-', '-' means 'x', 'x' means '÷' and '÷' means '+' then $2 \div 6 \times 6 \div 2 = ?$

- (1) 1 (2) 0
 (3) 10 (4) 5

(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (Ist Sitting))

6. If '+' stands for Multiplication, 'x' stands for Division, '-' stands for Addition and '÷' stands for Subtraction, what would the following equation stand for?

$$20 - 8 \times 4 \div 3 + 2 = ?$$

- (1) 41 (2) 19
 (3) 16 (4) 18

(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (IInd Sitting))

7. If 'x' means '-', '-' means '÷', '÷' means 'x' and '÷' means '+', then what will be the value of the following expression?

$$16 \times 8 \div 4 - 3 + 9 = ?$$

- (1) 10 (2) 19
 (3) 20 (4) 9

(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (Middle Zone))

8. If '+' means 'minus', '-' means 'multiplication', '÷' means 'plus', and 'x' means 'division', $15 - 3 + 10 \times 5 \div 5 = ?$

- (1) 52 (2) 48
 (3) 22 (4) 5

(SSC CPO Sub-Inspector Exam. 07.09.2003)

9. If '-' stands for division, '÷' stands for multiplication, '+' stands for subtraction and 'x' for addition, then which of the following equation is correct?

- (1) $20 + 8 - 7 \div 6 \times 4 = 25$
 (2) $20 - 5 \div 4 + 6 \times 5 = 15$
 (3) $20 \times 5 - 6 \div 7 + 4 = 28$
 (4) $20 \div 4 - 8 \times 10 + 6 = 36$

(SSC Combined Graduate Level Prelim Exam. 08.02.2004 (IInd Sitting))

10. If x stands for + and ÷ for -, find the value of the following equation.

$$39 \times 23 \div 21 \times 5$$

- (1) 46 (2) 36
 (3) 62 (4) 89

(SSC CPO Sub-Inspector Exam. 05.09.2004)

11. If '+' stands for division; '÷' stands for multiplication; 'x' stands of subtraction; '-' stands for addition which one of the following is correct?

- (1) $18 \div 6 - 7 + 5 \times 2 = 20$
 (2) $18 + 6 \div 7 \times 5 - 2 = 18$
 (3) $18 \times 6 + 7 \div 5 - 2 = 16$
 (4) $18 \div 6 \times 7 + 5 - 2 = 22$

(SSC CPO Sub-Inspector Exam. 03.09.2006)

12. If + stands for 'division', x stands for 'addition', - stands for 'multiplication' and ÷ stands for 'subtraction', then which of the following equations is correct?

- (1) $36 \times 6 + 7 \div 2 - 6 = 20$
 (2) $36 + 6 - 3 \times 5 \div 3 = 24$
 (3) $36 \div 6 + 3 \times 5 - 3 = 45$
 (4) $36 - 6 + 3 \times 5 \div 3 = 74$

(SSC CPO Sub-Inspector Exam. 03.09.2006)

13. If + means 'minus' — means 'multiplied by', ÷ means 'plus' and x means 'divided by', then

$$10 \times 5 \div 3 - 2 + 3 = ?$$

- (1) 5 (2) 21
 (3) $\frac{53}{3}$ (4) 18

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (Ist Sitting))

14. In the following question you have to identify the correct response from the given premises stated according to following symbols.

If + means ÷, - means x, ÷ means + and x means -, then

$$63 \times 24 + 8 \div 4 + 2 - 3 = ?$$

- (1) 54 (2) 66
 (3) 186 (4) 48

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (IInd Sitting))

15. The following equation becomes mathematically correct when you interchange either the sign or the numbers as indicated in the question. Find the correct alternative. Given equation :

$$(16 - 4) \times 6 \div 2 + 8 = 30$$

- (1) 4 and 2 (2) ÷ and -
 (3) 16 and 6 (4) - and +

(SSC CPO Sub-Inspector Exam. 16.12.2007)

| SYMBOLS & NOTATIONS |

- 16.** If '-' stands for division, '+' stands for subtraction, '÷' stands for multiplication, 'x' stands for addition, then which one of the following equations is correct?

- (1) $70 - 2 + 4 \div 5 \times 6 = 44$
 (2) $70 - 2 + 4 \div 5 \times 6 = 21$
 (3) $70 - 2 + 4 \div 5 \times 6 = 341$
 (4) $70 - 2 + 4 \div 5 \times 6 = 36$

(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (1st Sitting))

- 17.** If - stands for division, + for multiplication, ÷ for subtraction and x for addition, then which one of the following equations is correct ?

- (1) $19 + 5 - 4 \times 2 \div 4 = 11$
 (2) $19 \times 5 - 4 \div 2 + 4 = 16$
 (3) $19 \div 5 + 4 - 2 \times 4 = 13$
 (4) $19 \div 5 + 4 + 2 \div 4 = 20$

(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (1st Sitting))

- 18.** If '-' stands for '÷', '+' stands for 'x', '÷' for '-' and 'x' for '+', which one of the following equations is correct?

- (1) $30 - 6 + 5 \times 4 \div 2 = 27$
 (2) $30 + 6 - 5 \div 4 \times 2 = 30$
 (3) $30 \times 6 \div 5 - 4 + 2 = 32$
 (4) $30 \div 6 \times 5 + 4 - 2 = 40$

(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (1st Sitting))

- 19.** If 'x' means 'addition', '-' means 'division', '÷' means 'subtraction' and '+' means 'multiplication', then which of the following equations is correct ?

- (1) $16 + 5 - 10 \times 4 \div 3 = 9$
 (2) $16 - 5 \times 10 \div 4 + 3 = 12$
 (3) $16 + 5 \div 10 \times 4 - 3 = 9$
 (4) $16 \times 5 \div 10 \div 4 - 3 = 19$

(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (IInd Sitting))

- 20.** If '-' stands for division, '+' for multiplication, '÷' for subtraction and 'x' for addition, which one of the following equations is correct?

- (1) $24 \div 8 - 4 + 2 \times 3 = 16$
 (2) $24 - 8 + 4 \times 2 \div 3 = 12$
 (3) $24 \times 8 - 4 \div 2 + 3 = 17$
 (4) $24 + 8 - 4 \times 2 \div 3 = 47$

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (East Zone))

- 21.** If ÷ means plus, × means subtraction, then

$(15 \times 9) \div (12 \times 4) \times (4 \div 4)$ is equal to :

- (1) 96 (2) 6
 (3) $\frac{3}{128}$ (4) $\frac{143}{4}$

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (Raipur, Madhya Pradesh))

- 22.** If + means -, - means ×, ÷ means +, × means ÷, find the value of $15 \times 3 \div 4 - 6 + 7$?

- (1) 22 (2) 25
 (3) 9 (4) $175/3$

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (Middle Zone))

- 23.** If × stands for -, ÷ stands for +, + stands for ×, find the value of following equation:

$(16 \times 5) \div 5 + 3 = ?$

- (1) 62 (2) 10
 (3) 2 (4) 26

(SSC Combined Matric Level (PRE) Exam. 21.05.2000, 30.07.2006 (1st Sitting) (Middle Zone))

- 24.** If '÷' stands for addition, '-' stands for multiplication, 'x' stands for subtraction and '+' stands for division, which of the responses does not hold good?

- (1) $10 \times 4 = 06$ (2) $10 - 4 = 40$
 (3) $10 + 5 = 50$ (4) $10 - 5 = 15$

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (1st Sitting))

- 25.** If 'x' stands for '+', '÷' for '-', '÷' for 'x' and '+' for '÷', find the value of the following equation :

$54 \div 16 - 3 \times 6 + 2 = ?$

- (1) 9 (2) 12
 (3) 8 (4) 15

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (1st Sitting))

- 26.** If '+' stands for 'subtraction' and '÷' stands for 'addition' and '-' stands for 'multiplication' and 'x' stands for division, then which of the following equations is correct?

- (1) $56 + 12 \times 34 - 12 = 102$
 (2) $8 \div 44 - 5 + 25 = 203$
 (3) $112 \times 44 - 12 + 10 = 46$
 (4) $9 \div 64 - 2 \times 6 = 54$

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (IInd Sitting))

- 27.** If '-' stands for division, '+' stands for multiplication, '÷' stands for subtraction and 'x' stands for addition, then which one of the equation is correct?

- (1) $30 + 5 - 12 \div 8 \times 12 = 70$
 (2) $30 - 5 + 12 \div 8 \times 12 = 76$
 (3) $30 \times 5 - 12 + 8 \div 12 = 60$
 (4) $30 \div 5 \times 12 + 8 - 12 = 24$

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (IInd Sitting))

- 28.** If '+' stands for multiplication, '-' stands for division, '÷' stands for addition and '÷' stands for subtraction, what is the answer for the following equation?

$20 - 5 \div 18 \times (3 + 2) = ?$

- (1) 20 (2) 18
 (3) 108 (4) 22

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (IInd Sitting))

- 29.** If 'x' stands for minus, '+' stands for multiplication, '-' stands for plus, then which one of the following is correct?

$6 + (3 \times 1) + 5 = ?$

- (1) 58 (2) 64
 (3) 60 (4) 12

(SSC Combined Matric Level (PRE) Exam. 27.05.2001 (IInd Sitting) (East Zone))

- 30.** If '+' means '÷', '÷' means '-', '-' means 'x', 'x' means '+', then

$12 + 6 \div 3 - 2 \times 8 = ?$

- (1) -2 (2) 4
 (3) 2 (4) 8

(SSC Combined Matric Level (PRE) Exam. 27.05.2001 (IInd Sitting) (East Zone))

- 31.** If '+' stands for 'x', '-' for '÷', 'x' for '-' and '÷' for '+', then find the value of the following equation

$26 + 74 - 4 \times 5 \div 2 = ?$

- (1) 220 (2) 376
 (3) 478 (4) 488

(SSC Combined Matric Level (PRE) Exam. 27.05.2001 (IInd Sitting) (East Zone))

- 32.** If '+' stands for subtraction, '÷' stands for addition, '-' stands for multiplication and 'x' stands for division, then which one of the following equations is correct?

- (1) $46 - 10 + 10 \times 5 = 92$
 (2) $265 + 11 - 2 \times 14 = 22$
 (3) $66 \times 3 - 11 + 12 = 230$
 (4) $2 - 14 \times 4 \div 11 = 16$

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (1st Sitting) (Eastern Zone, Guwahati))

| SYMBOLS & NOTATIONS |

- 33.** In the following question, you have to identify the correct response from the given premises stated according to following symbols.
If + denotes -, - denotes =, = denotes +, ÷ denotes >, × denotes <, how do you solve the following problem?
 $8 + 4 - ?$
(1) 4 (2) 32
(3) 12 (4) 2
(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (1st Sitting) (North Zone, Delhi))
- 34.** If '-' means addition, '+' means subtraction, '×' means division and '÷' means multiplication, then $7 - 10 \times 5 \div 6 + 4 = ?$
(1) 3 (2) 12
(3) 15 (4) 9
(SSC Combined Matric Level (Pre) Exam. 30.03.2008 (1st Sitting))
- 35.** If '÷' means '×', '-' means '+', '×' means '-', '+' means '÷', then what will be the value of the following?
 $20 + 4 \times 6 - 5 \div 7$
(1) 28 (2) 32
(3) 34 (4) 36
(SSC Data Entry Operator Exam. 31.08.2008)
- 36.** If '+' means ×, '-' means +, '×' means ÷ and '÷' means -, then $10 + 5 \times 10 \div 2 - 5$ has a value of
(1) 35 (2) 45
(3) 30 (4) 8
(SSC Data Entry Operator Exam. 02.08.2009)
- 37.** If '+' stands for division, '÷' stands for multiplication, multiplication for addition; which one of the following equations is correct?
(1) $10 \div 5 + 4 = 6$
(2) $10 - 4 + 2 = 6$
(3) $10 + 2 - 5 = 6$
(4) $10 + 2 \times 1 = 6$
(SSC Stenographer (Grade 'C' & 'D') Exam. 26.09.2010)
- 38.** If '+' stands for '÷', '×' stands '+', '-' stands for '×', and '÷' stands for '-', then which of the following statements is correct?
(1) $36 \times 6 + 7 \div 2 - 6 = 20$
(2) $36 \div 6 + 3 \times 5 - 3 = 45$
(3) $36 + 6 - 3 \times 5 \div 3 = 24$
(4) $36 - 6 + 3 \times 5 \div 3 = 74$
(SSC HSL Data Entry Operator & LDC Exam. 27.11.2010 & SSC Combined Matric Level (Pre) Exam, 30.07.2006)
- 39.** Identify the correct response from the given premises stated according to following symbols.
If '-' stands for division
'+' stands for multiplication.
'÷' stands for subtraction
'×' stands for addition
Which one of the equations is correct?
(1) $18 + 14 - 24 \times 12 \div 16 = 12$
(2) $16 \times 14 - 24 \div 18 + 12 = -24$
(3) $24 - 12 + 12 \div 16 \times 18 = 26$
(4) $18 \div 16 + 12 \times 18 \div 12 = 24$
(SSC Higher Secondary Level Data Entry Operator & LDC Exam. 28.11.2010 (1st sitting))
- 40.** If '×' means '-', '-' means '×', '+' means '÷' and '÷' means '+', then $(15 - 10) \div (130 + 10) \times 50 = ?$
(1) 1800 (2) 113
(3) 2000 (4) 123
(SSC Multi-Tasking (Non-Technical) Staff Exam. 20.02.2011)
- 41.** If '+' means '÷', '-' means '×', '÷' means '+' and '×' means '-', then $36 \times 12 + 4 \div 6 + 2 - 3 = ?$
(1) 42 (2) 18
(3) 40 (4) 2
(SSC Multi-Tasking (Non-Technical) Staff Exam. 27.02.2011)
- 42.** If T means '×', U means '-', V means '÷' and W means '+', then what will be the value of the following expression:
 $(50 \text{ V } 2) \text{ W } (28 \text{ T } 4)$
(1) 142 (2) 158
(3) 137 (4) 163
(SSC Stenographer (Grade 'C' & 'D') Exam. 16.10.2011)
- 43.** If - stands for ÷, + stands for ×, ÷ stands for - and × stands for + find out which one is correct.
(1) $49 \times 7 + 3 \div 5 - 8 = 16$
(2) $49 \div 7 \times 3 + 5 - 8 = 26$
(3) $49 + 7 - 3 \times 5 \div 8 = 20$
(4) $49 - 7 + 3 \div 5 \times 8 = 24$
(SSC Level Data Entry Operator & LDC Exam. 21.10.2012 (1st Sitting))
- 44.** If '+' means '×', '-' means '÷', '×' means '+' and '÷' means '-' then $25 \times 5 - 3 \div 2 + 5 = ?$
(1) 20 (2) 50
(3) 30 (4) 40
(SSC Level Data Entry Operator & LDC Exam. 21.10.2012 (IInd Sitting))
- 45.** If '-' stands for addition, '+' for multiplication, '÷' for subtraction and '×' for division, which one of the following equation is wrong?
(1) $5 - 2 + 12 \times 6 \div 2 = 27$
(2) $5 + 2 - 12 \div 6 \times 2 = 13$
(3) $5 + 2 - 12 \times 6 \div 2 = 10$
(4) $5 \div 2 + 12 \times 6 - 2 = 3$
(SSC Level Data Entry Operator & LDC Exam. 28.10.2012 (1st Sitting))
- 46.** If × stands for -, + means ×, ÷ means + and - means ÷, then what is the value of the given expression?
 $175 - 25 \div 5 + 20 \times 3 + 10 = ?$
(1) 77 (2) 160
(3) 240 (4) 2370
(SSC Level Data Entry Operator & LDC Exam. 04.11.2012 (IInd Sitting))
- 47.** If '+' means '÷', '÷' means '-', '-' means '×', '×' means '+', then $8 + 2 \div 3 - 4 \times 6 = ?$
(1) -12 (2) -2
(3) -10 (4) -15
(SSC (10+2) Level DEO & LDC Exam. 04.11.2012, 1st Sitting)
- 48.** If '÷' means '-', '-' means '×', '×' means '+' and '+' means '÷' then $20 \times 60 \div 40 - 20 + 10 = ?$
(1) 80 (2) 60
(3) 40 (4) 0
(SSC Multi-Tasking Staff Exam. 17.03.2013, 1st Sitting)
- 49.** If '+' means '÷', '×' means '+', '-' means '×' and '÷' means '-', then which of the following equations is correct?
(1) $36 \times 6 + 3 - 2 < 20$
(2) $36 \times 6 + 3 \times 2 > 20$
(3) $36 + 6 \times 3 + 2 = 20$
(4) $36 + 6 - 3 \times 2 = 20$
(SSC Multi-Tasking Staff Exam. 17.03.2013, IInd Sitting)
- 50.** If - stands for addition, ÷ for multiplication, × for subtraction, and + for division, then which of the following is correct?
(1) $25 - 15 + 5 \div 4 \times 16 = 21$
(2) $25 + 11 - 4 \div 10 \times 6 = 20$
(3) $25 \times 12 - 14 \div 4 + 6 = 16$
(4) $25 - 12 + 14 \div 2 \times 4 = 15$
(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)
- 51.** Which of the following interchange of signs would make the given equation correct?
 $5 + 3 \times 8 - 12 \div 4 = 3$

| SYMBOLS & NOTATIONS |

- (1) – and ÷ (2) + and ×
(3) + and ÷ (4) + and –

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

- 52.** Put the correct mathematical signs in the following equation from the given alternatives.

$$33 \geq 11 \geq 3 \geq 6 ? = 115$$

- (1) –, ×, + (2) +, –, ×
(3) ×, ÷, – (4) ÷, ×, ×

(SSC Graduate Level Tier-I Exam. 19.05.2013, 1st Sitting)

- 53.** If '×' means '+', ÷ means '–', + means '÷' and '–' means '×' then what should be the value of the given equation?

$$14 \times 4 \div 70 + 10 - 2 = ?$$

- (1) 33 (2) 15
(3) 30 (4) 4

(SSC CAPFs SI & CISF ASI Exam. 23.06.2013)

- 54.** If + means ÷, – means ×, × means + and ÷ means –, then which of the alternatives is correct ?

- (1) $5 \times 8 - 5 + 5 \div 1 = 12$
(2) $55 - 2 + 10 \div 1 \times 5 = 16$
(3) $38 \div 10 - 5 + 7 \times 8 = 25$
(4) $10 - 12 + 2 \div 30 \times 1 = 10$

(SSC CGL Tier-I Re-Exam-2013, 27.04.2014)

- 55.** If '–' stands for '+', '+' stands for '×', '×' stands for '–' then which one of the following is not correct ?

- (1) $22 + 7 - 3 \times 9 = 148$
(2) $33 \times 5 - 10 + 20 = 228$
(3) $7 + 28 - 3 \times 52 = 127$
(4) $44 - 9 + 6 \times 11 = 87$

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam. 22.06.2014)

- 56.** If '+' stands for 'division'; '×' stands for 'addition'; '–' stands for 'multiplication'; '÷' stands for 'subtraction', which of the following equations is correct?

- (1) $5 - 3 + 2 \times 4 \div 8 = 2$
(2) $5 \times 3 + 2 - 4 \times 8 = 19$
(3) $5 \div 3 \times 2 - 4 + 8 = 8$
(4) $5 + 3 \times 2 \div 4 - 8 = 4$

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam. 22.06.2014)

- 57.** If '+' means '÷'; '–' means '+'; '×' means '–' and '÷' means '×', then, $8 \div 4 - 6 + 3 \times 4 = ?$

- (1) 4 (2) 14
(3) 28 (4) 30

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, 1st Sitting)

- 58.** If '+' means '–'; '–' means '×'; '×' means '÷' and '÷' means '+', then $25 \times 5 \div 30 + 8 - 2 = ?$

- (1) 54 (2) 15
(3) 18 (4) 19

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, 1st Sitting)

- 59.** If '–' stands for addition, '+' stands for subtraction, '÷' stands for multiplication and '×' stands for division, then which one of the following equation is correct ?

- (1) $50 \times 5 \div 2 - 30 + 25 = 25$
(2) $50 - 30 + 5 \div 2 \times 30 = 25$
(3) $40 + 35 \times 2 - 50 \div 30 = 95$
(4) $30 \times 2 - 25 + 50 \div 5 = 100$

(SSC GL Tier-I Exam. 19.10.2014 (1st Sitting))

- 60.** If + stands for division; × stands for addition; – stands for multiplication; ÷ stands for subtraction, which of the following is correct ?

- (1) $15 \div 5 \times 2 - 6 + 3 = 28$
(2) $15 \times 5 + 2 - 6 \div 3 = 56.5$
(3) $15 + 5 - 2 \div 6 \times 3 = 3$
(4) $15 - 5 + 2 \times 6 \div 3 = 41$

(SSC GL Tier-I Exam. 19.10.2014)

- 61.** If '–' stands for 'division', '+' stands for 'multiplication', '÷' stands for 'subtraction', '×' stands for 'addition', then which one of the following equations is correct ?

- (1) $36 \times 4 - 12 + 5 \div 3 = 420$
(2) $52 \div 4 + 5 \times 8 - 2 = 36$
(3) $36 - 12 \times 6 \div 3 + 4 = 60$
(4) $43 \times 7 \div 5 + 4 - 8 = 25$

(SSC GL Tier-I Exam. 26.10.2014)

- 62.** If a represents ÷, b represents +, c represents – and d represents × then

$$24 a 6 d 4 b 9 c 8 = ?$$

- (1) 20 (2) 6
(3) 17 (4) 19

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam, 21.06.2015 (1st Sitting) TF No. 8037731)

- 63.** If × means +, + means ÷, – means × and ÷ means –, then $6 \times 4 - 5 + 2 \div 1 = ?$

- (1) 11 (2) 15
(3) 10 (4) 12

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam, 21.06.2015 1st Sitting)

- 64.** If, + stands for division; × stands for addition; – stands for multiplication; ÷ stands for subtraction, which of the following is correct?

- (a) $46 \times 6 \div 4 - 5 + 3 = 74$
(b) $46 - 6 + 4 \times 5 \div 3 = 71$
(c) $46 \div 6 \times 4 - 5 + 3 = 75.5$
(d) $46 \times 6 - 4 + 5 \div 3 = 70.1$
(1) c (2) d
(3) b (4) a

(SSC CGL Tier-I Exam, 09.08.2015 (1st Sitting) TF No. 1443088)

- 65.** If + = ×, – = ÷, × = +, ÷ = –, then which is the correct equation out of the following?

- (1) $18 - 6 \times 7 \div 2 + 8 = 63$
(2) $18 \div 6 + 4 - 2 \div 3 = 22$
(3) $18 + 6 - 4 \times 2 \div 3 = 26$
(4) $18 \times 6 - 4 + 7 \times 8 = 47$

(SSC CGL Tier-I Exam, 09.08.2015 (1st Sitting) TF No. 1443088)

- 66.** If '+' means '×', '–' means '÷', '×' means '–' and '÷' means '+', then what will be the value of $16 \div 64 - 8 \times 4 + 2?$

- (1) 18 (2) 24
(3) 16 (4) 12

(SSC CGL Tier-I Exam, 16.08.2015 (1st Sitting) TF No. 3196279)

- 67.** If + stands for ×; – stands for ÷; × stands for @ and ÷ stands for %, then which of the following statements is correct?

- (1) $256\% 16 @ 5 \# 28 = 52$
(2) $256 \# 16 \% 5 \# 28 = 120$
(3) $256 @ 5 \% 16 * 28 = 408$
(4) $256 \# 16 @ 5 \% 28 = 80$

(SSC CGL Tier-I Re-Exam, 30.08.2015)

- 68.** If

÷ stands for ×

× stands for –

– stands for + and

+ stands for ÷, then

$$48 + 6 - 12 \div 2 + 10 = ?$$

(Do chronologically and not according to BODMAS rule)

- (1) 9 (2) 14
(3) 16 (4) 4

(SSC Constable (GD)

Exam, 04.10.2015, 1st Sitting)

- 69.** Select the set of symbols which can be fitted correctly in the equation.

$$5 _ 3 _ 8 _ 4 _ 2 = 21$$

| SYMBOLS & NOTATIONS |

- (1) +, ×, +, ÷ (2) ×, +, -, ÷
 (3) -, ×, +, ÷ (4) +, ×, -, ÷
 (SSC (10+2) LDC/DEO/PA/SA Exam. 01.11.2015 TF No. 1098066)
- 70.** If a means +, b means ×, c means ÷, d means -, then 20 a 10 b 45 c 5 d 12 = ?
 (1) 88 (2) 74
 (3) 98 (4) 68
 (SSC (10+2) LDC/DEO/PA/SA Exam. 01.11.2015 TF No. 1098066)
- 71.** If + stands for division; × stands for addition; - stands for multiplication; ÷ stands for subtraction, which of the following is correct?
 (1) $25 \times 3 - 7 \div 8 + 12 = 18$
 (2) $25 + 3 \times 7 - 8 \div 12 = 10.89$
 (3) $25 - 3 \div 7 \times 8 + 12 = 132$
 (4) $25 \div 3 \times 7 - 8 + 12 = 19.3$
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 01.11.2015, IInd Sitting)
- 72.** Insert the arithmetical operations in the following numerical figure-
 4_3_4 = 48
 (1) + + (2) + -
 (3) × + (4) × ×
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 01.11.2015, IInd Sitting)
- 73.** If the given interchanges are made in signs and numbers, which one of the following equation is true?
 (Signs : ÷ and +, Numbers : 6 and 5)
 (1) $18 + 6 \div 5 = 9.6$
 (2) $26 \div 5 + 6 = 6.4$
 (3) $5 \div 6 + 80 = 5.8$
 (4) $90 + 5 \div 6 = 8.6$
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (Ist Sitting) TF No. 6636838)
- 74.** If '-' stands for addition, '+' for multiplication, '÷' for subtraction and '×' for division, which one of the following equations is correct?
 (1) $5 + 2 - 12 \div 6 \times 2 = 13$
 (2) $5 + 2 - 12 \times 6 \div 2 = 10$
 (3) $5 \div 2 + 12 \times 6 - 2 = 4$
 (4) $5 - 2 + 12 \times 6 \div 2 = 27$
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (IInd Sitting) TF No. 7203752)
- 75.** If P denotes '÷', Q denotes '×', R denotes '+' and S denotes '-', then $16Q12P6R5S4 = ?$
 (1) 31 (2) 32
 (3) 33 (4) 30
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (IInd Sitting) TF No. 7203752)
- 76.** If × stands for ÷, ÷ stands for +, + stands for -, and - stands for ×, then what is the value of $(30 + 20) - 5(7 \div 3) \times 25 = ?$
 (1) 100 (2) 10
 (3) 20 (4) 25
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (Ist Sitting) TF No. 1375232)
- 77.** If + means ÷, ÷ means -, - means ×, × means +, then $12 - 8 \times 6 - 4 \div 6 + 3 = ?$
 (1) -112 (2) +118
 (3) -33 (4) +92
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (Ist Sitting) TF No. 1375232)
- 78.** If + means ÷, ÷ means ×, and × means +, then following will be:
 $64 + 8 \times 32 \div 4$
 (1) 128 (2) 160
 (3) 136 (4) 144
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (IInd Sitting) TF No. 3441135)
- 79.** If '-' denotes '+', '+' denotes '×', '÷' denotes '-', '×' denotes '÷' then, $27 \times 3 \div 6 + 9 - 8 = ?$
 (1) 15 (2) 14.5
 (3) 35 (4) 3.5
 (SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.01.2016 TF No. 3513283)
- 80.** What will be the correct mathematical signs that can be inserted in the following?
 $4_6_2_4_8 = 16$
 (1) - × + ÷ (2) ÷ + × -
 (3) + ÷ - × (4) × ÷ - +
 (SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.01.2016 TF No. 3513283)
- 81.** In a certain code 0, 1, 2 9 is coded as a,b,c,....., j then find $baf \div bf \times d$
 (1) cb (2) d
 (3) df (4) be
 (SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (Ist Sitting) TF No. 1375232)
- 82.** Which one of the following interchanges of signs would make the given equation correct?
 $5 + 3 \times 8 - 12 \div 4 = 3$
 (1) - and ÷ (2) + and ×
 (3) + and ÷ (4) + and -
 (SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.07.2016)
- 83.** If '+' stands for division; '×' stands for addition; '-' stands for multiplication; '÷' stands for subtraction, then which of the following equations is correct?
 (1) $33 \times 4 - 5 + 6 \div 2 = 26$
 (2) $33 \div 4 \times 5 + 6 - 2 = 30$
 (3) $33 - 4 + 5 \div 6 \times 2 = 24$
 (4) $33 - 4 \div 5 \times 6 + 2 = 130$
 (SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.07.2016)
- 84.** If '×' means addition, '-' means division, '÷' means subtraction and '+' means multiplication, then which of the equations is correct?
 (1) $16 \times 5 \div 10 + 4 - 3 = 19$
 (2) $16 + 5 \div 10 \times 4 - 3 = 9$
 (3) $16 + 5 - 10 \times 4 \div 3 = 9$
 (4) $16 - 5 \times 10 \div 4 + 3 = 12$
 (SSC CGL Tier-I (CBE) Exam.10.09.2016)
- 85.** If '+' is '×', '-' is '+', '×' is '÷' and '÷' is '-', then answer the following questions based on this information.
 $9 - 4 + 2 \div 16 \times 2$
 (1) 71 (2) 62
 (3) 9 (4) 24
 (SSC CPO SI, ASI Online Exam.05.06.2016) (IInd Sitting)
- 86.** Identify the symbols to be inserted to make the expression correct.
 (1) × + × (2) = × +
 (3) × ÷ = (4) + = ×
 (SSC CPO SI, ASI Online Exam.05.06.2016) (IInd Sitting)
- 87.** Study the following statements and answer the questions accordingly.
 '+' stands for division
 '÷' stands for multiplication
 '×' stands for addition
 Which one of the following is correct?
 (1) $30 \times 40 + 8 - 70 \div 40 = 180$
 (2) $30 + 40 \div 8 \times 70 - 40 = 340$
 (3) $30 - 40 \times 8 \div 70 + 40 = 180$
 (4) $30 + 40 + 8 \times 70 - 40 = 340$
 (SSC CPO SI, ASI Online Exam.05.06.2016) (IInd Sitting)

| SYMBOLS & NOTATIONS |

- 88.** If 'x' means 'addition', '-' means 'division', '/' means 'subtraction' and '+' means 'multiplication', then which of the equation is correct?
 (1) $25 + 10 - 5/10 \times 3 = 43$
 (2) $25 - 10 \times 5 + 10/3 = 72$
 (3) $25 \times 10/5 + 10 - 3 = 12$
 (4) $25/10 + 5 \times 10/3 = 18$
 (SSC CHSL (10+2) Tier-I (CBE) Exam. 08.09.2016) (1st Sitting)
- 89.** If '+' stands for multiplication, '-' stands for addition, 'x' stands for division, then what is the value of
 $128 + 9 - 16 \times 4 = ?$
 (1) 73 (2) 256
 (3) 1156 (4) 1352
 (SSC CGL Tier-I (CBE) Exam. 09.09.2016) (1st Sitting)
- 90.** If '+' stands for division, $\wedge$ stands for multiplication, $\wedge^*$ stands for subtraction and '-' stands for addition, which one of the following expressions is correct?
 (1) $18 \times 6 + 7 \div 5 - 2 = 16$
 (2) $18 \div 6 \times 7 + 5 - 2 = 22$
 (3) $18 \div 6 - 7 + 5 \times 2 = 20$
 (4) $18 + 6 \div 7 \times 5 - 2 = 18$
 (SSC CAPFs (CPO) SI & ASI, Delhi Police Exam. 20.03.2016) (1st Sitting)
- 91.** If '+' stands for multiplication, '-' stands for division, 'x' stands for addition, '+' stands for subtraction, then which one of the following equations is correct?
 (1) $12 \times 5 + 4 - 5 \div 4 = 20$
 (2) $12 \div 5 + 4 - 5 \times 4 = 18$
 (3) $12 + 5 - 4 \times 5 \div 4 = 16$
 (4) $12 \div 5 - 4 \times 5 + 4 = 22$
 (SSC CAPFs (CPO) SI & ASI, Delhi Police Exam. 20.03.2016) (1st Sitting)
- 92.** If $\div$ means +, + means x, x means - and - means $\div$, then what is the value of:
 $[(1440 - 36 \times 16) + 15] + 5 \div (144 - 12) + 25 = ?$
 (1) 1500 (2) 2100
 (3) 1200 (4) 4800
 (SSC CAPFs (CPO) SI & ASI, Delhi Police Exam. 05.06.2016) (1st Sitting)
- 93.** If + means x, - means +, x means $\div$, $\div$ means -, then what is the value of
 $50 + 10 - 50 \times 10 \div 125 = ?$
 (1) 380 (2) 56
 (3) 180 (4) -125
 (SSC CAPFs (CPO) SI & ASI, Delhi Police Exam. 05.06.2016) (1st Sitting)
- 94.** If 'x' means '-', '-' means ' $\div$ ', '+' means 'x' and ' $\div$ ' means '+', then what will be the value of the following expression?
 $16 \times 8 \div 4 - 3 + 9$
 (1) 10 (2) 19
 (3) 20 (4) 9
 (SSC CGL Tier-I (CBE) Exam. 27.08.2016) (1st Sitting)
- 95.** If '-' stands for division, '+' for multiplication, ' $\div$ ' for subtraction and 'x' for addition, which one of the following equations is correct?
 (1) $18 \div 3 \times 2 + 8 - 6 = 10$
 (2) $18 - 3 + 2 \times 8 \div 6 = 14$
 (3) $18 - 3 \div 2 \times 8 + 6 = 17$
 (4) $18 \times 3 + 2 \div 8 - 6 = 15$
 (SSC CGL Tier-I (CBE) Exam. 30.08.2016) (1st Sitting)
- 96.** If '+' means 'x', '-' means ' $\div$ ', 'x' means '-' and ' $\div$ ' means '+', then find the value of the following equation.
 $6 + 64 - 8 \div 45 \times 8$
 (1) 85 (2) 76
 (3) 87 (4) 75
 (SSC CGL Tier-I (CBE) Exam. 02.09.2016) (1st Sitting)
- 97.** If '+' means '-' and '-' means 'x', 'x' means ' $\div$ ' and ' $\div$ ' means '+' then
 $2 \div 6 \times 6 \div 2 = ?$
 (1) 1 (2) 0
 (3) 10 (4) 5
 (SSC CGL Tier-I (CBE) Exam. 03.09.2016) (1st Sitting)
- 98.** If + means $\div$, $\div$ means -, - means x and x means +, what will be the value of the following expression:
 $8 + 4 \div 3 \times 5 - 9 = ?$
 (1) 44 (2) 53
 (3) 62 (4) 64
 (SSC CGL Tier-I (CBE) Exam. 06.09.2016) (1st Sitting)
- 99.** If x means -, + means x, $\div$ means + and - means $\div$, then what is the value of the given expression?
 $175 - 25 \div 5 + 2 \times 3 + 10 = ?$
 (1) 77 (2) 160
 (3) 240 (4) 2370
 (SSC CGL Tier-I (CBE) Exam. 07.09.2016) (1st Sitting)
- 100.** If - stands for division, + for multiplication, $\div$ for subtraction and x for addition, then which of the following equations is correct?
 (1) $20 - 4 + 6 \div 9 \times 4 = 25$
 (2) $20 + 6 - 4 \times 9 \div 6 = 32$
 (3) $20 \div 9 \times 9 - 4 + 6 = 33$
 (4) $20 \times 4 - 6 - 4 + 9 = 20$
 (SSC CGL Tier-I (CBE) Exam. 01.09.2016) (1st Sitting)
- 101.** In a certain code language, '@' represents '+', '⊕' represents '-', 'a' represents ' $\div$ ' and 'θ' represents 'x'. Find out the answer to the following question:
 $8900a100 \oplus 504 \oplus 121a11 = ?$
 (1) 58 (2) 62
 (3) 158 (4) 205
 (SSC CGL Tier-I (CBE) Exam. 28.08.2016) (1st Sitting)
- 102.** If (-) stands for division, (+) stands for multiplication, ($\div$) stands for subtraction and (x) stands for addition, which one of the following equations is correct?
 (1) $100 + 5 - 10 \times 250 \div 200 = 100$
 (2) $200 + 10 - 20 \times 200 \div 100 = 150$
 (3) $50 \times 5 \div 10 + 100 - 75 = 50$
 (4) $300 + 5 - 20 \times 200 \div 100 = 200$
 (SSC CGL Tier-I (CBE) Exam. 03.09.2016) (1st Sitting)
- 103.** If '-' means 'x', 'x' means '+', '+' means ' $\div$ ' and ' $\div$ ' means '-', then what will be the value of $40 \times 12 + 3 - 6 \div 60 = ?$
 (1) 4 (2) 7
 (3) 16 (4) 44
 (SSC CGL Tier-I (CBE) Exam. 04.09.2016) (1st Sitting)
- 104.** If $25 + 5 \div 2 = 40$, and $35 + 5 \div 2 = 60$, then $45 + 5 \div 2 = ?$
 (1) 90 (2) 70
 (3) 60 (4) 80
 (SSC CGL Tier-I (CBE) Exam. 04.09.2016) (1st Sitting)
- 105.** If '+' means ' $\div$ ', ' $\div$ ' means '-', '-' means 'x' and 'x' means '+', then $48 + 16 \times 4 - 2 \div 8 = ?$
 (1) 3 (2) 6
 (3) 112 (4) -28
 (SSC CGL Tier-I (CBE) Exam. 09.09.2016) (1st Sitting)
- 106.** If '+' means minus, '-' means multiplication, ' $\div$ ' means plus, and 'x' means division, then $15 - 3 + 10 \times 5 \div 5 = ?$
 (1) 52 (2) 48
 (3) 22 (4) 5
 (SSC CGL Tier-I (CBE) Exam. 11.09.2016) (1st Sitting)
- 107.** If ' $\div$ ' stands for subtraction, '-' stands for addition, 'x' stands for division and '+' stands for multiplication, then which one of the following equation is correct?

| SYMBOLS & NOTATIONS |

- (1) $35 \div 4 - 25 \times 5 + 5 = 28$
 (2) $35 \div 4 - 25 \times 5 + 5 = 61$
 (3) $35 \div 4 - 25 \times 5 + 5 = 41$
 (4) $35 \div 4 - 25 \times 5 + 5 = 56$

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IInd Sitting)

- 108.** If '+' means 'x', '-' means '÷', 'x' means '-' and '÷' means '+', then what will be the value of $16 \div 4 \times 10 - 5 + 8 = ?$

- (1) 12 (2) 8
 (3) 4 (4) 2

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (IInd Sitting)

- 109.** If '+' means '÷', '-' means 'x', '÷' means '+' and 'x' means '-', then $36 \times 12 + 4 \div 6 + 2 - 3 = ?$

- (1) 2 (2) $6\frac{1}{2}$
 (3) 18 (4) 42

(SSC CGL Tier-I (CBE)

Exam. 08.09.2016) (IInd Sitting)

- 110.** If '+' means '/', '/' means '-', '-' means 'x', 'x' means '+', then $24 + 8/26 \times 6 = ?$

- (1) -10 (2) -3
 (3) 12 (4) 21

(SSC CGL Tier-I (CBE)

Exam. 09.09.2016) (IInd Sitting)

- 111.** If '+' means 'divided by', '-' means 'multiplied by', 'x' means 'minus' and '÷' means 'plus', which of the following will be the value of the expression :

- $16 \div 8 - 4 + 2 \times 4 = ?$
 (1) 16 (2) 28
 (3) 32 (4) 44

(SSC CGL Tier-I (CBE)

Exam. 10.09.2016) (IInd Sitting)

- 112.** If $\times$ means '+', + means '÷', - means 'x' and $\div$ means '-', then $8 \times 7 - 8 + 40 \div 2 = ?$

- (1) 1 (2) $\frac{37}{5}$
 (3) $8\frac{3}{5}$ (4) 44

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IInd Sitting)

- 113.** Which of the following interchanges of numbers would make the given equation correct?

- $8 \times 20 \div 3 + 9 - 5 = 38$
 (1) (8, 9) (2) (3, 5)
 (3) (3, 9) (4) (3, 8)

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (1st Sitting)

- 114.** If '+' stands for multiplication, '-' stands for addition, 'x' stands for

division and '÷' stands for subtraction, then which one of the following equations is correct ?

- (1) $9 + 8 - 4 \times 2 \div 18 = 56$
 (2) $9 \times 8 + 4 \div 2 - 18 = 26$
 (3) $9 \times 8 \div 4 + 2 - 18 = 200$
 (4) $9 - 8 \times 4 + 2 \div 18 = 203$

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (IInd Sitting)

- 115.** If - means ÷, + means x, ÷ means -, x means +, then which of the following equations is correct?

- (1) $43 \times 7 \div 5 + 4 - 8 = 25$
 (2) $48 \div 5 + 8 \times 10 - 2 = 03$
 (3) $36 \times 4 - 12 + 5 \div 3 = 420$
 (4) $42 + 5 \div 6 \times 8 - 3 = 28$

(SSC CGL Tier-I (CBE)

Exam. 31.08.2016) (IInd Sitting)

- 116.** If 'P' means '+', 'Q' means 'x', 'R' means '÷', and 'S' means '-', then $44Q9R12S6Q4P16 = ?$

- (1) 25 (2) 36
 (3) 112 (4) 12

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (IInd Sitting)

- 117.** If '+' means minus, '-' means multiplication, '÷' means plus, and 'x' means division, then

- $15 - 3 + 10 \times 5 \div 5 = ?$
 (1) 52 (2) 48
 (3) 22 (4) 5

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (1st Sitting)

- 118.** If '+' means 'x', '-' means '+', 'x' means '÷' and '÷' means '-', then $12 \times 2 + 6 - 7 \div 5 = ?$

- (1) 38 (2) 39
 (3) 40 (4) 37

(SSC Multi-Tasking Staff

Exam. 30.04.2017 1st Sitting)

TYPE-I (ii)

Directions (1) : In the following question you have to identify the correct response from the given premises stated according to following symbols.

- 1.** If $\rightarrow$ stands for addition, $\leftarrow$ stands for subtraction, $\uparrow$ stands for division, $\downarrow$ stands for multiplication, $\nearrow$ stands for equal to then which of the following alternatives is correct?

- (1) $2 \downarrow 5 \leftarrow 6 \rightarrow 2 \nearrow 6$
 (2) $5 \rightarrow 7 \leftarrow 3 \uparrow 2 \nearrow 4$
 (3) $3 \downarrow 6 \uparrow 2 \rightarrow 3 \leftarrow 6 \nearrow 5$
 (4) $7 \leftarrow 43 \uparrow 6 \downarrow 1 \nearrow 4$

(SSC Combined Graduate Level Prelim

Exam.11.05.2003 (1st Sitting)

TYPE-I (iii)

- 1.** If 'a' denotes '÷', 'b' denotes '+', 'c' denotes '-', 'd' denotes 'x', then $24a6d4b9c8 = ?$

- (1) 2 (2) 17
 (3) 34 (4) 19

(SSC Combined Matric Level (PRE)

Exam. 13.05.2001 (1st Sitting)

- 2.** A stands for 'addition', B for 'subtraction', C for 'division', D for 'multiplication', E for 'less than', F for 'greater than' and G for 'equal to'.

Out of the alternatives only one expression is correct according to the letter symbols. Identify that.

- (1) 18 C 2 A 4 B 6 G 9
 (2) 6 D 4 B 12 A 4 C 2 F 18
 (3) 10 C 2 D 4 B 6 E 12
 (4) 9 A 7 B 4 C 2 G 14

(SSC CPO Sub-Inspector

Exam. 03.09.2006)

- 3.** If P denotes +, Q denotes -, R denotes ÷ and S denotes x, then $18 S 36 R 12 Q 6 P 7 = ?$

- (1) 115 (2) 25
 (3) 55 (4) $\frac{648}{13}$

(SSC CPO Sub-Inspector

Exam. 03.09.2006)

Directions (4-5) : In the following questions you have to identify the correct response from the given premises stated according to following symbols

- 'A' stands for 'equal to'
 'B' stands for 'less than'
 'C' stands for 'greater than'
 'D' stands for 'not greater than'
 'E' stands for 'not equal to'
 'F' stands for 'not less than'

(SSC Combined Matric Level (PRE)

Exam. 24.10.1999 (1st, IInd Sitting)

- 4.** Permits (3XB2Y) and (2YDZ)

- (1) 3 X A Z (2) 3 X D Z
 (3) 3 X F Z (4) 3 X B Z

- 5.** Premises : (7X B 3Y) and (6Y D 2Z)

- (1) 7X C 2Z (2) 7X B 2Z
 (3) 7X D 2Z (4) 7X A 2Z

- 6.** If J represents +, K represents -, L represents ÷ and M represents x, then

$18 M 36 L 12 K 6 J 7 = ?$

| SYMBOLS & NOTATIONS |

(1) 115 (2) 55

(3) $\frac{648}{18}$ (4) 25

(SSC Combined Matric Level
(PRE) Exam. 21.05.2000
(1st Sitting (East Zone))

7.

+	-	×	÷	=	>	<
B	G	E	C	D	A	F

Of the four alternatives only one expression has the correct relationship. Identify that and indicate your answer.

- (1) 15 C 15 B 8 F 4 B 6 C 3
(2) 15 B 5 G 8 B 4 G 6 F 3
(3) 15 A 5 E 8 C 4 B 6 E 3
(4) 15 C 5 F 8 C 4 B 6 C 3

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)

8. If A represents +, B represents -, C represents × and D represents ÷, then which of the following statements is true?

- (1) 8B6D2A4C3 = 15
(2) 9C9B9D9A9 = 17
(3) 8A8B8C8 = - 48
(4) 3A3B3C3A3D3 = 4

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)

9. If A means 'x', D means '+' and G means '-', find the value of 7A4D4A3G2

- (1) 28 (2) 38
(3) 44 (4) 48

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (1st Sitting)

10. If L = +, M = -, N = ×, P = ÷, then 5 N 5 P 5 L 5 M 5 = ?

- (1) 0 (2) 5
(3) 10 (4) 15

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)

11. If L → +, M → -, N → ×, P → ÷, then 14 N 10 L 42 P 2 M 8 = ?

- (1) 153 (2) 216
(3) 248 (4) 251

(SSC Combined Matric Level
(PRE) Exam. 27.05.2001
(IInd Sitting (East Zone))

12. Some symbols are represented by alphabets as

+	-	×	÷	=	>	<
B	G	E	C	D	A	F

of the four alternatives only one expression has the correct relationship. Identify that :

(1) 18F 3B 6E 8G 4E 12

(2) 18C 3G 6B 8B 4D 12

(3) 18A 3E 6B 8G 4B 12

(4) 18C 3D 6B 8C 4G 12

SSC Combined Matric Level (Pre) Exam.
12.05.2002 (IInd Sitting)

13. If 'P' stands for '-', 'Q' stands for 'x', 'R' for '÷' and 'S' for '+', then what is the value of the given equation?

14 Q 3 P 12 S 4 R 2 = ?

- (1) 17 (2) 32
(3) 28 (4) 6

(SSC CISF Constable (GD)
Exam. 05.06.2011)

14. If L denotes ×, M denotes ÷, P denotes + and Q denotes -, then find the value of

16 P 24 M 8 Q 6 M 2 L 3 = ?

- (1) 6 (2) 8
(3) 10 (4) 12

(FCI Assistant Grade-II
Exam. 22.01.2012 Paper-I)

15. If A stands for 'addition', M for 'multiplication', D for 'division', G for 'greater than' and L for 'Lesser than' then which of the following will be logically correct?

- (1) 20A 4D 4L 4A 6D2
(2) 20 D 5G 8D 4A 6M3
(3) 20D 4A 4L 4A 2M3
(4) 20A 2G 10M 3A 12D 2

FCI Assistant Grade-III Exam. 05.02.2012
(Paper-I) East Zone (IInd Sitting)

16. X stands for +, Z stands for ÷, Y stands for -, and P stands for ×, then what is the value of 10 P 2 X 5 Y 5 ?

- (1) 10 (2) 15
(3) 20 (4) 25

FCI Assistant Grade-III Exam. 25.02.2012
(Paper-I) North Zone (1st Sitting)

17. If 'P' means '+', 'Q' means 'x', 'R' means '÷' and 'S' means '-', then :

44 Q 9 R 12 S 6 Q 4 P 16 = ?

- (1) 25 (2) 112
(3) 36 (4) 124

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

18. If 'R' stands for '-', 'A' stands for '+', 'B' stands for '÷' and 'C' stands for 'x', then what is the value, of the given equation? (BODMAS rule will not be applicable)

25 A 37 C 2 B 4 R 1 = ?

- (1) 32 (2) 35
(3) 30 (4) 27

(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)

19. If P denotes ÷, Q denotes ×, R denotes + and S denotes -, then 12 Q 15 P 3 R 4 S 6 = ?

- (1) 70 (2) 57
(3) 58 (4) 68

(SSC Graduate Level Tier-I
Exam. 19.05.2013, 1st Sitting)

20. You have to follow the symbolic interpretation to solve the question.

+ = Greater than

× = Equal to

- = Not less than

L = Not equal to

| = Less than

φ = Not greater than

Then if A - B φ C, which of the following is implied ?

- (1) A | B + C (2) A | B | C
(3) A + B - C (4) A φ B | C

(SSC Graduate Level Tier-I
Exam. 19.05.2013, 1st Sitting)

21. If A stands for +, Q stands for -, V stands for ×, R stands for ÷, then what is the value of the given equation ?

225 R 5 A 64 Q 13 V 6 = ?

- (1) 376 (2) 15
(3) 476 (4) 576

(SSC Graduate Level Tier-I
Exam. 19.05.2013, 1st Sitting)

22. If 'P' denotes 'multiplied by', 'T' denotes 'subtracted from', 'M' denotes 'added to' and 'B' denotes 'divided by' then : what should be the correct response of

12 P 6 M 15 T 16 B 4 ?

- (1) 70 (2) 75
(3) 83 (4) 110

(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)

23. If + = Greater than, φ = Not greater than, - = Not less than, × = Equal to, | = Less than and L = Not equal to, then of A | B × C which of the following is true ?

- (1) B + C | A (2) C - B + A
(3) B | A | C (4) A φ B | C

(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)

24. Identify one response which would be a correct inference from the given premises stated according to the following symbols :

'A' stands for not greater than

'B' stands for equal to

'C' stands for less than

| SYMBOLS & NOTATIONS |

- 'D' stands for not less than
'E' stands for not equal to
'F' stands for greater than
Premises (2 M B N) and (2N A 3K)
(1) 2M D 3K (2) 2M B 3K
(3) 2M C 3K (4) 2K B 3N
(SSC Graduate Level Tier-I
Exam. 19.05.2013, IIInd Sitting)
- 25.** 'B' stands for addition, 'G' stands for subtraction, 'E' stands for multiplication, 'C' stands for division, 'D' stands for equal to, 'A' stands for greater than, 'F' stands for less than. In each of the four alternatives, only one expression is correct according to the letter symbol. Identify that expression.
(1) 15 C 3 B 2 A 6 E 2
(2) 15 B 2 G 5 A 4 G 4
(3) 15 C 3 B 2 D 6 B 1
(4) 15 B 3 D 4 E 6
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IIInd Sitting)
- 26.** If A denotes +, B denotes – and C denotes $\times$, then
(10 C 4) A (4 C 4) B 6 = ?
(1) 46 (2) 50
(3) 55 (4) 58
(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014)
- 27.** If P denotes +, Q denotes –, R denotes $\div$, and S denotes $\times$, then: 18S36R12Q6P7 = ?
(1) 115 (2) 65
(3) 55 (4) 25
(SSC CGL Tier-I (CBE)
Exam. 11.09.2016) (Ist Sitting)
- 28.** If P denotes $\div$, Q denotes $\times$, R denotes + and S denotes – then what is the value of 18Q12P4R5S6 ?
(1) 64 (2) 53
(3) 81 (4) 24
(SSC CGL Tier-I (CBE)
Exam. 01.09.2016) (Ist Sitting)
- 29.** If A means +, B means $\times$, C means $\div$, D means –, then the value of given equation will be 9A2B6D4C2
(1) 16 (2) 19
(3) 27 (4) 30
(SSC CGL Tier-I (CBE)
Exam. 04.09.2016) (Ist Sitting)
- 30.** If D stands for $\times$, S stands for +, A stands for – and M stands for $\div$, what is the value of the given expression
28 D 6 S 34 M 2 A 8 D 6 ?

- (1) 558 (2) 3312
(3) 137 (4) 31
(SSC CGL Tier-I (CBE)
Exam. 30.08.2016) (IIInd Sitting)
- 31.** If 'A' stands for '+', 'B' stands for '-', 'C' stands for ' $\times$ ', what is the value of (10C4) A (4C4) B6 = ?
(1) 46 (2) 50
(3) 56 (4) 60
(SSC CGL Tier-I (CBE)
Exam. 31.08.2016) (IIInd Sitting)
- 32.** If a denotes $\times$, b denotes $\div$, c denotes + and d denotes –, then 8 a 3 c 24 b 12 d 19 = ?
(1) 17 (2) 7
(3) 14 (4) 8
(SSC CGL Tier-I (CBE)
Exam. 02.09.2016) (Ist Sitting)
- 33.** If A denotes –, C denotes $\times$, D denotes $\div$, E denotes +, then 14C3A12E4D2 = ?
(1) 6 (2) 17
(3) 28 (4) 32
(SSC CGL Tier-I (CBE)
Exam. 01.09.2016) (IIInd Sitting)
- 34.** If A denotes '+', B denotes '-', C denotes ' $\times$ ' and D denotes ' $\div$ ', then which of the following statement is true?
(1) 8B6D2A4C3 = 15
(2) 8A8B8C8 = – 48
(3) 9C9B9D9A9 = 17
(4) 3A3B3C3A3D3 = 4
(SSC CGL Tier-I (CBE)
Exam. 07.09.2016) (IIIrd Sitting)
- 35.** If P denotes multiply, T denotes subtraction, M denotes addition, B denotes division then 28 B 7 P 8 T 6 M 4 = ?
(1) 28 (2) 30
(3) 32 (4) 34
(SSC CGL Tier-I (CBE)
Exam. 08.09.2016) (IIIrd Sitting)
- 36.** If L denotes ' $\times$ ', M denotes ' $\div$ ', P denotes '+' and Q denotes '-', then 16 P 24 M 8 Q 6 M 2 L 3
(1) $\frac{13}{6}$ (2) $-\frac{1}{6}$
(3) $14\frac{1}{2}$ (4) 10
(SSC CGL Tier-I (CBE)
Exam. 10.09.2016) (IIIrd Sitting)
- 37.** If 'L' stands for '+', 'M' stands for '-', N stands for ' $\times$ ', P stands for ' $\div$ ' then 14 N 10 L 42 P 2 M 8 = ?

- (1) 153 (2) 216
(3) 248 (4) 251
(SSC CGL Tier-I (CBE)
Exam. 04.09.2016) (IIInd Sitting)
- 38.** If P denotes ' $\div$ ', Q denotes ' $\times$ ', R denotes '+' and S denotes '-', then 18 Q 12 P 4 R 5 S 6 = ?
(1) 53 (2) 54
(3) 57 (4) 95
(SSC CGL Tier-I (CBE)
Exam. 06.09.2016) (IIInd Sitting)
- 39.** If "K" means "subtracted from", "L" means "divided by", "M" means "added to" and "D" means "multiplied by", then 96 L 4 K 6 M 11 D 9 = ?
(1) 117 (2) 125
(3) 120 (4) 145
(SSC CHSL (10+2) Tier-I (CBE)
Exam. 16.01.2017) (IIInd Sitting)

TYPE-I (iv)

- 1.** If X stands for addition, V stands for subtraction, U stands for 'equal to', $\wedge$ stands for division, Σ stands for multiplication, $\circ$ stands for greater than and σ stands for less than. State which expression is true.
(1) 3 X 8 V 2 U 12 $\wedge$ 3
(2) 13 V 12 X 9 V 2 $\circ$ 5 Σ 1
(3) 2 Σ 3 Σ 4 σ 51 $\wedge$ 3
(4) 3 Σ 2 Σ 4 U 2 X 7 V 3
(SSC SAS Exam. 26.06.2010 (Paper-I))
- 2.** If +, –, $\times$, $\div$, =, > and < are represented as δ , γ , η , ω , β and α respectively, then which of the following is correct ?
(1) 3 γ 6 η 2 δ 8 $\cdot$ 4 ω 5
(2) 3 η 6 γ 2 δ 8 $\cdot$ 4 β 5
(3) 3 γ 6 $\cdot$ 2 δ 8 η 4 α 5
(4) 3 δ 6 $\cdot$ 2 γ 8 η 4 ω 5
(SSC CPO (SI, ASI & Intelligence Officer) Exam. 28.08.2011 (Paper-I))
- 3.** If 'S' is written as 'H', 'R' as '@', 'A' as ' ∇ ', 'M' as '#', 'T' as '\$' and 'E' as '%' then how is 'MASTER' written in that code ?
(1) #VH\$%@ (2) #HV\$%@
(3) #VH\$%@ (4) #VH%@\$
(SSC CGL Tier-I Exam. 19.10.2014
TF No. 022 MH 3)
- 4.** If # is the brother of @, $\neq$ is the daughter of @, \$ is the sister of #, and & is the brother of $\neq$, then who is the uncle of & ?

| SYMBOLS & NOTATIONS |

- (1) \$ (2) ≠
(3) @ (4) #

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam, 21.06.2015 IIInd Sitting)

5. In the question given below, the symbols used are as

Δ = equal to

⊗ = not equal to

+ = greater than

∅ = less than

– = not less than

O = not greater than

Read the question and mark the correct response.

If $X - Y - Z$ then it does not.

(1) $X \otimes Y \Delta Z$ (2) $X O Y + Z$

(3) $X \emptyset Y - Z$ (4) $X \emptyset Y + Z$

(SSC CGL Tier-I Exam, 09.08.2015 (IInd Sitting) TF No. 4239378)

6. If $x\% y = y^2 - x^2$, $x \$ y = x \div y^2$, $x \# y = 2xy$, then value of $\{(13 \% 5) \$ 6\} \# 15$ is

(1) 480 (2) 720

(3) -360 (4) -120

(SSC CAPFs (CPO) SI & ASI, Delhi Police Exam. 05.06.2016 (Ist Sitting))

7. In a certain code language, @ represents '+', ⊕ represents '-', α represents '÷' and 'Θ' represents '×'. Find out the answer to the following question.

$$107 \Theta 3 \oplus 64 \alpha 8 \oplus 2 \Theta 9 = ?$$

(1) 295 (2) 290

(3) 209 (4) 105

(SSC CGL Tier-I (CBE) Exam. 02.09.2016) (Ist Sitting)

8. If $324 \oplus 289 = 35$, $441 \oplus 484 = 43$, then $625 \oplus 400 = ?$

(1) 431 (2) 413

(3) 341 (4) 143

(SSC CGL Tier-I (CBE) Exam. 31.08.2016) (IInd Sitting)

9. If “#” means “subtraction”, “&” means “division”, “@” means “addition” and “%” means “multiplication”, then

$$217 \& 7 \# 3 @ 2 \% 7 = ?$$

(1) 21 (2) 19

(3) 22 (4) 42

(SSC CHSL (10+2) Tier-I (CBE) Exam. 15.01.2017) (IInd Sitting)

TYPE-I (v)

1. If $\square = 6$, $\triangle = 3$, $\diamond = 5$, $\lozenge = 4$, $\square = 8$, $\square = 10$, then

$$(\square \times \triangle) \div \diamond = ?$$

(1) $\square$ (2) $\diamond$

(3) $\square$ (4) $\triangle$

(SSC CPO (SI, ASI & Intelligence Officer) Exam. 28.08.2011 (Paper-I))

2. If $\square = 12$

$\triangle = 15$

$\square = 6$

$\nabla = 4$

$\bigcirc = 3$

$$\text{then } (\triangle + \square) \div \bigcirc = ?$$

(1) 8 (2) 9

(3) 7 (4) 6

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (Eastern Zone, Guwahati))

3. If $\square = 12$

$\triangle = 15$

$\square = 6$

$\nabla = 4$

$\bigcirc = 3$

$$\text{then } (\square \times \nabla) \div \square = ?$$

(1) 8 (2) 10

(3) 12 (4) 14

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (Ist Sitting))

4. If $\square = 12$, $\triangle = 15$, $\square = 6$, $\nabla = 4$, $\bigcirc = 3$, then write the answer in symbol :

$$\triangle + \bigcirc - \square = ?$$

(1) $\triangle$ (2) $\square$

(3) $\square$ (4) $\bigcirc$

(SSC Combined Matric Level (Pre) Exam. 12.05.2002 (IInd Sitting))

5. If $\square = 12$

$\triangle = 15$

$\square = 6$

$\nabla = 4$

$\bigcirc = 3$

then write the answer in symbol

$$\square \div \nabla = ?$$

(1) $\bigcirc$ (2) $\triangle$

(3) ∇ (4) $\square$

(SSC Combined Matric Level (Pre) Exam. 16.06.2002 (Re-Exam))

6. If rectangle = 12, triangle = 15, square = 6, parallelogram = 4 and circle = 3, solve the equation using the above values and answer in figures.

$$\frac{\text{rectangle} + \text{square}}{\text{triangle}} = ?$$

(1) $\frac{4}{5}$ (2) $\frac{3}{5}$

(3) $\frac{6}{5}$ (4) $\frac{2}{3}$

(SSC Multi-Tasking Staff Exam. 10.03.2013)

7. Certain numbers have symbols as given below.

1 2 3 4 5 6 7 8 9 0

$\bigcirc \square ([] () \backslash \cup$

What is the number indicated by these symbols ?

[] $\cup \cup$ []

(1) 5 6 9 0 7 (2) 4 5 0 9 6

(3) 4 5 9 0 6 (4) 4 7 0 9 5

(SSC Graduate Level Tier-I Exam. 21.04.2013, Ist Sitting)

8. Identify the symbols to be inserted to make the expression correct.

$$24 \triangle 4 \triangle 5 \triangle 4$$

(1) $\times + \times$ (2) $= \times +$

(3) $\times + =$ (4) $+ = \times$

(SSC Graduate Level Tier-I Exam. 21.04.2013, IInd Sitting)

9. Select the missing symbol from the given order :

$\square \bigcirc \nabla \triangle \square \bigcirc \nabla \triangle \square \bigcirc ?$

(1) $\square$ (2) $\triangle$

(3) $\bigcirc$ (4) ∇

(SSC Cabinet Secretariat RO (ECO), DFO (T) & DFO (GD) Tier-I Exam. 23.06.2013)

10. If # means <; ○ means >; □ means =; then which of the following follows definitely from a ○ b # c □ d?

| SYMBOLS & NOTATIONS |

- (1) $a \square c$ (2) $b \bigcirc d$
 (3) $b \square d$ (4) $b \# d$

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 20.12.2015 (1st Sitting) TF No. 9692918)

11. If ' * ' means subtraction; ' - ' means division, ' $\square$ ' means addition and ' % ' means multiplication, then find the value of :

$$13 \square 3 * 6 \% 8 - 4 \square 14 = ?$$

(1) 18 (2) 14
 (3) 12 (4) 8

(SSC CGL Tier-I (CBE) Exam. 06.09.2016) (IInd Sitting)

TYPE-I (vi)

Directions : In the following question you have to identify the correct response from the given premises stated according to following symbols.

1. If $> = \div$, $< = +$, $\wedge = -$, $\times = <$, $- = >$, $+ = =$ and $\vee = \times$

- (1) $6 > 3 < 2 \wedge 4 \vee 8 - 13$
 (2) $6 \wedge 3 < 2 > 4 \vee 8 + 13$
 (3) $6 \vee 3 \wedge 2 > 4 < 8 \times 13$
 (4) $6 \vee 3 > 2 < 4 \wedge 8 \times 13$

(SSC Combined Graduate Level Prelim Exam. 13.11.2005 (IInd Sitting))

2. If $> = \div$, $\vee = \times$, $< = +$, $\wedge = -$, $+ = =$, $\times = <$, $- = >$

- (1) $6 > 2 > 3 \wedge 8 \vee 4 + 13$
 (2) $6 \wedge 2 < 3 > 8 < 4 - 13$
 (3) $6 \vee 2 < 3 \wedge 8 > 4 \times 13$
 (4) $6 > 2 \vee 3 < 8 \wedge 4 + 13$

(SSC CPO Sub-Inspector Exam. 06.09.2009)

3. If $\div$ stands for 'greater than', $\times$ stands for 'addition', $+$ stands for 'division', $-$ stands for 'equal to', $>$ stands for 'multiplication', $=$ stands for 'less than', $<$ stands for 'minus' then which of the following alternatives is correct?

- (1) $5 > 2 < 1 - 3 \times 4 \times 1$
 (2) $5 < 2 \times 1 \div 3 > 4 \times 1$
 (3) $5 > 2 \times 1 - 3 > 4 < 1$
 (4) $5 + 2 \times 1 = 3 + 4 > 1$

SSC Combined Matric Level (Pre) Exam. 16.06.2002 (Re-Exam)

4. If $\times$ stands for addition, $<$ for subtraction, $+$ stands for division, $>$ stands for multiplication, $-$ stands for equation, $\div$ stands for greater than, and $=$ stands for less than, state which of the following is true?

- (1) $5 > 8 + 4 = 10 < 4 \times 8$
 (2) $3 \times 4 > 2 - 9 + 3 < 3$
 (3) $5 \times 3 < 3 \div 8 + 4 \times 1$
 (4) $3 \times 2 < 4 \div 16 > 2 + 4$

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

5. If '+' stands for 'multiplication', '<' stands for 'division', ' $\div$ ' stands for 'subtraction', '-' stands for 'addition' and ' $\times$ ' stands for 'greater than', identify which expression is correct.

- (1) $20 - 4 \div 4 + 8 < 2 \times 26$
 (2) $20 \times 8 + 15 < 5 \div 9 - 8$
 (3) $20 < 2 + 10 \div 4 - 6 \times 100$
 (4) $20 < 5 + 25 \div 10 - 2 \times 96$

(SSC Graduate Level Tier-I Exam. 21.04.2013, IInd Sitting)

6. In the following problem,

$=$ stands for $+$
 $+$ stands for $-$
 $\times$ stands for $=$
 $-$ stands for $>$
 $>$ stands for $+$
 $<$ stands for $\times$
 $\div$ stands for $<$

When these new symbols are substituted, only one will be wrong. Identify the **wrong** one.

- (1) $4 < 2 + 5 + 8 \times 5$
 (2) $4 = 2 + 5 > 8 \times 5$
 (3) $4 < 2 > 5 + 8 \times 5$
 (4) $4 > 2 < 5 + 8 - 5$

(SSC Graduate Level Tier-I Exam. 19.05.2013, 1st Sitting)

7. If $>$ denotes $+$, $<$ denotes $-$, $+$ denotes $\div$, $\wedge$ denotes $\times$, $-$ denotes $=$, $\times$ denotes $>$ and $=$ denotes $<$, choose the correct statement of the following.

- (1) $13 > 7 < 6 + 2 = 3 \wedge 4$
 (2) $9 > 5 > 4 - 18 + 9 > 16$
 (3) $9 < 3 < 2 > 1 \times 8 \wedge 2$
 (4) $28 + 4 \wedge 2 = 6 \wedge 4 + 2$

(SSC CGL Tier-I Re-Exam-2013, 27.04.2014)

8. You have to follow the symbolic interpretation to solve the problem

$+$ = greater than
 $-$ = not less than
 $|$ = less than
 ϕ = not greater than
 $\times$ = equal to
 $\angle$ = not equal to

If $x \angle y + z$, which of the following implied ?

- (1) $x \times y | z$ (2) $x - y \times z$
 (3) $x \angle y \phi z$ (4) $x - y \angle z$

(SSC CAPFs SI, CISF ASI & DP SI Exam, 21.06.2015 IInd Sitting)

9. Identify the correct response from the given premises stated according to following symbols.

'A' stands for not less than ($\nless$)

'B' stands for not equal to ($\neq$)

'C' stands for not greater than ($\ngtr$)

'D' stands for greater than ($>$)

'E' stands for less than ($<$)

'F' stands for equal to ($=$)

Premises : $4YF3X$ and $3XF6Z$

- (1) $2 Y D 3 Z$ (2) $2 Y E 3 Z$

- (3) $4 Y B 5 Z$ (4) $2 Y F 3 Z$

(SSC CGL Tier-I Exam, 16.08.2015 (1st Sitting) TF No. 3196279)

10. If '+' stands for ' $\div$ ', '-' stands for ' $=$ ', ' $\times$ ' stands for '+', ' $\div$ ' stands for 'greater than', ' $=$ ' stands for 'less than', ' $>$ ' stands for multiplication and ' $<$ ' stands for subtraction, then which of the following alternatives is correct?

- (1) $5 \div 2 \times 1 = 3 + 4 > 1$
 (2) $5 > 2 \times 1 - 3 > 4 < 1$
 (3) $5 \times 2 < 1 - 3 < 4 \times 1$
 (4) $5 < 2 \times 1 \div 3 > 4 \times 1$

(SSC CGL Tier-I (CBE) Exam. 27.08.2016) (1st Sitting)

11. If '+' means subtraction, ' $\div$ ' means addition, '<' means multiplication and '>' means division, then find the value of the given statement. The value of $9 \div 7 < 8 > (4 > 2) + 5$ will be

- (1) 32 (2) 18
 (3) 16 (4) 11

(SSC CGL Tier-I (CBE) Exam. 29.08.2016) (IInd Sitting)

12. If '+' means division, '-' means multiplication, ' $\div$ ' means subtraction, ' $\times$ ' means addition and '<' means less than, then which of the following is false?

- (1) $(10 + 2) \div 7 < (10 \div 7) + 2$
 (2) $(10 - 7) \times 2 < (10 \times 2) - 7$
 (3) $(10 \times 7) - 2 < (10 - 2) \times 7$
 (4) $(10 \div 2) + 7 < (10 + 7) \times 2$

(SSC CGL Tier-I (CBE) Exam. 31.08.2016) (1st Sitting)

TYPE-I (vii)

1. In the following question you have to identify the correct response from the given premises stated according to the following symbols.

If $\div$ stands for 'greater than', $\times$ stands for 'addition', $+$ stands for 'division', $-$ stands for 'equal to', $>$ stands for 'multiplication', $=$ stands for 'less than', $<$ stands for 'minus', then which of the following alternatives is correct?

- (1) $3 + 2 < 4 \div 6 > 3 \times 2$
 (2) $3 \times 2 < 4 \div 6 + 3 < 2$
 (3) $3 > 2 < 4 - 6 \times 3 \times 2$
 (4) $3 \times 2 \times 4 = 6 + 3 < 2$

(SSC Combined Graduate Level Prelim Exam. 11.05.2003 (IInd Sitting))

Direction (2) : In the following question you have to identify the correct response from the given premises stated according to following symbols.

2. If '+' stands for 'division', '-' stands for 'equal to', 'x' stands for 'addition', ' $\div$ ' stands for 'greater than', ' $\leq$ ' stands for 'less than', '>' stands for 'multiplication', and '<' stands for 'subtraction', then which of the following alternatives is correct ?

- (1) $5 + 2 \times 1 = 3 + 4 > 1$
 (2) $5 > 2 \times 1 - 3 > 4 < 1$
 (3) $5 \times 2 < 1 - 3 < 4 \times 1$
 (4) $5 < 2 \times 1 \div 3 > 4 \times 1$

(SSC Combined Graduate Level Prelim Exam. 08.02.2004 (Ist Sitting))

3. If $\oplus$ stands for +, $\neg$ stands for $\otimes$, $\ominus$ stands for $\div$, $\neg$ stands for $\times$, $\nearrow$ stands for =, which one is correct ?

- (1) $2 \neg 5 \neg 6 \oplus 2 \nearrow 6$
 (2) $5 \oplus 7 \neg 4 \quad 2 \nearrow 3$
 (3) $3 \neg 6 \quad 2 \oplus 3 \neg 6 \nearrow 5$
 (4) $7 \neg 43 \quad 6 \neg 1 \nearrow 4$

(SSC CPO Sub-Inspector Exam. 09.11.2008)

4. Some equations are solved on the basis of a certain system. Find out the correct answer for the unsolved equation on that basis.
 If $8 + 8 = 72$, $5 + 5 = 30$ and $7 + 7 = 56$, what is $6 + 6 = ?$

- (1) 40 (2) 42
 (3) 30 (4) 36

(FCI Assistant Grade-II Exam. 22.01.2012 Paper-I)

5. Some equations are solved on the basis of a certain system. Find out the correct answer for the unsolved equation on that basis.
 If $3 \div 5 = 5$, $4 \div 7 = 8$, $8 \div 7 = 6$ then, what should $9 \div 6$ be ?

- (1) 4 (2) 9
 (3) 5 (4) 6

(FCI Assistant Grade-III Exam. 25.02.2012 (Paper-I) North Zone (Ist Sitting))

6. Some equations are solved on the basis of a certain system. On the same basis find out the correct answer for the unsolved equation.
 If $8 \times 2 = 61$; $8 \times 5 = 04$, what is $8 \times 10 = ?$

- (1) 80 (2) 08
 (3) 8 (4) 0

(FCI Assistant Grade-III Exam. 05.02.2012 (Paper-I) East Zone (IInd Sitting))

7. If SENT is written as '+ ^ x -' and ANT is written as '* x -', then how is TEN written in that code ?

- (1) $x + -$ (2) $- ^ x$
 (3) $* - x$ (4) $- x ^$

(SSC Graduate Level Tier-I Exam. 19.05.2013, Ist Sitting)

8. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.
 $5 + 7 + 2 = 725$, $6 + 9 + 0 = 906$, $8 + 4 + 3 = ?$

- (1) 815 (2) 384
 (3) 438 (4) 834

(SSC Multi-Tasking (Non-Tech.) Staff Exam. 16.02.2014)

9. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$7 - 4 - 1 = 714$,
 $9 - 2 - 3 = 932$
 $8 - 0 - 4 = ?$

- (1) 804 (2) 840
 (3) 408 (4) 480

(SSC Multi-Tasking Staff (Patna) Exam. 16.02.2014)

10. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$678 = 366$, $567 = 255$, $946 = ?$
 (1) 334 (2) 499
 (3) 699 (4) 634

(SSC Multi-Tasking (Non-Tech.) Staff Exam. 23.02.2014, IInd Sitting)

11. If $1 + 4 = 9$, $2 + 8 = 18$, $3 + 6 = 15$ then $7 + 8 = ?$

- (1) 41 (2) 23
 (3) 30 (4) 32

(SSC CGL Tier-I Re-Exam-2013, 27.04.2014)

12. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$4 \times 5 = 42$,
 $5 \times 6 = 56$,
 $6 \times 7 = 72$,
 $7 \times 8 = ?$

- (1) 84 (2) 90
 (3) 92 (4) 102

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam. 22.06.2014)

13. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$58 \times 12 = 4$, $37 \times 96 = 5$,
 $11 \times 20 = 2$, $42 \times 12 = ?$

- (1) 2 (2) 3
 (3) 4 (4) 5

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, Ist Sitting)

14. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$5 \times 8 = 28$, $3 \times 7 = 12$, $8 \times 6 = 35$, $13 \times 13 = ?$

- (1) 169 (2) 130
 (3) 140 (4) 144

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, IInd Sitting)

Directions (15-16) : In each of the following questions, some equations are solved on the basis of certain system. Find out the correct answer for the unsolved equation on that basis.

(SSC GL Tier-I Exam. 19.10.2014, Ist Sitting)

15. $2 \times 4 \times 6 = 4$; $9 \times 3 \times 7 = 13$;
 $4 \times 7 \times 6 = 3$; $9 \times 7 \times 8 = ?$

- (1) 10 (2) 09
 (3) 08 (4) 07

16. $3 \times 5 \times 7 \times 2 = 24$, $2 \times 4 \times 6 \times 8 = 22$, $4 \times 4 \times 8 \times 9 = ?$

- (1) 33 (2) 25
 (3) 144 (4) 1152

| SYMBOLS & NOTATIONS |

17. If $7 \times 8 = 49$, $4 \times 4 = 12$ and $6 \times 4 = 18$, what will 9×6 be ?

(1) 54
(2) 50
(3) 45
(4) None of the above

(SSC GL Tier-I Exam. 19.10.2014)

18. Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$$5 \odot 3 = -7; 3 \odot 7 = -11; 7 \odot 11 = ?$$

(1) - 59 (2) 77
(3) - 15 (4) 18

(SSC GL Tier-I Exam. 26.10.2014)

19. Some equations are solved on the basis of a certain system. On the same basis, find out the correct answer for the unsolved equation.

$$2 \times 3 \times 4 = 432,$$

$$5 \times 6 \times 7 = 765$$

$$7 \times 8 \times 9 = 987$$

$$2 \times 5 \times 7 = ?$$

(1) 572 (2) 752
(3) 725 (4) 257

(SSC CHSL (10+2) DEO & LDC Exam. 09.11.2014)

20. If $2 \times 16 = 8$; $8 \times 8 = 1$; $6 \times 12 = 2$, then $12 \times 144 = ?$

(1) 11 (2) 12
(3) 16 (4) 24

(SSC CHSL (10+2) DEO & LDC Exam. 16.11.2014, Patna Region : (1st Sitting))

21. Some equations are solved on the basis of a certain system. Using the same, solve the unsolved equation.

$$\text{If } 10 - 3 = 12, 12 - 4 = 13, 14 - 5 = 14, \text{ then } 16 - 6 = ?$$

(1) 10 (2) 15
(3) 16 (4) 18

(SSC CHSL (10+2) DEO & LDC Exam. 16.11.2014, Patna Region : (1st Sitting))

22. If 1 candle in box number 1 is placed in box number 2, then box - 2 has twice the number of candles that box 1 has. If 1 candle from box-2 is placed in box-1, then box-2 and box-1 have the same number of candles.

How many candles were there in box-1 and box-2 ?

Box - 1 Box - 2

(1) 5 : 3

(2) 7 : 5

(3) 6 : 4

(4) 5 : 7

(SSC CGL Tier-I Exam. 19.10.2014 TF No. 022 MH 3)

23. If 0, 1, 2, 3, ..., 9 is written as $a, b, c, d, \dots, j$ then find $dc \times f - (bf - d) \times d$.

(1) bcf (2) abe
(3) abb (4) bce

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 20.12.2015 (1st Sitting) TF No. 9692918)

24. Some equations have been solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$$\text{If } 29 \times 13 = 14, 76 \times 26 = 34, \text{ then } 64 \times 14 = ?$$

(1) 39 (2) 32
(3) 26 (4) 54

(SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.07.2016)

25. If $4 \times 5 \times 2 = 524$, $3 \times 7 \times 2 = 723$ and $6 \times 8 \times 7 = 876$ then $9 \times 4 \times 5 = ?$

(1) 495 (2) 459
(3) 549 (4) 954

(SSC CGL Tier-I (CBE) Exam.10.09.2016)

26. Some equations have been solved on the basis of certain system. Find the correct answer for the unsolved equation on that basis.

$$\text{If } 98 - 39 - 27 = 31, 87 - 38 - 34 = 20, \text{ then } 79 - 25 - 12 = ?$$

(1) 51 (2) 22
(3) 42 (4) 15

(SSC CGL Tier-I (CBE) Exam.11.09.2016) (1st Sitting)

27. In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$$53 - 34 = 5334$$

$$65 - 46 = 6456$$

$$75 - 24 = ?$$

(1) 7542 (2) 7524
(3) 7452 (4) 7254

(SSC CHSL (10+2) Tier-I (CBE) Exam. 08.09.2016) (1st Sitting)

28. In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$$6 \times 2 \times 9 = 269$$

$$8 \times 7 \times 1 = 781$$

$$4 \times 1 \times 3 = ?$$

(1) 431 (2) 413
(3) 341 (4) 143

(SSC CGL Tier-I (CBE) Exam. 09.09.2016) (1st Sitting)

29. In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$$3 \times 4 \times 5 = 435$$

$$4 \times 3 \times 2 = 342$$

$$2 \times 3 \times 4 = ?$$

(1) 324 (2) 342
(3) 432 (4) 243

(SSC CGL Tier-I (CBE) Exam. 27.08.2016) (1st Sitting)

30. If $17 + 17 = 2895$

$$18 + 18 = 3245$$

$$19 + 19 = 3615$$

$$\text{then, } 23 + 23 = ?$$

(1) 5765 (2) 5295
(3) 2565 (4) 4005

(SSC CGL Tier-I (CBE) Exam. 27.08.2016) (IInd Sitting)

31. If $24 \times 2 = 84$, and $32 \times 3 = 69$, then $13 \times 3 = ?$

(1) 38 (2) 93
(3) 16 (4) 10

(SSC CGL Tier-I (CBE) Exam. 31.08.2016) (1st Sitting)

32. If $4 + 3 = 25$ and $8 + 4 = 80$, then, $3 + 2 = ?$

(1) 15 (2) 10
(3) 13 (4) 12

(SSC CGL Tier-I (CBE) Exam. 01.09.2016) (1st Sitting)

33. If $12 \times 16 = 188$ and $14 \times 18 = 248$, then find the value of $16 \times 20 = ?$

(1) 320 (2) 360
(3) 316 (4) 318

(SSC CGL Tier-I (CBE) Exam. 02.09.2016) (1st Sitting)

34. If $64 + 7 = 460$

$$43 + 8 = ?$$

$$25 + 8 = 212$$

(1) 360 (2) 376
(3) 332 (4) 356

(SSC CGL Tier-I (CBE) Exam. 02.09.2016) (IInd Sitting)

| SYMBOLS & NOTATIONS |

- 35.** If $6 \times 9 \times 3 = 963$ and $4 \times 8 \times 5 = 845$, then $9 \times 4 \times 7 = ?$

(1) 974 (2) 479
(3) 497 (4) 749

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IInd Sitting)

- 36.** If, $4 \times 3 = 14$; $5 \times 4 = 18$; $6 \times 5 = 22$, then find the value of 7×6 .

(1) 20 (2) 26
(3) 30 (4) 42

(SSC CGL Tier-I (CBE)

Exam. 04.09.2016) (Ist Sitting)

- 37.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$8 \times 5 \times 0 = 805$, $7 \times 4 \times 6 = 764$, $6 \times 8 \times 9 = ?$

(1) 689 (2) 698
(3) 968 (4) 986

(SSC CGL Tier-I (CBE)

Exam. 06.09.2016) (Ist Sitting)

- 38.** If $34 = 39304$, $27 = 19683$, then $13 = ?$

(1) 2197 (2) 10648
(3) 56743 (4) 17576

(SSC CAPFs (CPO) SI & ASI,
Delhi Police Exam. 05.06.2016)
(Ist Sitting)

- 39.** If $1 \times 3 \times 5 = 1925$ and $7 \times 9 \times 11 = 4981121$, then find the value of $19 \times 21 \times 23 = ?$

(1) 361529441 (2) 361441289
(3) 441361289 (4) 361441529

(SSC CGL Tier-I (CBE)

Exam. 01.09.2016) (Ist Sitting)

- 40.** Given equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis : $2 + 4 + 6 = 48$ and $3 + 2 + 8 = 48$, then $2 + 5 + 7 = ?$

(1) 48 (2) 70
(3) 14 (4) 59

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (Ist Sitting)

- 41.** Some equations are solved on the basis of a certain system. On the same basis, find out the correct answer from amongst the four alternatives to the unsolved equation.

$1 \times 2 \times 3 = 231$
 $3 \times 4 \times 5 = 453$
 $5 \times 6 \times 7 = ?$

(1) 657 (2) 675
(3) 756 (4) 765

(SSC CGL Tier-I (CBE)

Exam. 28.08.2016) (Ist Sitting)

- 42.** If $5 \times 4 \times 0 = 405$

$3 \times 2 \times 8 = 283$

then $1 \times 7 \times 6 = ?$

(1) 617 (2) 716
(3) 167 (4) 761

(SSC CGL Tier-I (CBE)

Exam. 30.08.2016) (IInd Sitting)

- 43.** If $56 \times 11 = 9$, $37 \times 13 = 6$, $42 \times 12 = 3$, then find the value of 87×77 .

(1) 1 (2) 2
(3) 3 (4) 4

(SSC CGL Tier-I (CBE)

Exam. 30.08.2016) (IInd Sitting)

- 44.** Some equations are solved on the basis of certain system. Find out the correct answer for unsolved equation on that basis :

$4 \times 5 \times 8 = 584$

$7 \times 3 \times 9 = 397$

$9 \times 7 \times 3 = ?$

(1) 397 (2) 793
(3) 973 (4) 739

(SSC CGL Tier-I (CBE)

Exam. 31.08.2016) (IInd Sitting)

- 45.** If $879 = 8$, $625 = 1$, $586 = 9$, then $785 = ?$

(1) 6 (2) 7
(3) 8 (4) 9

(SSC CGL Tier-I (CBE)

Exam. 01.09.2016) (IInd Sitting)

- 46.** If $84 + 96 = 4842$,

then $36 + 78 = ?$

(1) 3918 (2) 3678
(3) 3819 (4) 1839

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (IInd Sitting)

- 47.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$462 = 551$

$398 = 487$

$856 = ?$

(1) 745 (2) 773
(3) 945 (4) 743

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (IIIrd Sitting)

- 48.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$7 \times 5 \times 6 = 576$,

$4 \times 2 \times 5 = 245$,

$8 \times 2 \times 5 = ?$

(1) 258 (2) 285

(3) 582 (4) 852

(SSC CGL Tier-I (CBE)

Exam. 08.09.2016) (IIIrd Sitting)

- 49.** A certain system is followed to solve the problem . Accordingly find out the correct answer from the alternatives for the unsolved equation.

$7 \times 4 \times 9 = 479$

$9 \times 5 \times 2 = 592$

$6 \times 9 \times 5 = 965$

$8 \times 6 \times 2 = ?$

(1) 286 (2) 682

(3) 628 (4) 268

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IInd Sitting)

- 50.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$15 \times 26 = 6512$

$29 \times 36 = 6923$, $46 \times 54 = ?$

(1) 5464 (2) 4645

(3) 4564 (4) 4465

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IInd Sitting)

- 51.** If $23 \times 16 = 184$, $37 \times 10 = 185$, then $85 \times 12 = ?$

(1) 511 (2) 610

(3) 510 (4) 410

(SSC CGL Tier-I (CBE)

Exam. 06.09.2016) (IInd Sitting)

- 52.** Some equations are solved on the basis of certain system. Find the correct answer for the unsolved equation on that basis.

$5 \times 4 \times 3 = 70$,

$6 \times 5 \times 4 = 140$

$7 \times 6 \times 5 = ?$

(1) 210 (2) 220

(3) 230 (4) 240

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (IInd Sitting)

- 53.** Some equations have been solved on the basis of a certain pattern. Find the correct answer for the unsolved equation on that basis.

$8 \times 7 \times 6 = 765$

$5 \times 3 \times 2 = 421$

$9 \times 6 \times 4 = ?$

(1) 583 (2) 853

(3) 841 (4) 481

(SSC CGL Tier-I (CBE)

Exam. 08.09.2016) (IInd Sitting)

- 54.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$$7 \times 6 \times 8 = 678$$

$$8 \times 9 \times 7 = 987$$

$$6 \times 5 \times 7 = 567$$

$$5 \times 4 \times 6 = ?$$

(1) 456 (2) 564

(3) 645 (4) 654

(SSC CGL Tier-I (CBE)

Exam. 09.09.2016) (IInd Sitting)

- 55.** If, $6 \times 4 = 12$

$$4 \times 12 = 24$$

$$12 \times 6 = 36 \text{ then } 6 \times 9 = ?$$

(1) 35 (2) 24

(3) 27 (4) 31

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IInd Sitting)

- 56.** Some equations have been solved on the basis of certain system. Find out the correct answer for the unsolved equation on that basis.

$$9 \times 7 \times 4 = 794, 3 \times 4 \times 6 = 436,$$

$$4 \times 2 \times 7 = ?$$

(1) 742 (2) 247

(3) 724 (4) 472

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (Ist Sitting)

- 57.** In this question, some equations are solved on the basis of a certain system. On the same basis find out the correct answer from amongst the four alternatives for the unsolved equation.

$$8 + 5 - 5 = 45$$

$$12 + 6 - 5 = 77$$

$$14 + 5 - 10 = ?$$

(1) 60 (2) 80

(3) 58 (4) 76

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (IInd Sitting)

- 58.** A certain system is followed to solve the problem. Accordingly find out the correct answer from the alternatives for the unsolved equation.

$$7 \times 4 \times 9 = 479$$

$$9 \times 5 \times 2 = 592$$

$$6 \times 9 \times 5 = 965$$

$$8 \times 6 \times 2 = ?$$

(1) 286 (2) 682

(3) 628 (4) 268

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (Ist Sitting)

TYPE-II

- 1.** Which one of the following is correct?

$$6 * 3 * 4 * 45$$

(1) $\div, +, >$ (2) $\div, >, +$

(3) $>, \div, +$ (4) $+, >, \div$

(SSC Combined Graduate Level Prelim Exam. 27.07.2008
(Second- Sitting)

- 2.** In the following question, * stands for any of the mathematical signs at different places, which are given as choices under each question. Select the choice with the correct sequence of signs which when substituted makes the question as a correct equation.

$$24 * 4 * 5 * 4$$

(1) $\times + =$ (2) $= \times +$

(3) $+ \times =$ (4) $= + \times$

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010
(Ist Sitting)

- 3.** $25 * 2 * 6 = 4 * 11 * 0$

Which set of symbols can replace * ?

(1) $\times, -, \times, +$ (2) $+, -, \times, +$

(3) $\times, +, \times, -$ (4) $\times, +, +, \times$

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010
(IInd Sitting)

- 4.** Which one of the following responses is correct ?

$$8 * 5 * 27 * 3 * 16$$

(1) $\times, =, -, +$ (2) $-, =, \times, +$

(3) $\times, =, +, -$ (4) $+, -, =, \times$

(SSC CISF ASI

Exam. 29.08.2010 (Paper-I)

- 5.** Which one of the following is correct ?

$$96 * 6 * 8 * 2$$

(1) $\div, =, \times$ (2) $\times, =, \div$

(3) $=, \div, \times$ (4) $=, \times, \div$

(SSC CPO Sub-Inspector

Exam. 12.12.2010 (Paper-I)

Directions (6-7) : What should be the correct signs of the equation to arrive at the given answer ?

(SSC Combined Matric Level (PRE) Exam. 21.05.2000
(Ist Sitting (East Zone)

- 6.** $17 * 3 * 6 = 45$

(1) $\times, =, -$ (2) $-, \times, =$

(3) $=, \times, -$ (4) $\times, -, =$

- 7.** $3 * 2 * 1 * 7$

(1) $\times, +, =$ (2) $+, \times, =$

(3) $=, \times, +$ (4) $\times, =, +$

Direction (8) : In the following question you have to identify the correct response from the given premises stated according to following symbols.
(SSC Combined Matric Level (PRE) Exam.
13.05.2001 (Ist Sitting)

- 8.** $12 * 3 * 4 * 8 * 0$

(1) $- + +$ (2) $\div + +$

(3) $- - -$ (4) $\div + -$

- 9.** Which alternative clearly indicates the rule followed in the following set of numbers?

$$7 * 4 * 8 * 2 * 24$$

(1) $-, \times, \times$ (2) $-, \times, \div$

(3) $\times, -, \div$ (4) $\times, \div, -$

(SSC Combined Matric Level (PRE) Exam.
13.05.2001 (IInd Sitting)

- 10.** Select the correct set of symbols which will fit in the given equation $5 * 0 * 3 * 5 * 20$

(1) $+ - \times$ (2) $\times + \times$

(3) $- + \times$ (4) $\times \times \times$

(SSC Combined Matric Level (PRE) Exam. 27.05.2001
(IInd Sitting (East Zone)

Directions (11-12) : Select the correct set of symbols which will fit in the given equation.

SSC Combined Matric Level (Pre) Exam.
05.05.2002 (IInd Sitting)
(North Zone Delhi)

- 11.** $23 * 26 * 27$

(1) $+ 3 =; \times 1 =$

(2) $\times 3 =; \times 1 =$

(3) $+ 3 =; + 1 =$

(4) $\times 3 =; + 1 =$

- 12.** $65 * 40 * 11 * 36$

(1) $- \text{ and } +$ (2) $\times \text{ and } \div$

(3) $\div \text{ and } +$ (4) $+ \text{ and } \times$

- 13.** Choose proper signs for sequential operations to produce the resultant figure:

$$31 * 1 * 2 * 1 \rightarrow 16$$

(1) $\times \div \times =$ (2) $- + \div =$

(3) $+ - \times =$ (4) $- \div + =$

SSC Combined Matric Level (Pre) Exam.
12.05.2002 (Ist Sitting)

- 14.** The symbols for addition (+), subtraction (-), multiplication (x) and division ($\div$) will have to be inserted in the blank * in order to get answer 3 as shown in the equation :

$$20 \boxed{*} 5 \boxed{*} 8 \boxed{*} 2 \boxed{*} 17 = 3$$

| SYMBOLS & NOTATIONS |

Find out which set below is correct.

- (1) $- + \div \times$ (2) $+ - \times \div$
 (3) $\times - \div +$ (4) $\div + \times -$

SSC Combined Matric Level
 (Pre) Exam. 30.07.2006
 (IInd Sitting) (Central Zone)

- 15.** In the following equation, select correct combination of mathematical signs to replace * signs and to balance the equation

$$16 * 4 * 5 * 9 * 1$$

- (1) $+ \div = \times$ (2) $\div + = \times$
 (3) $\times = + -$ (4) $+ \times = \div$

(SSC Higher Secondary Level
 Data Entry Operator & LDC
 Exam. 27.11.2010)

- 16.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation.

$$7 * 5 * 5 * 4 * 10$$

- (1) $+ \div - =$
 (2) $\times - = \times$
 (3) $\times + = \times$
 (4) $+ \times \div =$

(SSC Higher Secondary Level
 Data Entry Operator & LDC
 Exam. 28.11.2010 (1st sitting))

- 17.** If $34 * 12 = 23$, $28 * 76 = 52$, $97 * 39 = 68$ then what should $37 * 73$ be ?

- (1) 32 (2) 25
 (3) 86 (4) 55

(SSC Higher Secondary Level
 Data Entry Operator & LDC
 Exam. 28.11.2010 (1st sitting))

- 18.** Select the correct combination of mathematical signs to replace * signs so as to balance the equation.

$$8 * 8 * 1 * 11 * 11$$

- (1) $+ = \div -$ (2) $\times + = \div$
 (3) $\div \times + =$ (4) $- + = \div$

(SSC Higher Secondary Level
 Data Entry Operator & LDC
 Exam. 28.11.2010 (IInd sitting))

- 19.** Substitute the correct mathematical symbols in place of * in the following equation:

$$16 * 4 * 5 * 14 * 6$$

- (1) $\div - = \times$ (2) $- \times + =$
 (3) $\div \times = +$ (4) $\div + = -$

(SSC Multi-Tasking (Non-Technical) Staff
 Exam. 20.02.2011)

- 20.** Substitute the arithmetical signs in the place of * in the following equation :

$$7 * 7 * 2 * 1 = 12$$

- (1) $\times - \div$ (2) $+ - \times$
 (3) $\times - +$ (4) $+ \times -$

(SSC Multi-Tasking (Non-Technical)
 Staff Exam. 27.02.2011)

- 21.** Some equations have been solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis. If $9 * 7 = 32$, $13 * 7 = 120$, $17 * 9 = 208$, then $19 * 11 = ?$

- (1) 150 (2) 180
 (3) 210 (4) 240

(SSC Combined Graduate Level Tier-1
 Exam. 26.06.2011 (1st Sitting))

- 22.** Which sequence of mathematical symbols can replace * in the given equation :

$$8 * 5 * 9 * 31$$

- (1) $- \times =$ (2) $- = \times$
 (3) $= \times -$ (4) $\times - =$

(SSC Stenographer (Grade 'C' & 'D')
 Exam. 16.10.2011)

- 23.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$4 * 6 * 6 * 2 * 20$$

- (1) $+ \div = \div$ (2) $\times - + =$
 (3) $+ - = \div$ (4) $- + = \div$

SSC (10+2) Level Data Entry Operator
 & LDC Exam. 04.12.2011 (1st Sitting
 (North Zone))

- 24.** Select the correct combination of mathematical signs to replace signs and to balance the given equation.

$$8 \quad 5 \quad 2 \quad 72 \quad 4$$

- (1) $= \times + \div$ (2) $\times = + \div$
 (3) $\times + = \div$ (4) $+ \times = \div$

SSC (10+2) Level Data Entry Operator
 & LDC Exam. 04.12.2011 (IInd Sitting
 (North Zone))

- 25.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$15 * 3 * 5 * 20 * 2$$

- (1) $+ - = \div$ (2) $\times - = \times$
 (3) $+ = + \times$ (4) $\times - = \div$

SSC (10+2) Level Data Entry Operator
 & LDC Exam. 04.12.2011
 (1st Sitting (East Zone))

- 26.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$2 * 3 * 2 * 4 * 8$$

- (1) $+ - + =$ (2) $\times - + =$
 (3) $- + \times =$ (4) $\times + - =$

SSC (10+2) Level Data Entry
 Operator & LDC Exam. 04.12.2011
 (IInd Sitting (East Zone))

- 27.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$16 * 2 * 24 * 3 * 6$$

- (1) $+ = - \div$ (2) $\times - + =$
 (3) $+ \div = \div$ (4) $- - \div =$

SSC (10+2) Level Data Entry
 Operator & LDC Exam. 11.12.2011
 (1st Sitting (Delhi Zone))

- 28.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation

$$16 * 4 * 3 * 4 * 13$$

- (1) $\div \times - =$ (2) $\times - + =$
 (3) $+ = \div \times$ (4) $- \times \div =$

SSC (10+2) Level Data Entry
 Operator & LDC Exam. 11.12.2011
 (IInd Sitting (Delhi Zone))

- 29.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$6 * 15 * 10 * 3 * 12$$

- (1) $\div + = \times$ (2) $+ \div \times =$
 (3) $\times \div + =$ (4) $+ - = \div$

SSC (10+2) Level Data Entry
 Operator & LDC Exam. 11.12.2011
 (1st Sitting (East Zone))

- 30.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$18 * 6 * 3 * 12 * 24$$

- (1) $\div - = \times$ (2) $\times \div - =$
 (3) $+ \div \times =$ (4) $\times = \div +$

SSC (10+2) Level Data Entry
 Operator & LDC Exam. 11.12.2011
 (IInd Sitting (East Zone))

- 31.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$16 * 4 * 64 * 4$$

- (1) $\times, <, \div$ (2) $\times, >, \div$
 (3) $\div, >, \times$ (4) $\times, >, +$

(FCI Assistant Grade-II
 Exam. 22.01.2012 Paper-I)

| SYMBOLS & NOTATIONS |

- 32.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation :

$$28 * 4 * 9 * 16$$

- (1) $\div + =$ (2) $+ \div =$
(3) $- \times +$ (4) $- = \times$

FCI Assistant Grade-III Exam. 25.02.2012
(Paper-I) North Zone (Ist Sitting)

- 33.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$16 * 6 * 4 * 24$$

- (1) $\div = \times$ (2) $\times = \div$
(3) $= \div \div$ (4) $\times \div =$

FCI Assistant Grade-III Exam. 05.02.2012
(Paper-I) East Zone (IInd Sitting)

- 34.** Choose the appropriate combination of signs to solve.

$$16 * 8 * 1 * 8$$

- (1) $= - \div$ (2) $- \div =$
(3) $\div - =$ (4) $\div = -$

(SSC (10+2) Level Data Entry Operator
& LDC Exam. 04.11.2012, Ist Sitting)

- 35.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation —

$$9 * 3 * 3 * 3 * 6$$

- (1) $\div \times - =$ (2) $+ - \times =$
(3) $- + + =$ (4) $\times + - =$

(SSC Constable (GD)

Exam. 12.05.2013 Ist Sitting)

- 36.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$8 * 6 * 96 * 2 = 0$$

- (1) $\times \div -$ (2) $\times - \div$
(3) $- \times \div$ (4) $\div - \times$

(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)

- 37.** If $264 * 2 = 6$, $870 * 3 = 11$, then what should $735 * 5$ be ?

- (1) 05 (2) 12
(3) 16 (4) 03

(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)

- 38.** Find the correct group of signs to solve the equation.

$$24 * 16 * 8 * 32$$

- (1) $+ - =$ (2) $\div - =$
(3) $- + =$ (4) $\times \div =$

(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)

- 39.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$15 * 24 * 3 * 6 * 17$$

- (1) $- \div + =$ (2) $+ \div - =$
(3) $+ \times = \div$ (4) $- \times = +$

(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)

- 40.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$5 * 5 * 5 * 3 * 10$$

- (1) $\times + = \times$ (2) $+ - \times =$
(3) $+ \div = \times$ (4) $+ \div \times =$

(SSC CAPFs SI & CISF ASI
Exam. 23.06.2013)

- 41.** Select the correct combination of mathematical signs to replace signs and to balance the following equation:

$$21 * 7 * 6 * 9$$

- (1) $+ \div =$ (2) $\div + =$
(3) $= + \div$ (4) $\div = +$

(SSC Multi-Tasking (Non-Tech.)
Staff Exam. 16.02.2014)

- 42.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation :

$$6 * 4 * 12 * 12$$

- (1) $\div - =$ (2) $+ - \div$
(3) $= - \div$ (4) $\times - =$

(SSC Multi-Tasking Staff
(Patna) Exam. 16.02.2014)

- 43.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation :

$$8 * 8 * 1 * 7 = 8$$

- (1) $\times \div +$ (2) $+ \div \times$
(3) $\div \times +$ (4) $+ \times \div$

(SSC Multi-Tasking (Non-Tech.) Staff
Exam. 23.02.2014, IInd Sitting)

- 44.** Insert the arithmetic operations in the following numerical figures :

$$13 * 3 * 4 * 3 = 4$$

- (1) $\div \times +$ (2) $- \times +$
(3) $+ \times \div$ (4) $+ \div \times$

(SSC CGL Tier-I
Re-Exam-2013, 27.04.2014)

- 45.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.

$$2 * 4 * 3 * 4 * 9$$

- (1) $+ \times = -$ (2) $\times \div - =$
(3) $\times - + =$ (4) $+ - = \div$

(SSC CGL Tier-I

Re-Exam-2013, 27.04.2014)

- 46.** Some equations are solved on the basis of a certain system. Find the correct answer for the unsolved equation on that basis.

$$5 * 6 = 35, 8 * 4 = 28, 6 * 8 = ?$$

- (1) 46 (2) 34
(3) 23 (4) 38

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

- 47.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation.

$$12 * 3 * 4 = 6 * 8 * 8$$

- (1) $+, \times, -, \times$ (2) $\times, +, -, \times$
(3) $\times, +, \times, -$ (4) $\times, -, \times, +$

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

- 48.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation :

$$5 * 9 * 3 * 6 * 8$$

- (1) $\times + = \times$ (2) $\times - = \times$
(3) $+ \div - =$ (4) $+ \times \div =$

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

- 49.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation

$$33 * 11 * 3 * 6 = 115$$

- (1) $+ - \times$ (2) $\times \div -$
(3) $\div \times -$ (4) $- \times \div$

(SSC GL Tier-I Re-Exam. (2013)
20.07.2014, Ist Sitting)

- 50.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation :

$$13 * 12 * 5 * 4$$

- (1) $= - +$ (2) $= + -$
(3) $+ - =$ (4) $- = +$

(SSC GL Tier-I Re-Exam. (2013)
20.07.2014, IInd Sitting)

- 51.** Some equations have been solved on the basis of a certain operation. Find the correct answer for the unsolved equation on that basis.

If $73 * 17 = 45$ and $68 * 40 = 54$, then $83 * 15 = ?$

| SYMBOLS & NOTATIONS |

- (1) 49 (2) 64
(3) 69 (4) 79
(SSC GL Tier-I Exam. 19.10.2014)
- 52.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation :
- $$(\sqrt{121} * 9) * 5 * 4 * 1$$
- (1) $- + \times =$ (2) $+ \div \times =$
(3) $+ \times \div$ (4) $- \times + =$
(SSC GL Tier-I Exam. 19.10.2014)
- 53.** Select the correct combination of mathematical signs to replace the * signs and to balance the following equation :
- $$45 * 3 * 6 * 2 * 16$$
- (1) $+ \times \div =$ (2) $+ \div \times =$
(3) $+ \times - =$ (4) $+ + - =$
(SSC CGL Tier-I Exam. 19.10.2014 TF No. 022 MH 3)
- 54.** Select the correct combination of mathematical signs to replace * signs and to balance the following equation :
- $$8 * 5 * 10 * 2 * 25$$
- (1) $+ \times \div =$ (2) $+ \div - =$
(3) $\times + = \times$ (4) $\times - = \times$
(SSC CGL Tier-I Exam. 19.10.2014 TF No. 022 MH 3)
- 55.** Select the correct combination of mathematical signs to replace * signs and to balance the given equation.
- $$5 * 3 * 3 * 5 * 0$$
- (1) $\times \div - =$ (2) $+ - \div =$
(3) $- - + =$ (4) $- \times \div =$
(SSC CHSL (10+2) DEO & LDC Exam. 16.11.2014, IIInd Sitting TF No. 545 QP 6)
- 56.** If '+' means '-', '-' means 'x', 'x' means '÷' and '÷' means '+', which combination will give you the value of '0' in $42 * 4 * 12 * 20 * 9$?
- (1) $- \div \times +$ (2) $\div + - \times$
(3) $+ - \div \times$ (4) $\times - \div +$
(SSC CGL Tier-I Exam, 16.08.2015 (IIInd Sitting) TF No. 2176783)
- 57.** Choose the appropriate combination of signs to solve the equation.
- $$(16 + 18) * (21 - 11) * 32 * 8$$
- (1) $- \times =$ (2) $- = -$
(3) $+ = -$ (4) $\div - =$
(SSC CGL Tier-I Re-Exam, 30.08.2015)
- 58.** A*B means multiply A by B; A@B means divide A by B, A? B means add B to A and A=B means subtract B from A. Then find the value of
- $$10*10 = 5*10 ? 50@10$$
- (1) 100 (2) 45
(3) 1000 (4) 55
(SSC CGL Tier-I (CBE) Exam. 28.08.2016) (IIInd Sitting)
- 59.** If $37 * 14 = 17$,
 $69 * 33 = 34$,
 $91 * 125 = 72$
then what should be $28 * 56 = ?$
- (1) 26 (2) 42
(3) 34 (4) 28
(SSC CGL Tier-I (CBE) Exam. 28.08.2016) (IIInd Sitting)
- 60.** If $5 * 3 = 19$ and $8 * 5 = 49$, then what should $6 * 4$ be ?
- (1) 24 (2) 28
(3) 18 (4) 16
(SSC CGL Tier-I (CBE) Exam. 29.08.2016) (IIInd Sitting)
- 61.** Complete the third equation on the basis of a certain system followed in the first two equations.
- (1) $5 * 4 * 2 * 1 = 1425$
(2) $7 * 8 * 1 * 6 = 6817$
(3) $9 * 3 * 7 * 5 = ?$
(1) 3795 (2) 5397
(3) 5973 (4) 5379
(SSC CGL Tier-I (CBE) Exam. 30.08.2016) (Ist Sitting)
- 62.** If $4 * 2 = 3$ and $8 * 4 = 3$, then $21 * 7 = ?$
- (1) 4 (2) 3
(3) 8 (4) 16
(SSC CGL Tier-I (CBE) Exam. 07.09.2016) (Ist Sitting)
- 63.** If $3 * 4 = 10$, $5 * 8 = 18$, $7 * 7 = ?$
- (1) 26 (2) 21
(3) 28 (4) 49
(SSC CGL Tier-I (CBE) Exam. 30.08.2016) (IIInd Sitting)
- 64.** Select the correct combination of mathematical signs to replace * signs and to balance the equation.
- $$48 * 4 * 6 * 3 * 30$$
- (1) $-, +, =, \times$ (2) $\div, =, \times, +$
(3) $\div, +, \times, =$ (4) $-, =, \times, +$
(SSC CGL Tier-I (CBE) Exam. 29.08.2016) (Ist Sitting)
- 65.** If, $1 * 2 = 1$, $2 * 3 = -1$ and $3 * 4 = -5$, then find the value of $7 * 9 = ?$
- (1) $- 47$ (2) $- 29$
(3) $- 2$ (4) $- 9$
(SSC CGL Tier-I (CBE) Exam. 03.09.2016) (IIInd Sitting)
- 66.** Some equations are solved on the basis of certain system. Find out the correct answer for the unsolved equation on that basis.
- If $3*2*8*4 = 632$,
 $2*4*4*4 = 816$
then $3*3*5*1 = ?$
- (1) 95 (2) 45
(3) 315 (4) 184
(SSC CGL Tier-I (CBE) Exam. 06.09.2016) (IIInd Sitting)
- 67.** Some equations have been solved on the basis of a certain pattern. Find the correct answer for the unsolved equation on that basis :
- $7 * 4 * 3 = 437$
 $8 * 6 * 4 = 648$
 $4 * 3 * 6 = ?$
- (1) 346 (2) 364
(3) 643 (4) 463
(SSC CGL Tier-I (CBE) Exam. 09.09.2016) (IIInd Sitting)
- 68.** If $13 * 45 = 29$, $24 * 58 = 41$, $74 * 32 = 53$, what should $97 * 47$ be ?
- (1) 73 (2) 72
(3) 63 (4) 64
(SSC CGL Tier-I (CBE) Exam. 04.09.2016) (IIInd Sitting)
- 69.** Some equations have been solved on the basis of a certain pattern. Find the correct answer for the unsolved equation on that basis :
- $2 * 3 * 5 = 523$
 $6 * 4 * 1 = 164$
 $8 * 2 * 4 = 482$,
then $6 * 8 * 2 = ?$
- (1) 826 (2) 268
(3) 286 (4) 628
(SSC CGL Tier-I (CBE) Exam. 10.09.2016) (IIInd Sitting)
- 70.** Some equations are solved on the basis of a certain system. Find out the correct answer for the unsolved equation on that basis.
- $9 * 8 = 63$; $7 * 8 = 49$,
 $5 * 6 = 25$, $11 * 7 = ?$
- (1) 77 (2) 70
(3) 66 (4) 121
(SSC Multi-Tasking Staff Exam. 30.04.2017 Ist Sitting)

TYPE-III

1. After interchanging $\div$ and $+$, 12 and 18, which one of the following equations becomes correct ?

(1) $(90 \times 18) + 18 = 60$
 (2) $(18 + 6) \div 12 = 2$
 (3) $(72 \div 18) \times 18 = 72$
 (4) $(12 + 6) \times 18 = 36$

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (1st Sitting))

2. After interchanging $\div$ and $=$, 2 and 3 which one of the following statements becomes correct ?

(1) $15 = 2 \div 3$ (2) $5 \div 15 = 2$
 (3) $2 = 15 \div 3$ (4) $3 = 2 \div 15$

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (IInd Sitting))

3. Which of the following interchange of signs would make the given equation correct ?

$(12 \div 6) + 3 \times 7 = 42$
 (1) $+$ and $\times$ (2) 6 and 7
 (3) $\div$ and $+$ (4) 12 and 3

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.12.2011 (1st Sitting (North Zone)))

4. Which interchange of signs will make the following equation correct ?

$35 + 7 \times 5 \div 5 - 6 = 24$
 (1) $+$ and $-$ (2) $+$ and $\times$
 (3) $\div$ and $+$ (4) $-$ and $\div$

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.12.2011 (IInd Sitting (North Zone)))

5. Which of the following interchanges of signs would make the given equation correct?

$24 + 6 \times 3 \div 3 - 1 = 14$
 (1) $+$ and $\times$ (2) $\times$ and $-$
 (3) $\div$ and $+$ (4) $-$ and $\div$

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.12.2011 (1st Sitting (East Zone)))

6. Which of the following interchange of signs or figures would make the given equation correct?

$(5 + 2) \times 2 - 10 = 16$
 (1) $+$ and $\times$ (2) 5 and 10
 (3) $+$ and $-$ (4) 5 and 2

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.12.2011 (IInd Sitting (East Zone)))

7. Which interchange of signs will make the following equation correct?

$30 - 6 \div 4 + 2 \times 3 = 7$
 (1) $+$ and $\times$ (2) $-$ and $+$
 (3) $-$ and $\div$ (4) $+$ and $-$

(SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (1st Sitting (Delhi Zone)))

8. Which of the following interchanges of signs would make the given equation correct ?

$5 + 6 \div 3 - 12 \times 2 = 17$
 (1) $\div$ and $\times$ (2) $+$ and $\times$
 (3) $+$ and $\div$ (4) $+$ and $-$

(SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (IInd Sitting (Delhi Zone)))

9. Which interchange of signs or numbers will make the following equation correct ?

$(7 + 2) \times 3 \times 4 - 1 = 20$
 (1) 2 and 3 (2) $\times$ and $-$
 (3) 7 and 3 (4) $+$ and $\times$

(SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (1st Sitting (East Zone)))

10. Which interchange of signs will make the following equation correct ?

$(16 - 4) \times 6 \div 2 + 8 = 30$
 (1) $\div$ and $-$ (2) 4 and 2
 (3) $-$ and $+$ (4) 16 and 6

(SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (IInd Sitting (East Zone)))

11. Which one of the four interchanges in signs and numbers would make the given equation correct ?

$6 \times 4 + 2 = 16$
 (1) $+$ and $\times$, and 4
 (2) $+$ and $\times$, 2 and 4
 (3) $+$ and $\times$, 4 and 6
 (4) None of these

(SSC Assistant Grade-III Exam. 11.11.2012 (IInd Sitting))

12. Which of the following interchange of sign would make the given equation correct ?

$(20 - 4) \times 4 + 16 = 36$
 (1) $+$ and $-$ (2) 5 and 5
 (3) 16 and 6 (4) $\div$ and $+$

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

13. Which of the following interchange of signs would make the given equation correct ?

$2 \times 3 + 6 - 12 \div 4 = 17$
 (1) $\times$ and $+$ (2) $+$ and $-$
 (3) $+$ and $\div$ (4) $-$ and $\div$

(SSC Graduate Level Tier-I Exam. 21.04.2013, IInd Sitting)

- Directions (14 – 15) :** In each of the following questions, which of the following interchange of signs would make the given equation correct?

(SSC Graduate Level Tier-I Exam. 21.04.2013, IInd Sitting)

14. $10 + 10 \div 10 - 10 \times 10 = 10$

(1) $+$ and $-$ (2) $+$ and $\div$
 (3) $+$ and $\times$ (4) $\div$ and $+$

15. $(8 - 8) + 8 \times 32 = 64$

(1) $\times$, $+$, $-$ (2) $-$, $\div$, $+$
 (3) $+$, $\div$, $+$ (4) $+$, $\div$, $\times$

16. Which of the following interchanges of numbers would make the given equation correct ?

$8 \times 20 \div 3 + 9 - 5 = 38$
 (1) 3 , 9 (2) 3 , 8
 (3) 8 , 9 (4) 3 , 5

(SSC Graduate Level Tier-I Exam. 19.05.2013, 2nd Sitting)

17. Which of the following interchange of signs or numbers would make the given equation correct ?

$(18 \div 9) + 3 \times 5 = 45$
 (1) $\times \div$ (2) $+$
 (3) 18 and 5 (4) 3 and 9

(SSC Cabinet Secretariat RO (ECO), DFO (T) & DFO (GD) Tier-I Exam. 23.06.2013)

18. Which of the following interchange of signs would make the equation correct ?

$8 \times 6 + 2 = 22$
 (1) $+$, $\times$, 2 and 6
 (2) $+$, $\times$, 2 and 8
 (3) $+$, $\times$, 6 and 8
 (4) $+$, $\times$, 2 and 22

(SSC GL Tier-I Exam. 19.10.2014, 1st Sitting)

19. Which of the following interchange of signs would make the given equation correct ?

$64 - 8 \times 9 \times 8 = 64$
 (1) $+$ and $-$ (2) $\div$ and $\times$
 (3) $+$ and $\div$ (4) $-$ and $\times$

(SSC GL Tier-I Exam. 19.10.2014)

| SYMBOLS & NOTATIONS |

20. After interchanging $\div$ and $\times$, 10 and 5, which one of the following becomes a correct equation ?

- (1) $(30 \div 5) \times 10 = 24$
 (2) $(30 \times 10) \div 5 = 60$
 (3) $(30 \div 10) \times 5 = 18$
 (4) $(10 \div 30) \times 5 = 70$

(SSC CHSL (10+2) DEO & LDC Exam. 02.11.2014, Patna Region : (1st Sitting))

21. Which of the following interchange of signs would make the given equation correct ?

$$(6 + 3) + (4 \times 7) = 29$$

- (1) + and - (2) $\div$ and +
 (3) $\times$ and + (4) $\div$ and $\times$

(SSC CHSL (10+2) DEO & LDC Exam. 02.11.2014, IInd Sitting)

22. Change the sign to find the equation $28 - 7 + 2 \times 2 = 0$

- (1) Change + into $\times$
 (2) Change $\times$ into +
 (3) Change - into +
 (4) Change + into -

(SSC CHSL (10+2) DEO & LDC Exam. 02.11.2014, IInd Sitting)

23. What sign should be changed to make the equation $5 + 6, 3 - 12 \times 2 = 17$ correct ?

- (1) + (2) $\times$
 (3) - + (4) None of these

(SSC CHSL (10+2) DEO & LDC Exam. 16.11.2014)

24. Which of the following interchange of signs would make the equation correct ?

$$6 \times 4 + 2 = 16$$

- (1) + and $\times$, 2 & 4
 (2) + and $\times$, 4 & 6
 (3) + and $\times$, 2 & 6
 (4) + and $\times$, 3 & 4

(SSC CGL Tier-I Exam. 19.10.2014 TF No. 022 MH 3)

25. Interchange of signs and numbers would make the given equation correct.

$$6 + 2 - 3 = 16$$

- (1) + and -, 2 and 3
 (2) $\times$ and +, 3 and 2
 (3) $\div$ and -, 3 and 2
 (4) $\times$ and -, 2 and 3

(SSC CAPFs SI, CISF ASI & DP SI Exam. 21.06.2015, IInd Sitting)

26. After interchanging + and -, 8 and 7, which one of the following becomes correct?

- (1) $8 - 7 + 3 \times 5 = 35$
 (2) $7 \times 8 + 6 - 9 = 25$

$$(3) 6 + 8 \times 2 - 7 = 0$$

$$(4) 8 \times 2 + 7 - 6 = 9$$

(SSC CGL Tier-I Exam, 16.08.2015 (IInd Sitting) TF No. 2176783)

27. Which of the following interchange of signs would make the equation correct?

$$5 + 3 \times 8 - 12 \div 4 = 3$$

- (1) - and $\div$ (2) + and $\times$
 (3) + and $\div$ (4) + and -

(SSC CGL Tier-I (CBE) Exam. 29.08.2016) (1st Sitting)

ANSWERS

TYPE-I (i)

1. (2)	2. (2)	3. (3)	4. (1)
5. (4)	6. (3)	7. (3)	8. (2)
9. (2)	10. (1)	11. (2)	12. (4)
13. (1)	14. (2)	15. (2)	16. (2)
17. (3)	18. (1)	19. (1)	20. (4)
21. (2)	22. (1)	23. (4)	24. (3)
25. (1)	26. (2)	27. (2)	28. (4)
29. (3)	30. (2)	31. (3)	32. (3)
33. (1)	34. (3)	35. (3)	36. (4)
37. (4)	38. (4)	39. (3)	40. (2)
41. (1)	42. (3)	43. (4)	44. (*)
45. (*)	46. (1)	47. (2)	48. (4)
49. (*)	50. (1)	51. (1)	52. (3)
53. (4)	54. (1)	55. (3)	56. (2)
57. (4)	58. (4)	59. (1)	60. (3)
61. (2)	62. (3)	63. (2)	64. (3)
65. (3)	66. (3)	67. (1)	68. (4)
69. (2)	70. (3)	71. (*)	72. (4)
73. (1)	74. (2)	75. (3)	76. (3)
77. (2)	78. (3)	79. (*)	80. (4)
81. (1)	82. (1)	83. (4)	84. (3)
85. (3)	86. (3)	87. (*)	88. (1)
89. (3)	90. (4)	91. (3)	92. (2)
93. (1)	94. (3)	95. (2)	96. (1)
97. (4)	98. (1)	99. (*)	100. (1)
101. (1)	102. (1)	103. (1)	104. (4)
105. (1)	106. (2)	107. (4)	108. (3)
109. (4)	110. (2)	111. (2)	112. (2)
113. (2)	114. (1)	115. (*)	116. (1)
117. (2)	118. (1)		

TYPE-I (ii)

1. (2)			
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TYPE-I (iii)

1. (2)	2. (4)	3. (3)	4. (4)
5. (2)	6. (2)	7. (4)	8. (3)
9. (2)	10. (2)	11. (1)	12. (2)
13. (2)	14. (3)	15. (3)	16. (3)
17. (1)	18. (1)	19. (3)	20. (*)
21. (*)	22. (3)	23. (2)	24. (3)
25. (*)	26. (2)	27. (3)	28. (2)
29. (2)	30. (3)	31. (2)	32. (2)
33. (4)	34. (2)	35. (2)	36. (4)
37. (1)	38. (1)	39. (1)	

TYPE-I (iv)

1. (2)	2. (4)	3. (1)	4. (4)
5. (*)	6. (4)	7. (1)	8. (*)
9. (4)			

TYPE-I (v)

1. (1)	2. (2)	3. (1)	4. (3)
5. (1)	6. (3)	7. (3)	8. (2)
9. (4)	10. (4)	11. (1)	

TYPE-I (vi)

1. (4)	2. (4)	3. (3)	4. (*)
5. (3)	6. (1)	7. (2)	8. (4)
9. (4)	10. (2)	11. (1)	12. (3)

TYPE-I (vii)

1. (2)	2. (2)	3. (1)	4. (2)
5. (1)	6. (2)	7. (2)	8. (3)
9. (2)	10. (4)	11. (2)	12. (2)
13. (2)	14. (4)	15. (1)	16. (1)
17. (3)	18. (1)	19. (2)	20. (2)
21. (2)	22. (4)	23. (4)	24. (3)
25. (2)	26. (1)	27. (4)	28. (4)
29. (1)	30. (2)	31. (2)	32. (3)
33. (3)	34. (4)	35. (3)	36. (2)

37. (2)	38. (1)	39. (4)	40. (2)
41. (2)	42. (4)	43. (1)	44. (4)
45. (1)	46. (1)	47. (3)	48. (2)
49. (2)	50. (2)	51. (3)	52. (4)
53. (2)	54. (1)	55. (3)	56. (2)
57. (2)	58. (2)		

TYPE-II

1. (1)	2. (*)	3. (1)	4. (1)
5. (1)	6. (4)	7. (1)	8. (4)
9. (3)	10. (3)	11. (3)	12. (1)
13. (4)	14. (4)	15. (2)	16. (3)
17. (4)	18. (4)	19. (3)	20. (2)
21. (4)	22. (4)	23. (2)	24. (4)
25. (2)	26. (2)	27. (4)	28. (4)
29. (3)	30. (2)	31. (2)	32. (1)
33. (4)	34. (2)	35. (1)	36. (2)
37. (2)	38. (1)	39. (2)	40. (1)
41. (2)	42. (4)	43. (3)	44. (2)
45. (3)	46. (1)	47. (3)	48. (1)
49. (2)	50. (2)	51. (1)	52. (*)
53. (2)	54. (3)	55. (1)	56. (*)
57. (2)	58. (4)	59. (4)	60. (2)
61. (4)	62. (1)	63. (2)	64. (3)
65. (1)	66. (1)	67. (2)	68. (2)
69. (2)	70. (3)		

TYPE-III

1. (4)	2. (2)	3. (3)	4. (3)
5. (3)	6. (3)	7. (3)	8. (1)
9. (4)	10. (1)	11. (3)	12. (4)
13. (1)	14. (3)	15. (4)	16. (4)
17. (2)	18. (3)	19. (3)	20. (2)
21. (3)	22. (1)	23. (*)	24. (2)
25. (4)	26. (3)	27. (1)	

EXPLANATIONS

TYPE-I (i)

1. (2)

$+$	$\rightarrow$	$\div$	$\div$	$\rightarrow$	$\times$
$\times$	$\rightarrow$	$-$	$-$	$\rightarrow$	$+$

Option (1)

$$18 \div 6 - 7 + 5 \times 2 = 20$$

After conversion

$$18 \times 6 + 7 \div 5 - 2 = 20$$

$$\text{or, } 108 + \frac{7}{5} - 2 \neq 20$$

Option (2)

$$18 + 6 \div 7 \times 5 - 2 = 18$$

After conversion

$$18 \div 6 \times 7 - 5 + 2 = 18$$

$$\text{or, } 3 \times 7 - 5 + 2 = 18$$

$$\text{or, } 21 - 5 + 2 = 18$$

$$\text{or, } 23 - 5 = 18$$

$$\text{or, } 18 = 18$$

2. (2)

$-$	$\rightarrow$	$\div$	$+$	$\rightarrow$	$\times$
$\div$	$\rightarrow$	$-$	$\times$	$\rightarrow$	$+$

Option (1)

$$18 \div 3 \times 2 + 8 - 6 = 10$$

After conversion

$$18 - 3 + 2 \times 8 \div 6 = 10$$

$$\text{or, } 18 - 3 + 2 \times \frac{8}{6} = 10$$

$$\text{or, } 18 - 3 + \frac{8}{3} \neq 10$$

Option (2)

$$18 - 3 + 2 \times 8 \div 6 = 14$$

After conversion

$$18 \div 3 \times 2 + 8 - 6 = 14$$

$$\text{or, } 6 \times 2 + 8 - 6 = 14$$

$$\text{or, } 12 + 8 - 6 = 14$$

3. (3)

$+$	$\Rightarrow$	$\times$	$\times$	$\Rightarrow$	$-$
$\div$	$\Rightarrow$	$+$	$-$	$\Rightarrow$	$\div$

Given expression

$$175 - 25 \div 5 + 20 \times 3 + 10 = ?$$

After conversion

$$? = 175 \div 25 + 5 \times 20 - 3 \times 10$$

$$\text{or, } ? = 7 + 100 - 30 = 77$$

4. (1)

$-$	$\Rightarrow$	$\div$	$+$	$\Rightarrow$	$\times$
$\div$	$\Rightarrow$	$-$	$\times$	$\Rightarrow$	$+$

Option (1)

Given expression

$$6 \div 20 \times 12 + 7 - 1 = 70$$

After conversion

$$6 - 20 + 12 \times 7 \div 1 = 70$$

$$\text{or, } 6 - 20 + 84 = 70$$

$$\text{or, } 90 - 20 = 70$$

$$\text{or, } 70 = 70$$

5. (4)

$+$	$\Rightarrow$	$-$	$-$	$\Rightarrow$	$\times$
$\times$	$\Rightarrow$	$\div$	$\div$	$\Rightarrow$	$+$

Given expression

$$2 \div 6 \times 6 \div 2 = ?$$

After changing the signs

$$? = 2 + 6 \div 6 + 2$$

$$\text{or, } ? = 2 + 1 + 2 = 5$$

6. (3)

$+$	$\Rightarrow$	$\times$	$\times$	$\Rightarrow$	$\div$
$-$	$\Rightarrow$	$+$	$\div$	$\Rightarrow$	$-$

Given expression

$$20 - 8 \times 4 \div 3 + 2 = ?$$

After conversion

$$? = 20 + 8 \div 4 - 3 \times 2$$

$$\text{or, } ? = 20 + 2 - 6$$

$$\text{or, } ? = 22 - 6 = 16$$

7. (3)

$\times$	$\mathbb{R}$	$-$	$-$	$\mathbb{R}$	$\div$
$+$	$\mathbb{R}$	$\times$	$\div$	$\mathbb{R}$	$+$

Given expression

$$16 \times 8 \div 4 - 3 + 9 = ?$$

After conversion

$$? = 16 - 8 + 4 \div 3 \times 9$$

$$\text{or, } ? = 16 - 8 + \frac{4}{3} \times 9$$

$$\text{or, } ? = 16 - 8 + 12 = 20$$

8. (2)

$+$	$\Rightarrow$	$-$	$-$	$\Rightarrow$	$\times$
$\div$	$\Rightarrow$	$+$	$\times$	$\Rightarrow$	$\div$

Given expression

$$15 - 3 + 10 \times 5 \div 5 = ?$$

After conversion

$$? = 15 \times 3 - 10 \div 5 + 5$$

$$\text{or, } ? = 45 - 2 + 5$$

$$\text{or, } ? = 50 - 2 = 48$$

9. (2)

$-$	$\Rightarrow$	$\div$	$\div$	$\Rightarrow$	$\times$
$+$	$\Rightarrow$	$-$	$\times$	$\Rightarrow$	$+$

Option (1)

$$20 + 8 - 7 \div 6 \times 4 = 20$$

$$\text{or, } 20 - 8 \div 7 \times 6 + 4 = 20$$

$$\text{or, } 20 - \frac{8}{7} \times 6 + 4 \neq 20$$

Option (2)

$$20 - 5 \div 4 + 6 \times 5 = 15$$

or, $20 \div 5 \times 4 - 6 + 5 = 15$

or, $4 \times 4 - 6 + 5 = 15$

or, $16 - 6 + 5 = 15$

10. (1)

$\times \Rightarrow +$	$\div \Rightarrow -$
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$? = 39 \times 23 \div 21 \times 5$

or, $? = 39 + 23 - 21 + 5$

or, $? = 67 - 21 = 46$

11. (2)

$+\Rightarrow\div$	$\div\Rightarrow\times$
$\times\Rightarrow-$	$-\Rightarrow+$

Option (1)

$18 \div 6 - 7 + 5 \times 2 = 20$

or, $18 \times 6 + 7 \div 5 - 2 = 20$

or, $108 + \frac{7}{5} - 2 = 20$

or, $\frac{540 + 7 - 10}{5} = 20$

or, $547 - 10 \neq 20 \times 5$

Option (2)

$18 + 6 \div 7 \times 5 - 2 = 18$

or, $18 \div 6 \times 7 - 5 + 2 = 18$

or, $3 \times 7 - 5 + 2 = 18$

or, $21 - 5 + 2 = 18$

12. (4)

$+\Rightarrow\div$	$\times\Rightarrow+$
$-\Rightarrow\times$	$\div\Rightarrow-$

$36 - 6 + 3 \times 5 \div 3 = 74$

or, $36 \times 6 \div 3 + 5 - 3 = 74$

or, $36 \times 2 + 5 - 3 = 74$

or, $72 + 5 - 3 = 74$

13. (1)

$+\Rightarrow-$	$-\Rightarrow\times$
$\div\Rightarrow+$	$\times\Rightarrow\div$

$10 \times 5 \div 3 - 2 + 3 = ?$

or, $? = 10 \div 5 + 3 \times 2 - 3$

or, $? = 2 + 6 - 3 = \boxed{5}$

14. (2)

$+\Rightarrow\div$	$-\Rightarrow\times$
$\div\Rightarrow+$	$\times\Rightarrow-$

$63 \times 24 + 8 \div 4 + 2 - 3 = ?$

or, $? = 63 - 24 \div 8 + 4 \div 2 \times 3$

or, $? = 63 - 3 + 2 \times 3$

or, $? = 63 - 3 + 6$

or, $? = 66$

15. (2) $(16 - 4) \times 6 \div 2 + 8 = 30$

$\Rightarrow (16 \div 4) \times 6 - 2 + 8 = 30$

$\Rightarrow 4 \times 6 - 2 + 8 = 30$

$\Rightarrow 24 - 2 + 8 = 30$

16. (2)

$-\Rightarrow\div$	$+\Rightarrow-$
$\div\Rightarrow\times$	$\times\Rightarrow+$

Option (1)

$70 - 2 + 4 \div 5 \times 6 = 44$

$\Rightarrow 70 \div 2 - 4 \times 5 + 6 = 44$

$\Rightarrow 35 - 20 + 6 = 44$

Option (2)

$70 - 2 + 4 \div 5 \times 6 = 21$

$70 \div 2 - 4 \times 5 + 6 = 21$

$\Rightarrow 35 - 20 + 6 = 21$

$\Rightarrow 41 - 20 = 21$

17. (3)

$-\Rightarrow\div$	$+\Rightarrow\times$
$\div\Rightarrow-$	$\times\Rightarrow+$

Option (1)

$19 + 5 - 4 \times 2 \div 4 = 11$

$\Rightarrow 19 \times 5 \div 4 + 2 - 4 = 11$

$\Rightarrow \frac{95}{4} + 2 - 4 \neq 11$

Option (2)

$19 \times 5 - 4 \div 2 + 4 = 16$

$\Rightarrow 19 + 5 \div 4 - 2 \times 4 = 16$

$\Rightarrow 19 + \frac{5}{4} - 8 \neq 16$

Option (3)

$19 \div 5 + 4 - 2 \times 4 = 13$

$\Rightarrow 19 - 5 \times 4 \div 2 + 4 = 13$

$\Rightarrow 19 - 5 \times 2 + 4 = 13$

$\Rightarrow 19 - 10 + 4 = 13$

18. (1)

$-\Rightarrow\div$	$+\Rightarrow\times$
$\div\Rightarrow-$	$\times\Rightarrow+$

Option (1)

$30 - 6 + 5 \times 4 \div 2 = 27$

$\Rightarrow 30 \div 6 \times 5 + 4 - 2 = 27$

$\Rightarrow 5 \times 5 + 4 - 2 = 27$

$\Rightarrow 25 + 4 - 2 = 27$

Option (2)

$30 + 6 - 5 \div 4 \times 2 = 30$

$\Rightarrow 30 \times 6 \div 5 - 4 + 2 = 30$

$\Rightarrow 36 - 4 + 2 \neq 30$

Option (3)

$30 \times 6 \div 5 - 4 + 2 = 32$

$\Rightarrow 30 + 6 - 5 \div 4 \times 2 \neq 32$

19. (1)

$\times\Rightarrow+$	$-\Rightarrow\div$
$+\Rightarrow\times$	$\div\Rightarrow-$

Option (1)

$16 + 5 - 10 \times 4 \div 3 = 9$

$\Rightarrow 16 \times 5 \div 10 + 4 - 3 = 9$

$\Rightarrow 8 + 4 - 3 = 9$

20. (4)

$-\Rightarrow\div$	$+\Rightarrow\times$
$\div\Rightarrow-$	$\times\Rightarrow+$

Option (1)

$24 \div 8 - 4 + 2 \times 3 = 16$

After conversion

$24 - 8 \div 4 \times 2 + 3 = 16$

or, $24 - 2 \times 2 + 3 = 16$

or, $24 - 4 + 3 = 16$

or, $27 - 4 = 16$

or, $23 \neq 16$

Option (2)

$24 - 8 + 4 \times 2 \div 3 = 12$

After conversion

$24 \div 8 \times 4 + 2 - 3 = 12$

or, $3 \times 4 + 2 - 3 = 12$

or, $12 + 2 - 3 = 12$

or, $11 \neq 12$

Option (3)

$24 \times 8 - 4 \div 2 + 3 = 17$

After conversion

$24 + 8 \div 4 - 2 \times 3 = 17$

or, $24 + 2 - 6 = 17$

or, $26 - 6 = 17$

or, $20 \neq 17$

Option (4)

$24 + 8 - 4 \times 2 \div 3 = 47$

After conversion

or, $24 \times 8 \div 4 + 2 - 3 = 47$

or, $24 \times 2 + 2 - 3 = 47$

or, $48 + 2 - 3 = 47$

or, $50 - 3 = 47$

or, $47 = 47$

21. (2)

$\div\Rightarrow+$	$\times\Rightarrow-$
--------------------	----------------------

Given expression

$(15 \times 9) \div (12 \times 4) \times (4 \div 4) = ?$

After changing the sign,

$? = (15 - 9) + (12 - 4) - (4 + 4)$

or, $? = (6) + (8) - (8)$

or, $? = 6 + 8 - 8 = 6$

22. (1)

$+\Rightarrow-$	$-\Rightarrow\times$
$\div\Rightarrow+$	$\times\Rightarrow\div$

Given expression

$15 \times 3 \div 4 - 6 + 7 = ?$

After conversion

$? = 15 \div 3 + 4 \times 6 - 7$

or, $? = 5 + 24 - 7$

or, $? = 29 - 7 = \boxed{22}$

23. (4)

$\times\Rightarrow-$	$\div\Rightarrow+$
$+\Rightarrow\times$	

Given expression

$(16 \times 5) \div 5 + 3 = ?$

After conversion

$? = (16 - 5) + 5 \times 3$

or, $? = 11 + 5 \times 3$

or, $? = 11 + 15 = 26$

24. (3)

$\div\Rightarrow+$	$-\Rightarrow\times$
$\times\Rightarrow-$	$+\Rightarrow\div$

Option (1)

$$10 \times 4 = 06 \Rightarrow 10 - 4 = 06$$

Option (2)

$$10 - 4 = 40 \Rightarrow 10 \times 4 = 40$$

Option (3)

$$10 + 5 = 50 \Rightarrow 10 \div 5 = 2$$

and $2 \neq 50$

Option (4)

$$10 \div 5 = 15 \Rightarrow 10 + 5 = 15$$

25. (1)

$\times \rightarrow +$	$\div \rightarrow -$
$- \rightarrow \times$	$+ \rightarrow \div$

Given expression

$$54 \div 16 - 3 \times 6 + 2 = ?$$

After conversion

$$? = 54 - 16 \times 3 + 6 \div 2$$

$$\text{or, } ? = 54 - 48 + 3 = \boxed{9}$$

26. (2)

$+ \rightarrow -$	$\div \rightarrow +$
$- \rightarrow \times$	$\times \rightarrow \div$

Option (1)

$$56 + 12 \times 34 - 12 = 102$$

After changing the signs

$$56 - 12 \div 34 \times 12 = 102$$

$$\text{or, } 56 - \frac{12}{34} \times 12 = 102$$

$$\text{or, } 56 - \frac{144}{34} \times 12 \neq 102$$

Option (2)

$$8 \div 44 - 5 + 25 = 203$$

After changing the signs

$$8 + 44 \times 5 - 25 = 203$$

$$\text{or, } 8 + 220 - 25 = 203$$

$$\text{or, } 228 - 25 = 203$$

$$\text{or, } 203 = 203$$

Option (3)

$$112 \times 44 - 12 + 10 = 46$$

After changing the signs

$$112 \div 44 \times 12 - 10 = 46$$

$$\text{or, } \frac{112}{44} \times 12 - 10 = 46$$

$$\text{or, } \frac{28}{11} \times 12 - 10 = 46$$

$$\text{or, } \frac{336}{11} - 10 = 46$$

$$\text{or, } \frac{336 - 110}{11} = 46$$

$$\text{or, } 226 = 46 \times 11$$

$$\text{or, } 226 \neq 506$$

Option (4)

$$9 \div 64 - 2 \times 6 = 54$$

After changing the signs

$$9 + 64 \times 2 \div 6 = 54$$

$$\text{or, } 9 + 64 \times \frac{2}{6} = 54$$

$$\text{or, } 9 + \frac{64}{3} = 54$$

$$\text{or, } 27 + 64 = 54 \times 3$$

$$\text{or, } 91 \neq 162$$

27. (2)

$- \rightarrow \div$	$+ \rightarrow \times$
$\div \rightarrow -$	$\times \rightarrow +$

Option (1)

$$30 + 5 - 12 \div 8 \times 12 = 70$$

After changing the signs

$$30 \times 5 \div 12 \div 8 + 12 = 70$$

$$\text{or, } 30 \times \frac{5}{12} - 8 + 12 = 70$$

$$\text{or, } \frac{25}{2} - 8 + 12 = 70$$

$$\text{or, } 25 - 16 + 24 = 70 \times 2$$

$$\text{or, } 33 \neq 140$$

Option (2)

$$30 - 5 + 12 \div 8 \times 12 = 76$$

After changing the signs

$$30 \div 5 + 12 \div 8 + 12 = 76$$

$$\text{or, } 6 \times 12 - 8 + 12 = 76$$

$$\text{or, } 72 - 8 + 12 = 76$$

$$\text{or, } 84 - 8 = 76$$

$$\text{or, } 76 = 76$$

Option (3)

$$30 \times 5 - 12 + 8 \div 12 = 60$$

After changing the signs

$$30 + 5 \div 12 \times 8 - 12 = 60$$

$$\text{or, } 30 + \frac{5}{12} \times 8 - 12 = 60$$

$$\text{or, } 30 + \frac{5}{3} \times 2 - 12 = 60$$

$$\text{or, } 30 + \frac{10}{3} - 12 = 60$$

$$\text{or, } 90 + 10 - 36 = 60 \times 3$$

$$\text{or, } 64 \neq 180$$

Option (4)

$$30 \div 5 \times 12 + 8 - 12 = 24$$

After changing the signs

$$30 - 5 + 12 \times 8 \div 12 = 24$$

$$\text{or, } 30 - 5 + 12 \times \frac{8}{12} = 24$$

$$\text{or, } 30 - 5 + 8 = 24$$

$$\text{or, } 33 \neq 24$$

28. (4)

$+ \rightarrow \times$	$\times \rightarrow \div$
$- \rightarrow +$	$\div \rightarrow -$

Given expression

$$20 - 5 \div 18 \times (3 + 2) = ?$$

After changing the signs

$$? = 20 + 5 - 18 \div (3 \times 2)$$

$$\text{or, } ? = 20 + 5 - 18 \div 6$$

$$\text{or, } ? = 20 + 5 - 3$$

$$\text{or, } ? = 25 - 3 = 22$$

29. (3)

$\times \Rightarrow -$	$+ \Rightarrow \times$
$- \Rightarrow +$	

Given expression

$$6 + (3 \times 1) + 5 = ?$$

After changing the signs

$$? = 6 \times (3 - 1) \times 5$$

$$\text{or, } ? = 6 \times 2 \times 5 = 60$$

30. (2)

$+ \Rightarrow \div$	$\div \Rightarrow -$
$- \Rightarrow \times$	$\times \Rightarrow +$

Give expression

$$12 + 6 \div 3 - 2 \times 8 = ?$$

After changing the signs

$$? = 12 \div 6 - 3 \times 2 + 8$$

$$\text{or, } ? = 2 - 6 + 8$$

$$\text{or, } ? = 10 - 6 = \boxed{4}$$

31. (3)

$+ \Rightarrow \times$	$- \Rightarrow \div$
$\times \Rightarrow -$	$\div \Rightarrow +$

Give expression

$$26 + 74 - 4 \times 5 \div 2 = ?$$

After changing the signs

$$? = 26 \times 74 \div 4 - 5 + 2$$

$$? = 481 - 5 + 2$$

$$\text{or, } ? = 483 - 5 = \boxed{478}$$

32. (3)

$+ \Rightarrow -$	$\div \Rightarrow +$
$- \Rightarrow \times$	$\times \Rightarrow \div$

Option (1)

$$46 - 10 + 10 \times 5 = 92$$

$$\text{or, } 46 \times 10 - 10 \div 5 = 92$$

$$\text{or, } 460 - 2 \neq 92$$

Option (2)

$$265 + 11 - 2 \times 14 = 22$$

$$\text{or, } 265 - 11 \times 2 \div 14 = 22$$

$$\text{or, } 265 - \frac{22}{14} \neq 22$$

Option (3)

$$66 \times 3 - 11 + 12 = 230$$

$$\text{or, } 66 \div 3 \times 11 - 12 = 230$$

$$\text{or, } 22 \times 11 - 12 = 230$$

$$\text{or, } 242 - 12 = 230$$

33. (1)

$+ \Rightarrow -$	$- \Rightarrow =$	$= \Rightarrow +$
$\div \Rightarrow >$	$\times \Rightarrow <$	

$$8 + 4 - ?$$

$$\Rightarrow 8 - 4 = ?$$

$$\therefore ? = 4$$

34. (3)

$- \Rightarrow +$	$+ \Rightarrow -$
$\times \Rightarrow \div$	$\div \Rightarrow \times$

$$\begin{aligned} ? &= 7 - 10 \times 5 \div 6 + 4 \\ \Rightarrow ? &= 7 + 10 \div 5 \times 6 - 4 \\ \Rightarrow ? &= 7 + 2 \times 6 - 4 \\ \Rightarrow ? &= 7 + 12 - 4 = 15 \end{aligned}$$

35. (3)

$\div \Rightarrow \times$	$- \Rightarrow +$
$\times \Rightarrow -$	$+ \Rightarrow \div$

$$\begin{aligned} ? &= 20 + 4 \times 6 - 5 \div 7 \\ \Rightarrow ? &= 20 \div 4 - 6 + 5 \times 7 \\ \Rightarrow ? &= 5 - 6 + 35 = 34 \end{aligned}$$

36. (4)

$+ \Rightarrow \times$	$- \Rightarrow +$
$\times \Rightarrow \div$	$\div \Rightarrow -$

$$\begin{aligned} 10 + 5 \times 10 \div 2 - 5 \\ \Rightarrow ? &= 10 \times 5 \div 10 - 2 + 5 \\ \Rightarrow ? &= 5 - 2 + 5 = 8 \end{aligned}$$

37. (4) Option (1)

$$\begin{aligned} 10 \div 5 + 4 &= 6 \\ \Rightarrow 10 \times 5 \div 4 &= 6 \\ \Rightarrow \frac{10 \times 5}{4} &\neq 6 \end{aligned}$$

Option (2)

$$\begin{aligned} 10 - 4 + 2 &= 6 \\ \Rightarrow 10 - 4 \div 2 &= 6 \\ \Rightarrow 10 - 2 &\neq 6 \end{aligned}$$

Option (3)

$$\begin{aligned} 10 + 2 - 5 &= 6 \\ \Rightarrow 10 \div 2 - 5 &\neq 6 \end{aligned}$$

Option (4)

$$\begin{aligned} 10 + 2 \times 1 &= 6 \\ \Rightarrow 10 \div 2 + 1 &= 6 \\ \Rightarrow 5 + 1 &= 6 \end{aligned}$$

38. (4) Option (1)

$$\begin{aligned} 36 \times 6 + 7 \div 2 - 6 &= 20 \\ \Rightarrow 36 + 6 \div 7 - 2 \times 6 &= 20 \\ \Rightarrow 36 + \frac{6}{7} - 12 &\neq 20 \end{aligned}$$

Option (2)

$$\begin{aligned} 36 \div 6 + 3 \times 5 - 3 &= 45 \\ \Rightarrow 36 - 6 \div 3 + 5 \times 3 &= 45 \\ \Rightarrow 36 - 2 + 15 &\neq 45 \end{aligned}$$

Option (3)

$$\begin{aligned} 36 + 6 - 3 \times 5 \div 3 &= 24 \\ \Rightarrow 36 \div 6 \times 3 + 5 - 3 &= 25 \\ \Rightarrow 6 \times 3 + 5 - 3 &= 24 \\ \Rightarrow 18 + 5 - 3 &\neq 24 \end{aligned}$$

Option (4)

$$\begin{aligned} 36 - 6 + 3 \times 5 \div 3 &= 74 \\ \Rightarrow 36 \times 6 \div 3 + 5 - 3 &= 74 \\ \Rightarrow 72 + 5 - 3 &= 74 \end{aligned}$$

39. (3) Option (1)

$$\begin{aligned} 18 + 14 - 24 \times 12 \div 16 &= 12 \\ \Rightarrow 18 \times 14 \div 24 + 12 - 16 &= 12 \\ \Rightarrow \frac{18 \times 14}{24} + 12 - 16 &= 12 \\ \Rightarrow \frac{21}{2} + 12 - 16 &\neq 12 \end{aligned}$$

Option (2)

$$\begin{aligned} 16 \times 14 - 24 \div 18 + 12 &= -24 \\ \Rightarrow 16 + 14 \div 24 - 18 \times 12 &= -24 \\ \Rightarrow 16 + \frac{14}{24} - 18 \times 12 &\neq -24 \end{aligned}$$

Option (3)

$$\begin{aligned} 24 - 12 + 12 \div 16 \times 18 &= 26 \\ \Rightarrow 24 \div 12 \times 12 - 16 + 18 &= 26 \\ \Rightarrow 24 - 16 \times 18 &= 26 \end{aligned}$$

40. (2)

$\times \Rightarrow -$	$- \Rightarrow \times$
$+ \Rightarrow \div$	$\div \Rightarrow +$

$$\begin{aligned} ? &= (15 - 10) \div (130 + 10) \times 50 \\ \Rightarrow ? &= (15 \times 10) + (130 \div 10) - 50 \\ \Rightarrow ? &= (150) + (13) - 50 \\ \Rightarrow ? &= 163 - 50 = \boxed{113} \end{aligned}$$

41. (1)

$+ \Rightarrow \div$	$- \Rightarrow \times$
$\div \Rightarrow +$	$\times \Rightarrow -$

$$\begin{aligned} 36 \times 12 + 4 \div 6 + 2 - 3 &= ? \\ \Rightarrow ? &= 36 - 12 \div 4 + 6 \div 2 \times 3 \\ \Rightarrow ? &= 36 - 3 + 9 \\ \Rightarrow ? &= 45 - 3 = \boxed{42} \end{aligned}$$

42. (3)

$T \Rightarrow \times$	$U \Rightarrow -$
$V \Rightarrow \div$	$W \Rightarrow +$

$$\begin{aligned} (50 \vee 2) W (28 T 4) \\ \Rightarrow (50 \div 2) + (28 \times 4) \\ \Rightarrow 25 + 112 = 137 \end{aligned}$$

43. (4)

$- \Rightarrow \div$	$+ \Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow +$

Option (1)

$$\begin{aligned} 49 \times 7 + 3 \div 5 - 8 &= 16 \\ \Rightarrow 49 + 7 \times 3 - 5 \div 8 &= 16 \\ \Rightarrow 49 + 21 - \frac{5}{8} &= 16 \end{aligned}$$

$$\Rightarrow 392 + 168 - 5 \neq 128$$

Option (2)

$$49 \div 7 \times 3 + 5 - 8 = 26$$

$$\Rightarrow 49 - 7 + 3 \times 5 \div 8 = 26$$

$$\Rightarrow 49 - 7 + \frac{15}{8} = 26$$

$$\Rightarrow 392 - 56 + 15 = 208$$

$$\Rightarrow 351 \neq 208$$

Option (3)

$$49 + 7 - 3 \times 5 \div 8 = 20$$

$$\Rightarrow 49 \times 7 \div 3 + 5 - 8 = 20$$

$$\Rightarrow \frac{49 \times 7}{3} + 5 - 8 \neq 20$$

Option (4)

$$49 - 7 + 3 \div 5 \times 8 = 24$$

$$\Rightarrow 49 \div 7 \times 3 - 5 + 8 = 24$$

$$\Rightarrow 7 \times 3 - 5 + 8 = 24$$

$$\Rightarrow 21 - 5 + 8 = 24$$

44. (*)

$+ \Rightarrow \times$	$- \Rightarrow \div$
$\times \Rightarrow +$	$\div \Rightarrow -$

$$25 \times 5 - 3 \div 2 + 5 = ?$$

$$\Rightarrow ? = 25 + 5 \div 3 - 2 \times 5$$

$$\Rightarrow ? = 25 + \frac{5}{3} - 10$$

$$\Rightarrow ? = \frac{75 + 5 - 30}{3} = \frac{80 - 30}{3}$$

$$= \frac{50}{3} = 16\frac{2}{3}$$

45. (*)

$- \Rightarrow +$	$+ \Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow \div$

Option (1)

$$5 - 2 + 12 \times 6 \div 2 = 27$$

$$\Rightarrow 5 + 2 \times 12 \div 6 - 2 = 27$$

$$\Rightarrow 5 + 2 \times 2 - 2 = 27$$

$$\Rightarrow 5 + 4 - 2 \neq 27$$

Option (2)

$$5 + 2 - 12 \div 6 \times 2 = 13$$

$$\Rightarrow 5 \times 2 + 12 - 6 \div 2 = 13$$

$$\Rightarrow 10 + 12 - 3 = 13$$

$$\Rightarrow 19 \neq 13$$

Option (3)

$$5 + 2 - 12 \times 6 \div 2 = 10$$

$$\Rightarrow 5 \times 2 + 12 \div 6 - 2 = 10$$

$$\Rightarrow 10 + 2 - 2 = 10$$

Option (4)

$$5 \div 2 + 12 \times 6 - 2 = 3$$

$$\Rightarrow 5 - 2 \times 12 \div 6 + 2 = 3$$

$$\Rightarrow 5 - 2 \times 2 + 2 = 3$$

$$\Rightarrow 5 - 4 + 2 = 3$$

Options (1) and (2) are wrong.

46. (1)

$\times \Rightarrow -$	$+ \Rightarrow \times$
$\div \Rightarrow +$	$- \Rightarrow \div$

$$175 - 25 \div 5 + 20 \times 3 + 10 = ?$$

$$\Rightarrow ?$$

$$= 175 \div 25 + 5 \times 20 - 3 \times 10$$

$$\Rightarrow ? = 7 + 100 - 30 = 77$$

47. (2)

$+ \Rightarrow \div$	$\div \Rightarrow -$
$- \Rightarrow \times$	$\times \Rightarrow +$

$$8 + 2 \div 3 - 4 \times 6 = ?$$

$$\Rightarrow ? = 8 \div 2 - 3 \times 4 + 6$$

$$\Rightarrow ? = 4 - 12 + 6 = -2$$

48. (4)

$\div \Rightarrow -$	$- \Rightarrow \times$
$\times \Rightarrow +$	$+ \Rightarrow \div$

$$20 \times 60 \div 40 - 20 + 10 = ?$$

$$\Rightarrow ? = 20 + 60 - 40 \times 20 \div 10$$

$$\Rightarrow ? = 20 + 60 - 40 \times 2$$

$$\Rightarrow ? = 80 - 80 = 0$$

49. (*)

$+ \Rightarrow \div$	$\times \Rightarrow +$
$- \Rightarrow \times$	$\div \Rightarrow -$

Option (1)

$$36 \times 6 + 3 - 2 < 20$$

$$\Rightarrow 36 + 6 \div 3 \times 2 < 20$$

$$\Rightarrow 36 + 2 \times 2 < 20$$

Option (2)

$$36 \times 6 + 3 \times 2 > 20$$

$$\Rightarrow 36 + 6 \div 3 + 2 > 20$$

$$\Rightarrow 36 + 2 + 2 > 20$$

Option (3)

$$36 + 6 \times 3 + 2 = 20$$

$$\Rightarrow 36 \div 6 + 3 \div 2 = 20$$

$$\Rightarrow 6 + \frac{3}{2} \neq 20$$

Option (4)

$$36 + 6 - 3 \times 2 = 20$$

$$\Rightarrow 36 \div 6 \times 3 + 2 = 20$$

$$\Rightarrow 6 \times 3 + 2 = 20$$

$$\Rightarrow 18 + 2 = 20$$

Both the Options (2) and (4) are correct.

50. (1)

$- \Rightarrow +$	$\div \Rightarrow \times$
$\times \Rightarrow -$	$+ \Rightarrow \div$

Option (1)

$$25 - 15 + 5 \div 4 \times 16 = 21$$

$$\Rightarrow 25 + 15 \div 5 \times 4 - 16 = 21$$

$$\Rightarrow 25 + 3 \times 4 - 16 = 21$$

$$\Rightarrow 25 + 12 - 16 = 21$$

$$\Rightarrow 37 - 16 = 21$$

Option (2)

$$25 + 11 + 4 \div 10 \times 6 = 20$$

$$\Rightarrow 25 \div 11 + 4 \times 10 - 6 = 20$$

$$\Rightarrow \frac{25}{11} + 40 - 6 \neq 20$$

Option (3)

$$25 \times 12 - 14 \div 4 + 6 = 16$$

$$\Rightarrow 25 - 12 + 14 \times 4 \div 6 = 16$$

$$\Rightarrow 25 - 12 + 14 \times \frac{2}{3} = 16$$

$$\Rightarrow 25 - 12 + \frac{28}{3} \neq 16$$

Option (4)

$$25 - 12 + 14 \div 2 \times 4 = 15$$

$$\Rightarrow 25 + 12 \div 14 \times 2 - 4 = 15$$

$$\Rightarrow 25 + \frac{6}{7} \times 2 - 4 \neq 15$$

51. (1)

$$5 + 3 \times 8 - 12 \div 4 = 3$$

$$\Rightarrow 5 + 3 \times 8 \div 12 - 4 = 3$$

$$\Rightarrow 5 + 2 - 4 = 3$$

52. (3)

$$33 \times 11 \div 3 - 6 = 115$$

$$\Rightarrow \left(\frac{363}{3} \right) - 6 = 115$$

$$\Rightarrow 121 - 6 = 115$$

53. (4)

$\times \Rightarrow +$	$\div \Rightarrow -$
$+ \Rightarrow \div$	$- \Rightarrow \times$

$$14 \times 4 \div 70 + 10 - 2 = ?$$

$$\Rightarrow ? = 14 + 4 - 70 \div 10 \times 2$$

$$\Rightarrow ? = 14 + 4 - 7 \times 2$$

$$\Rightarrow ? = 18 - 14 = 4$$

54. (1)

$+ \Rightarrow \div$	$- \Rightarrow \times$
$\times \Rightarrow +$	$\div \Rightarrow -$

Option (1)

$$5 \times 8 - 5 + 5 \div 1 = 12$$

$$\Rightarrow 5 + 8 \times 5 \div 5 - 1 = 12$$

$$\Rightarrow 5 + 8 \times 1 - 1 = 12$$

$$\Rightarrow 5 + 8 - 1 = 12$$

Option (2)

$$55 - 2 + 10 \div 1 \times 5 = 16$$

$$\Rightarrow 55 \times 2 \div 10 - 1 + 5 = 16$$

$$\Rightarrow \frac{55 \times 2}{10} - 1 + 5 = 16$$

$$\Rightarrow 11 - 1 + 5 \neq 16$$

Option (3)

$$38 \div 10 - 5 + 7 \times 8 = 25$$

$$\Rightarrow 38 - 10 \times 5 \div 7 + 8 = 25$$

$$\Rightarrow 38 - \frac{10 \times 5}{7} + 8 = 25$$

$$\Rightarrow 38 - \frac{50}{7} + 8 \neq 25$$

Option (4)

$$10 - 12 + 2 \div 30 \times 1 = 10$$

$$\Rightarrow 10 \times 12 \div 2 - 30 + 1 = 10$$

$$\Rightarrow 10 \times 6 - 30 + 1 = 10$$

$$\Rightarrow 60 - 30 + 1 \neq 10$$

55. (3)

$- \Rightarrow +$	$+ \Rightarrow \times$
$\times \Rightarrow -$	

Option (1)

$$22 + 7 - 3 \times 9 = 148$$

$$\Rightarrow 22 \times 7 + 3 - 9 = 148$$

$$\Rightarrow 154 + 3 - 9 = 148$$

Option (2)

$$33 \times 5 - 10 + 20 = 228$$

$$\Rightarrow 33 - 5 + 10 \times 20 = 228$$

$$\Rightarrow 33 - 5 + 200 = 228$$

$$\Rightarrow 233 - 5 = 228$$

Option (3)

$$7 \times 28 - 3 \times 52 = 127$$

$$\Rightarrow 7 \times 28 + 3 - 52 = 127$$

$$\Rightarrow 196 + 3 - 52 = 127$$

$$\Rightarrow 199 - 52 \neq 127$$

Option (4)

$$44 - 9 + 6 \times 11 = 87$$

$$\Rightarrow 44 + 9 \times 6 - 11 = 87$$

$$\Rightarrow 44 + 54 - 11 = 87$$

$$\Rightarrow 98 - 11 = 87$$

56. (2)

$+ \Rightarrow \div$	$\times \Rightarrow +$
$- \Rightarrow \times$	$\div \Rightarrow -$

Option (1)

$$5 - 3 + 2 \times 4 \div 8 = 2$$

$$\Rightarrow 5 \times 3 \div 2 + 4 - 8 = 2$$

$$\Rightarrow 5 \times \frac{3}{2} + 4 - 8 = 2$$

$$\Rightarrow \frac{15}{2} + 4 - 8 = 2$$

$$\Rightarrow \frac{15 + 8 - 16}{2} \neq 2$$

Option (2)

$$5 \times 3 + 2 - 4 \times 8 = 19$$

$$\Rightarrow 5 + 3 \div 2 \times 4 + 8 = 19$$

$$\Rightarrow 5 + \frac{3}{2} \times 4 + 8 = 19$$

$$\Rightarrow 5 + 3 \times 2 + 8 = 19$$

$$\Rightarrow 5 + 6 + 8 = 19$$

Option (3)

$$5 \div 3 \times 2 - 4 + 8 = 8$$

$$\Rightarrow 5 - 3 + 2 \times 4 \div 8 = 8$$

$$\Rightarrow 5 - 3 + 2 \times \frac{4}{8} = 8$$

$$\Rightarrow 5 - 3 + 1 \neq 8$$

Option (4)

$$5 + 3 \times 2 \div 4 - 8 = 4$$

$$\Rightarrow 5 \div 3 + 2 - 4 \times 8 = 4$$

$$\Rightarrow \frac{5}{3} + 2 - 32 \neq 4$$

57. (4)

$+\Rightarrow\div$	$-\Rightarrow+$
$\times\Rightarrow-$	$\div\Rightarrow\times$

$$8 \div 4 - 6 + 3 \times 4 = ?$$

$$? = 8 \times 4 + 6 \div 3 - 4$$

$$\Rightarrow ? = 32 + 2 - 4 = \boxed{30}$$

58. (4)

$+\Rightarrow-$	$-\Rightarrow\times$
$\times\Rightarrow\div$	$\div\Rightarrow+$

$$25 \times 5 \div 30 + 8 - 2 = ?$$

$$? = 25 \div 5 + 30 - 8 \times 2$$

$$\Rightarrow ? = 5 + 30 - 16 = 19$$

59. (1)

$-\Rightarrow+$	$+\Rightarrow-$
$\div\Rightarrow\times$	$\times\Rightarrow\div$

Option (1)

$$50 \times 5 \div 2 - 30 + 25 = 25$$

$$\Rightarrow 50 \div 5 \times 2 + 30 - 25 = 25$$

$$\Rightarrow 10 \times 2 + 30 - 25 = 25$$

$$\Rightarrow 20 + 30 - 25 = 25$$

Option (2)

$$50 - 30 + 5 \div 2 \times 30 = 25$$

$$\Rightarrow 50 + 30 - 5 \times 2 \div 30 = 25$$

$$\Rightarrow 50 + 30 - \frac{1}{3} = 25$$

$$\Rightarrow 80 - \frac{1}{3} \neq 25$$

Option (3)

$$40 + 35 \times 2 - 50 \div 30 = 95$$

$$\Rightarrow 40 - 35 \div 2 + 50 \times 30 = 95$$

$$\Rightarrow 40 - \frac{35}{2} + 1500 \neq 95$$

Option (4)

$$30 \times 2 - 25 + 50 \div 5 = 100$$

$$\Rightarrow 30 \div 2 + 25 - 50 \times 5 = 100$$

$$\Rightarrow 15 + 25 - 250 \neq 100$$

60. (3)

$+\Rightarrow\div$	$\times\Rightarrow+$
$-\Rightarrow\times$	$\div\Rightarrow-$

$$(1) 15 \div 5 \times 2 - 6 + 3 = 28$$

$$\Rightarrow 15 - 5 + 2 \times 6 \div 3 = 28$$

$$\Rightarrow 15 - 5 + 2 \times 2 = 28$$

$$\Rightarrow 15 - 5 + 4 \neq 28$$

$$(2) 15 \times 5 + 2 - 6 \div 3 = 56.5$$

$$\Rightarrow 15 + 5 \div 2 \times 6 - 3 = 56.5$$

$$\Rightarrow 15 + 2.5 \times 6 - 3 = 56.5$$

$$\Rightarrow 15 + 15 - 3 \neq 56.5$$

$$(3) 15 + 5 - 2 \div 6 \times 3 = 3$$

$$\Rightarrow 15 \div 5 \times 2 - 6 + 3 = 3$$

$$\Rightarrow 3 \times 2 - 6 + 3 = 3$$

$$\Rightarrow 6 - 6 + 3 = 3$$

$$(4) 15 - 5 + 2 \times 6 \div 3 = 41$$

$$\Rightarrow 15 \times 5 \div 2 + 6 - 3 = 41$$

$$\Rightarrow 15 \times 2.5 + 6 - 3 = 41$$

$$\Rightarrow 37.5 + 6 - 3 \neq 41$$

61. (2)

$-\Rightarrow\div$	$+\Rightarrow\times$
$\div\Rightarrow-$	$\times\Rightarrow+$

Option (1)

$$36 \times 4 - 12 + 5 \div 3 = 420$$

$$\Rightarrow 36 + 4 \div 12 \times 5 - 3 = 420$$

$$\Rightarrow 36 + \frac{4}{12} \times 5 - 3 = 420$$

$$\Rightarrow 36 + \frac{5}{3} - 3 = 420$$

$$\Rightarrow \frac{108 + 5 - 9}{3} = 420$$

$$\Rightarrow \frac{104}{3} \neq 420$$

Option (2)

$$52 \div 4 + 5 \times 8 - 2 = 36$$

$$\Rightarrow 52 - 4 \times 5 + 8 \div 2 = 36$$

$$\Rightarrow 52 - 4 \times 5 + 4 = 36$$

$$\Rightarrow 52 - 20 + 4 = 36$$

$$\Rightarrow 56 - 20 = 36$$

Option (3)

$$36 - 12 \times 6 \div 3 + 4 = 60$$

$$\Rightarrow 36 \div 12 + 6 - 3 \times 4 = 60$$

$$\Rightarrow 3 + 6 - 12 \neq 60$$

Option (4)

$$43 \times 7 \div 5 + 4 - 8 = 25$$

$$\Rightarrow 43 + 7 - 5 \times 4 \div 8 = 25$$

$$\Rightarrow 43 + 7 - \frac{5}{2} = 25$$

$$\Rightarrow 50 - \frac{5}{2} \neq 25$$

62. (3)

$a \Rightarrow \div$	$b \Rightarrow +$
$c \Rightarrow -$	$d \Rightarrow \times$

$$24 a 6 d 4 b 9 c 8 = ?$$

$$\Rightarrow ? = 24 \div 6 \times 4 + 9 - 8$$

$$\Rightarrow ? = 4 \times 4 + 9 - 8$$

$$\Rightarrow ? = 16 + 9 - 8 = 17$$

63. (2)

$\times \Rightarrow +$	$+\Rightarrow\div$
$-\Rightarrow\times$	$\div\Rightarrow-$

$$6 \times 4 - 5 + 2 \div 1 = ?$$

$$\Rightarrow ? = 6 + 4 \times 5 \div 2 - 1$$

$$\Rightarrow ? = 6 + 10 - 1 = 15$$

64. (3)

$+\Rightarrow\div$	$\times\Rightarrow+$
$-\Rightarrow\times$	$\div\Rightarrow-$

$$(a) 46 \times 6 \div 4 - 5 + 3 = 74$$

$$\Rightarrow 46 + 6 - 4 \times 5 \div 3 = 74$$

$$\Rightarrow 46 + 6 - 4 \times \frac{5}{3} = 74$$

$$\Rightarrow 46 + 6 - \frac{20}{3} = 74$$

$$\Rightarrow 52 - \frac{20}{3} = 74$$

$$\Rightarrow \frac{156 - 20}{3} = 74$$

$$\Rightarrow 136 \neq 74 \times 3$$

$$(b) 46 - 6 + 4 \times 5 \div 3 = 71$$

$$\Rightarrow 46 \times 6 \div 4 + 5 - 3 = 71$$

$$\Rightarrow 46 \times \frac{6}{4} + 5 - 3 = 71$$

$$\Rightarrow 69 + 5 - 3 = 71$$

$$(c) 46 \div 6 \times 4 - 5 + 3 = 75.5$$

$$\Rightarrow 46 - 6 + 4 \times 5 \div 3 = 75.5$$

$$\Rightarrow 46 - 6 + 4 \times \frac{5}{3} = 75.5$$

$$\Rightarrow 46 - 6 + \frac{20}{3} \neq 75.5$$

$$(d) 46 \times 6 - 4 + 5 \div 3 = 70.1$$

$$\Rightarrow 46 + 6 \times 4 \div 5 - 3 = 70.1$$

$$\Rightarrow 46 + 6 \times \frac{4}{5} - 3 = 70.1$$

$$\Rightarrow 46 + \frac{24}{5} - 3 = 70.1$$

$$\Rightarrow \frac{230 + 24 - 15}{5} = 70.1$$

$$\Rightarrow \frac{239}{5} \neq 70.1$$

65. (3)

$+\Rightarrow\times$	$-\Rightarrow\div$
$\times\Rightarrow+$	$\div\Rightarrow-$

Option (1)

$$18 - 6 \times 7 \div 2 + 8 = 63$$

$$\Rightarrow 18 \div 6 + 7 - 2 \times 8 = 63$$

$$\Rightarrow 3 + 7 - 16 \neq 63$$

Option (2)

$$\begin{aligned} 18 \div 6 + 4 - 2 \div 3 &= 22 \\ \Rightarrow 18 - 6 \times 4 \div 2 - 3 &= 22 \\ \Rightarrow 18 - 6 \times 2 - 3 &= 22 \\ \Rightarrow 18 - 12 - 3 &\neq 22 \end{aligned}$$

Option (3)

$$\begin{aligned} 18 + 6 - 4 \times 2 \div 3 &= 26 \\ \Rightarrow 18 \times 6 \div 4 + 2 - 3 &= 26 \\ \Rightarrow 27 + 2 - 3 &= 26 \end{aligned}$$

Option (4)

$$\begin{aligned} 18 \times 6 - 4 + 7 \times 8 &= 47 \\ \Rightarrow 18 + 6 \div 4 \times 7 + 8 &= 47 \end{aligned}$$

$$\Rightarrow 18 + \frac{3}{2} \times 7 + 8 = 47$$

$$\Rightarrow 18 + \frac{21}{2} + 8 = 47$$

$$\Rightarrow \frac{36 + 21 + 16}{2} = 47$$

$$\Rightarrow 73 \neq 47 \times 2$$

66. (3)

$+$	$\Rightarrow$	$\times$	$-$	$\Rightarrow$	$\div$
$\times$	$\Rightarrow$	$-$	$\div$	$\Rightarrow$	$+$

$$\begin{aligned} 16 \div 64 - 8 \times 4 + 2 &= ? \\ \Rightarrow ? &= 16 + 64 \div 8 - 4 \times 2 \\ \Rightarrow ? &= 16 + 8 - 8 = 16 \end{aligned}$$

67. (1)

$*$	$\Rightarrow$	$+$	$\#$	$\Rightarrow$	$-$
$@$	$\Rightarrow$	$\times$	$\%$	$\Rightarrow$	$\div$

Option (1)

$$\begin{aligned} 256 \% 16 @ 5 \# 28 &= 52 \\ \Rightarrow 256 \div 16 \times 5 - 28 &= 52 \\ \Rightarrow 16 \times 5 - 28 &= 52 \\ \Rightarrow 80 - 28 &= 52 \end{aligned}$$

Option (2)

$$\begin{aligned} 256 \# 16 \% 5 \# 28 &= 120 \\ \Rightarrow 256 - 16 \div 5 - 28 &= 120 \\ \Rightarrow 256 - \frac{16}{5} - 28 &\neq 120 \end{aligned}$$

Option (3)

$$\begin{aligned} 256 @ 5 \% 16 * 28 &= 408 \\ \Rightarrow 256 \times 5 \div 16 + 28 &= 408 \\ \Rightarrow \frac{256 \times 5}{16} + 28 &\neq 408 \end{aligned}$$

Option (4)

$$\begin{aligned} 256 \# 16 @ 5 \% 28 &= 80 \\ \Rightarrow 256 - 16 + 5 \div 28 &= 80 \\ \Rightarrow 256 - 16 + \frac{5}{28} &\neq 80 \end{aligned}$$

68. (4)

$\div \Rightarrow \times$	$\times \Rightarrow -$
$- \Rightarrow +$	$+ \Rightarrow \div$

$$\begin{aligned} 48 + 6 - 12 \div 2 + 10 &= ? \\ \Rightarrow ? &= 48 \div 6 + 12 \times 2 \div 10 \\ \Rightarrow ? &= 8 + 12 \times \frac{2}{10} \end{aligned}$$

$$\Rightarrow ? = 20 \times \frac{2}{10} = 4$$

69. (2) $5 \times 3 + 8 - 4 \div 2 = 21$
 $\Rightarrow 15 + 8 - 2 = 21$

70. (3)

$a \Rightarrow +$	$b \Rightarrow \times$
$c \Rightarrow \div$	$d \Rightarrow -$

$$\begin{aligned} 20 a 10 b 45 c 5 d 12 &= ? \\ \Rightarrow ? &= 20 + 10 \times 45 \div 5 - 12 \\ \Rightarrow ? &= 20 + 10 \times 9 - 12 \\ \Rightarrow ? &= 20 + 90 - 12 = 98 \end{aligned}$$

71. (*)

$+$	$\Rightarrow$	$\div$	$\times$	$\Rightarrow$	$+$
$-$	$\Rightarrow$	$\times$	$\div$	$\Rightarrow$	$-$

Option (1)

$$\begin{aligned} 25 \times 3 - 7 \div 8 + 12 &= 18 \\ \Rightarrow 25 + 3 \times 7 - 8 \div 12 &= 18 \\ \Rightarrow 25 + 3 \times 7 - \frac{8}{12} &= 18 \end{aligned}$$

$$\Rightarrow 25 + 21 - \frac{2}{3} = 18$$

$$\Rightarrow 46 - \frac{2}{3} = 18$$

$$\Rightarrow \frac{138 - 2}{3} = 18$$

$$\Rightarrow 136 \neq 18 \times 3$$

Option (2)

$$\begin{aligned} 25 + 3 \times 7 - 8 \div 12 &= 10.89 \\ \Rightarrow 25 \div 3 + 7 \times 8 - 12 &= 10.89 \\ \Rightarrow \frac{25}{3} + 56 - 12 &= 10.89 \end{aligned}$$

$$\Rightarrow \frac{25 + 168 - 36}{3} = 10.89$$

$$\Rightarrow 157 \neq 10.89 \times 3$$

Option (3)

$$\begin{aligned} 25 - 3 \div 7 \times 8 + 12 &= 132 \\ \Rightarrow 25 \times 3 - 7 + 8 \div 12 &= 132 \end{aligned}$$

$$\Rightarrow 75 - 7 + \frac{8}{12} = 132$$

$$\Rightarrow 75 - 7 + \frac{2}{3} = 132$$

$$\Rightarrow \frac{225 - 21 + 2}{3} = 132$$

$$\Rightarrow 206 \neq 132 \times 3$$

Option (4)

$$\begin{aligned} 25 \div 3 \times 7 - 8 + 12 &= 19.3 \\ \Rightarrow 25 - 3 + 7 \times 8 \div 12 &= 19.3 \end{aligned}$$

$$\Rightarrow 25 - 3 + \frac{56}{12} = 19.3$$

$$\Rightarrow 25 - 3 + \frac{14}{3} = 19.3$$

$$\Rightarrow \frac{75 - 9 + 14}{3} = 19.3$$

$$\Rightarrow 80 \neq 19.3 \times 3$$

72. (4) $4 \times 3 \times 4 = 48$

73. (1) **Option (1)**

$$\begin{aligned} 18 + 6 \div 5 &= 9.6 \\ \Rightarrow 18 \div 5 + 6 &= 9.6 \\ \Rightarrow 3.6 + 6 &= 9.6 \end{aligned}$$

Option (2)

$$\begin{aligned} 26 \div 5 + 6 &= 6.4 \\ \Rightarrow 26 + 6 \div 5 &= 6.4 \\ \Rightarrow 26 + 1.2 &\neq 6.4 \end{aligned}$$

Option (3)

$$\begin{aligned} 5 \div 6 + 80 &= 5.8 \\ \Rightarrow 6 + 5 \div 80 &= 5.8 \\ \Rightarrow 6 + 0.0625 &\neq 5.8 \end{aligned}$$

Option (4)

$$\begin{aligned} 90 + 5 \div 6 &= 8.6 \\ \Rightarrow 90 \div 6 + 5 &= 8.6 \\ \Rightarrow 15 + 5 &\neq 8.6 \end{aligned}$$

74. (2)

$- \Rightarrow +$	$+$	$\Rightarrow$	$\times$
$\div \Rightarrow -$	$\times$	$\Rightarrow$	$\div$

Option (1)

$$\begin{aligned} 5 + 2 - 12 \div 6 \times 2 &= 13 \\ \Rightarrow 5 \times 2 + 12 - 6 \div 2 &= 13 \\ \Rightarrow 10 + 12 - 3 &= 13 \\ \Rightarrow 22 - 3 &\neq 13 \end{aligned}$$

Option (2)

$$\begin{aligned} 5 + 2 - 12 \times 6 \div 2 &= 10 \\ \Rightarrow 5 \times 2 + 12 \div 6 - 2 &= 10 \\ \Rightarrow 10 + 2 - 2 &= 10 \end{aligned}$$

Option (3)

$$\begin{aligned} 5 \div 2 + 12 \times 6 - 2 &= 4 \\ \Rightarrow 5 - 2 \times 12 \div 6 + 2 &= 4 \\ \Rightarrow 5 - 2 \times 2 + 2 &= 4 \\ \Rightarrow 5 - 4 + 2 &= 4 \\ \Rightarrow 7 - 4 &\neq 4 \end{aligned}$$

Option (4)

$$\begin{aligned} 5 - 2 + 12 \times 6 \div 2 &= 27 \\ \Rightarrow 5 + 2 \times 12 \div 6 - 2 &= 27 \\ \Rightarrow 5 + 2 \times 2 - 2 &= 27 \\ \Rightarrow 5 + 4 - 2 &\neq 27 \end{aligned}$$

75. (3)

P $\Rightarrow \div$	Q $\Rightarrow \times$
R $\Rightarrow +$	S $\Rightarrow -$

$$\begin{aligned} 16 \text{ Q } 12 \text{ P } 6 \text{ R } 5 \text{ S } 4 &= ? \\ \Rightarrow ? &= 16 \times 12 \div 6 + 5 - 4 \\ \Rightarrow ? &= 16 \times 2 + 5 - 4 \\ \Rightarrow ? &= 32 + 5 - 4 \\ \Rightarrow ? &= 37 - 4 = 33 \end{aligned}$$

76. (3)

+ $\Rightarrow \div$	$\div \Rightarrow +$
$\times \Rightarrow -$	$- \Rightarrow \times$

$$\begin{aligned} (30 + 20) - 5 (7 \div 3) \times 25 &= ? \\ \Rightarrow ? &= (30 - 20) \times 5 (7 + 3) \div 25 \\ \Rightarrow ? &= 10 \times 5 \times 10 \div 25 = 20 \end{aligned}$$

77. (2)

+ $\Rightarrow \div$	$\div \Rightarrow -$
$- \Rightarrow \times$	$\times \Rightarrow +$

$$\begin{aligned} 12 - 8 \times 6 - 4 \div 6 + 3 &= ? \\ \Rightarrow ? &= 12 \times 8 + 6 \times 4 - 6 \div 3 \\ \Rightarrow ? &= 96 + 24 - 2 \\ \Rightarrow ? &= 120 - 2 = 118 \end{aligned}$$

78. (3)

+ $\Rightarrow \div$	$\div \Rightarrow \times$
$\times \Rightarrow +$	

$$\begin{aligned} 64 + 8 \times 32 \div 4 &= ? \\ \Rightarrow ? &= 64 \div 8 + 32 \times 4 \\ \Rightarrow ? &= 8 + 128 = 136 \end{aligned}$$

79. (*)

$- \Rightarrow +$	+ $\Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow \div$

$$\begin{aligned} 27 \times 3 \div 6 + 9 - 8 &= ? \\ \Rightarrow ? &= 27 \div 3 - 6 \times 9 + 8 \\ \Rightarrow ? &= 9 - 54 + 8 \\ \Rightarrow ? &= 17 - 54 = -37 \end{aligned}$$

80. (4) $4 \times 6 \div 2 - 4 + 8 = 16$

$$\begin{aligned} \Rightarrow 4 \times 3 - 4 + 8 &= 16 \\ \Rightarrow 12 - 4 + 8 &= 16 \end{aligned}$$

81. (1) $b \text{ af } \div b \text{ f } \times d$

$$\begin{aligned} \Rightarrow 105 \div 15 \times 3 \\ \Rightarrow 7 \times 3 = 21 \Rightarrow cb \end{aligned}$$

82. (1) $5 + 3 \times 8 - 12 \div 4 = 3$

$$\begin{aligned} \Rightarrow 5 + 3 \times 8 \div 12 - 4 &= 3 \\ \Rightarrow 5 + \frac{3 \times 8}{12} - 4 &= 3 \\ \Rightarrow 5 + 2 - 4 &= 3 \end{aligned}$$

83. (4)

+ $\Rightarrow \div$	$\times \Rightarrow +$
$- \Rightarrow \times$	$\div \Rightarrow -$

Option (1)

$$\begin{aligned} 33 \times 4 - 5 + 6 \div 2 &= 26 \\ \Rightarrow 33 + 4 \times 5 \div 6 - 2 &= 26 \\ \Rightarrow 33 + \frac{10}{3} - 2 &= 26 \end{aligned}$$

$$\Rightarrow \frac{99 + 10 - 6}{3} \neq 26$$

Option (2)

$$\begin{aligned} 33 \div 4 \times 5 + 6 - 2 &= 30 \\ \Rightarrow 33 - 4 + 5 \div 6 \times 2 &= 30 \end{aligned}$$

$$\Rightarrow 33 - 4 + \frac{5}{6} \times 2 = 30$$

$$\Rightarrow 33 - 4 + \frac{5}{3} = 30$$

$$\Rightarrow \frac{99 - 12 + 5}{3} \neq 30$$

Option (3)

$$\begin{aligned} 33 - 4 + 5 \div 6 \times 2 &= 24 \\ \Rightarrow 33 \times 4 \div 5 - 6 + 2 &= 24 \end{aligned}$$

$$\Rightarrow \frac{132}{5} - 6 + 2 = 24$$

$$\Rightarrow \frac{132 - 30 + 10}{5} = 24$$

$$\Rightarrow 112 \neq 120$$

Option (4)

$$\begin{aligned} 33 - 4 \div 5 \times 6 + 2 &= 130 \\ \Rightarrow 33 \times 4 - 5 + 6 \div 2 &= 130 \\ \Rightarrow 132 - 5 + 3 &= 130 \\ \Rightarrow 135 - 5 &= 130 \end{aligned}$$

84. (3)

$\times \Rightarrow +$	$- \Rightarrow \div$
$\div \Rightarrow -$	+ $\Rightarrow \times$

Option (1)

$$\begin{aligned} 16 \times 5 \div 10 + 4 - 3 &= 19 \\ \Rightarrow 16 + 5 - 10 \times 4 \div 3 &= 19 \end{aligned}$$

$$\Rightarrow 16 + 5 - \frac{10 \times 4}{3} = 19$$

$$\Rightarrow 21 - \frac{40}{3} = 19$$

$$\Rightarrow \frac{63 - 40}{3} = 19$$

$$\Rightarrow \frac{23}{3} \neq 19$$

Option (2)

$$\begin{aligned} 16 + 5 \div 10 \times 4 - 3 &= 9 \\ \Rightarrow 16 \times 5 - 10 + 4 \div 3 &= 9 \end{aligned}$$

$$\Rightarrow 80 - 10 + \frac{4}{3} = 9$$

$$\Rightarrow \frac{240 - 30 + 4}{3} = 9$$

$$\Rightarrow \frac{214}{3} \neq 9$$

Option (3)

$$\begin{aligned} 16 + 5 - 10 \times 4 \div 3 &= 9 \\ \Rightarrow 16 \times 5 \div 10 + 4 - 3 &= 9 \\ \Rightarrow 8 + 4 - 3 &= 9 \end{aligned}$$

Option (4)

$$\begin{aligned} 16 - 5 \times 10 \div 4 + 3 &= 12 \\ \Rightarrow 16 \div 5 + 10 - 4 \times 3 &= 12 \end{aligned}$$

$$\Rightarrow \frac{16}{5} + 10 - 12 = 12$$

$$\Rightarrow \frac{16 + 50 - 60}{5} = 12$$

$$\Rightarrow \frac{66 - 60}{5} \neq 12$$

85. (3)

+ $\Rightarrow \times$	$- \Rightarrow +$
$\times \Rightarrow \div$	$\div \Rightarrow -$

$$\begin{aligned} 9 - 4 + 2 \div 16 \times 2 &= ? \\ \Rightarrow ? &= 9 + 4 \times 2 - 16 \div 2 \\ \Rightarrow ? &= 9 + 4 \times 2 - 8 \end{aligned}$$

$$\Rightarrow ? = 9 + 8 - 8 = \boxed{9}$$

86. (3) $92 \times 4 \div 2 = 184$

$$\Rightarrow 92 \times 2 = 184$$

87. (*)

+ $\Rightarrow \div$	$\div \Rightarrow \times$
$\times \Rightarrow -$	$- \Rightarrow +$

Option (1)

$$\begin{aligned} 30 \times 40 + 8 - 70 \div 40 &= 180 \\ \Rightarrow 30 - 40 \div 8 + 70 \times 40 &= 180 \\ \Rightarrow 30 - 5 + 2800 &= 180 \\ \Rightarrow 2830 - 5 &\neq 180 \end{aligned}$$

Option (2)

$$\begin{aligned} 30 + 40 \div 8 \times 70 - 40 &= 340 \\ \Rightarrow 30 \div 40 \times 8 - 70 + 40 &= 340 \\ \Rightarrow 6 - 70 + 40 &= 340 \\ \Rightarrow 46 - 70 &\neq 340 \end{aligned}$$

Option (3)

$$\begin{aligned} 30 - 40 \times 8 \div 70 + 40 &= 180 \\ \Rightarrow 30 + 40 - 8 \times 70 \div 40 &= 180 \\ \Rightarrow 30 + 40 - 14 &\neq 180 \end{aligned}$$

Option (4)

$$30 + 40 + 8 \times 70 - 40 = 340$$

$$\Rightarrow 30 \div 40 \div 8 - 70 + 40 = 340$$

$$\Rightarrow \frac{3}{48} - 70 + 40 \neq 340$$

88. (1)

$\times \Rightarrow +$	$- \Rightarrow \div$
$/ \Rightarrow -$	$+ \Rightarrow \times$

Option (1)

$$25 + 10 - 5/10 \times 3 = 43$$

$$\Rightarrow 25 \times 10 \div 5 - 10 + 3 = 43$$

$$\Rightarrow 25 \times 2 - 10 + 3 = 43$$

$$\Rightarrow 50 - 10 + 3 = 43$$

$$\Rightarrow 53 - 10 = 43$$

Option (2)

$$25 - 10 \times 5 + 10/3 = 72$$

$$\Rightarrow 25 \div 10 + 5 \times 10 - 3 = 72$$

$$\Rightarrow 2.5 + 50 - 3 = 72$$

$$\Rightarrow 52.5 - 3 \neq 72$$

Option (3)

$$25 \times 10/5 + 10 - 3 = 12$$

$$\Rightarrow 25 + 10 - 5 \times 10 \div 3 = 12$$

$$\Rightarrow 25 + 10 - \frac{5 \times 10}{3} = 12$$

$$\Rightarrow 25 + 10 - 16.66 = 12$$

$$\Rightarrow 35 - 16.66 \neq 12$$

Option (4)

$$25/10 + 5 \times 10/3 = 18$$

$$\Rightarrow 25 - 10 \times 5 + 10 - 3 = 18$$

$$\Rightarrow 25 - 50 + 10 - 3 = 18$$

$$\Rightarrow 35 - 53 \neq 18$$

89. (3)

$+ \Rightarrow \times$	$- \Rightarrow +$
$\times \Rightarrow \div$	

$$128 + 9 - 16 \times 4 = ?$$

$$\Rightarrow ? = 128 \times 9 + 16 \div 4$$

$$\Rightarrow ? = 128 \times 9 + 4$$

$$\Rightarrow ? = 1152 + 4 = 1156$$

90. (4)

$+ \Rightarrow \div$	$\div \Rightarrow \times$
$\times \Rightarrow -$	$- \Rightarrow +$

Option (1)

$$18 \times 6 + 7 \div 5 - 2 = 16$$

$$\Rightarrow 18 - 6 \div 7 \times 5 + 2 = 16$$

$$\Rightarrow 18 - \frac{6}{7} \times 5 + 2 = 16$$

$$\Rightarrow 18 - \frac{30}{7} + 2 = 16$$

$$\Rightarrow \frac{126 - 30 + 14}{7} = 16$$

$$\Rightarrow \frac{110}{7} \neq 16$$

Option (2)

$$18 \div 6 \times 7 + 5 - 2 = 22$$

$$\Rightarrow 18 \times 6 - 7 \div 5 + 2 = 22$$

$$\Rightarrow 108 - \frac{7}{5} + 2 \neq 22$$

Option (3)

$$18 \div 6 - 7 + 5 \times 2 = 20$$

$$\Rightarrow 18 \times 6 + 7 \div 5 - 2 = 20$$

$$\Rightarrow 108 + \frac{7}{5} - 2 \neq 20$$

Option (4)

$$18 + 6 \div 7 \times 5 - 2 = 18$$

$$\Rightarrow 18 \div 6 \times 7 - 5 + 2 = 18$$

$$\Rightarrow 3 \times 7 - 5 + 2 = 18$$

$$\Rightarrow 21 - 5 + 2 = 18$$

91. (3)

$+ \Rightarrow \times$	$- \Rightarrow \div$
$\times \Rightarrow +$	$\div \Rightarrow -$

Option (1)

$$12 \times 5 + 4 - 5 \div 4 = 20$$

$$\Rightarrow 12 + 5 \times 4 \div 5 - 4 = 20$$

$$\Rightarrow 12 + 5 \times \frac{4}{5} - 4 = 20$$

$$\Rightarrow 12 + 4 - 4 \neq 20$$

Option (2)

$$12 \div 5 + 4 - 5 \times 4 = 18$$

$$\Rightarrow 12 - 5 \times 4 \div 5 + 4 = 18$$

$$\Rightarrow 12 - \frac{5 \times 4}{5} + 4 = 18$$

$$\Rightarrow 12 - 4 + 4 \neq 18$$

Option (3)

$$12 + 5 - 4 \times 5 \div 4 = 16$$

$$\Rightarrow 12 \times 5 \div 4 + 5 - 4 = 16$$

$$\Rightarrow 12 \times \frac{5}{4} + 5 - 4 = 16$$

$$\Rightarrow 15 + 5 - 4 = 16$$

Option (4)

$$12 \div 5 - 4 \times 5 + 4 = 22$$

$$\Rightarrow 12 - 5 \div 4 + 5 \times 4 = 22$$

$$\Rightarrow 12 - \frac{5}{4} + 20 = 22$$

$$\Rightarrow \frac{48 - 5 + 80}{4} = 22$$

$$\Rightarrow \frac{123}{4} \neq 20$$

92. (2)

$\div \Rightarrow +$	$+ \Rightarrow \times$
$\times \Rightarrow -$	$- \Rightarrow \div$

$$[(1440 - 36 \times 16) + 15] + 5 \div (144 - 12) + 25 = ?$$

$$\Rightarrow ? = [(1440 \div 36 - 16) \times 15] \times 5 + (144 \div 12) \times 25$$

$$\Rightarrow ? = [(40 - 16) \times 15] \times 5 + (12) \times 25$$

$$\Rightarrow ? = [24 \times 15] \times 5 + 300$$

$$\Rightarrow ? = 360 \times 5 + 300$$

$$\Rightarrow ? = 1800 + 300 = 2100$$

93. (1)

$+ \Rightarrow \times$	$- \Rightarrow +$
$\times \Rightarrow \div$	$\div \Rightarrow -$

$$50 + 10 - 50 \times 10 \div 125 = ?$$

$$\Rightarrow ? = 50 \times 10 + 50 \div 10 - 125$$

$$\Rightarrow ? = 500 + 5 - 125$$

$$\Rightarrow ? = 505 - 125 = 380$$

94. (3)

$\times \Rightarrow -$	$- \Rightarrow \div$
$+ \Rightarrow \times$	$\div \Rightarrow +$

$$16 \times 8 \div 4 - 3 + 9 = ?$$

$$\Rightarrow ? = 16 - 8 + 4 \div 3 \times 9$$

$$\Rightarrow ? = 16 - 8 + \frac{4}{3} \times 9$$

$$\Rightarrow ? = 16 - 8 + 12$$

$$\Rightarrow ? = 28 - 8 = 20$$

95. (2)

$- \Rightarrow \div$	$+ \Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow +$

Option (1)

$$18 \div 3 \times 2 + 8 - 6 = 10$$

$$\Rightarrow 18 - 3 + 2 \times 8 \div 6 = 10$$

$$\Rightarrow 18 - 3 + 2 \times \frac{8}{6} = 10$$

$$\Rightarrow 18 - 3 + \frac{8}{3} \neq 10$$

Option (2)

$$18 - 3 + 2 \times 8 \div 6 = 14$$

$$\Rightarrow 18 \div 3 \times 2 + 8 - 6 = 14$$

$$\Rightarrow 6 \times 2 + 8 - 6 = 14$$

$$\Rightarrow 12 + 8 - 6 = 14$$

Option (3)

$$18 - 3 \div 2 \times 8 + 6 = 17$$

$$\Rightarrow 18 \div 3 - 2 + 8 \times 6 = 17$$

$$\Rightarrow 6 - 2 + 48 \neq 17$$

Option (4)

$$18 \times 3 + 2 \div 8 - 6 = 15$$

$$\Rightarrow 18 + 3 \times 2 - 8 \div 6 = 15$$

$$\Rightarrow 18 + 6 - \frac{8}{6} \neq 15$$

96. (1)

$+\Rightarrow \times$	$-\Rightarrow \div$
$\times \Rightarrow -$	$\div \Rightarrow +$

$$\begin{aligned} 6 + 64 - 8 \div 45 \times 8 &= ? \\ \Rightarrow ? &= 6 \times 64 \div 8 + 45 - 8 \\ \Rightarrow ? &= 6 \times 8 + 45 - 8 \\ \Rightarrow ? &= 48 + 45 - 8 \\ \Rightarrow ? &= 93 - 8 = 85 \end{aligned}$$

97. (4)

$+\Rightarrow -$	$-\Rightarrow \times$
$\times \Rightarrow \div$	$\div \Rightarrow +$

$$\begin{aligned} 2 \div 6 \times 6 \div 2 &= ? \\ \Rightarrow ? &= 2 + 6 \div 6 + 2 \\ \Rightarrow ? &= 2 + 1 + 2 = 5 \end{aligned}$$

98. (1)

$+\Rightarrow \div$	$\div \Rightarrow -$
$-\Rightarrow \times$	$\times \Rightarrow +$

$$\begin{aligned} 8 + 4 \div 3 \times 5 - 9 &= ? \\ \Rightarrow ? &= 8 \div 4 - 3 + 5 \times 9 \\ \Rightarrow ? &= 2 - 3 + 45 \\ \Rightarrow ? &= 47 - 3 = \boxed{44} \end{aligned}$$

99. (*)

$\times \Rightarrow -$	$+\Rightarrow \times$
$\div \Rightarrow +$	$-\Rightarrow \div$

$$\begin{aligned} 175 - 25 \div 5 + 2 \times 3 + 10 &= ? \\ \Rightarrow ? &= 175 \div 25 + 5 \times 2 - 3 \times 10 \\ \Rightarrow ? &= 7 + 10 - 30 \\ \Rightarrow ? &= 17 - 30 = -13 \end{aligned}$$

100. (1)

$-\Rightarrow \div$	$+\Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow +$

Option (1)

$$\begin{aligned} 20 - 4 + 6 \div 9 \times 4 &= 25 \\ \Rightarrow 20 \div 4 \times 6 - 9 + 4 &= 25 \\ \Rightarrow 5 \times 6 - 9 + 4 &= 25 \\ \Rightarrow 30 - 9 + 4 &= 25 \end{aligned}$$

Option (2)

$$\begin{aligned} 20 + 6 - 4 \times 9 \div 6 &= 32 \\ \Rightarrow 20 \times 6 \div 4 + 9 - 6 &= 32 \\ \Rightarrow 30 + 9 - 6 &\neq 32 \end{aligned}$$

Option (3)

$$\begin{aligned} 20 \div 9 \times 9 - 4 + 6 &= 33 \\ \Rightarrow 20 - 9 + 9 \div 4 \times 6 &= 33 \end{aligned}$$

$$\Rightarrow 20 - 9 + \frac{9}{4} \times 6 = 33$$

$$\Rightarrow 20 - 9 + \frac{27}{2} = 33$$

$$\Rightarrow \frac{40 - 18 + 27}{2} = 33$$

$$\Rightarrow 49 \neq 2 \times 33$$

Option (4)

$$\begin{aligned} 20 \times 4 - 6 - 4 + 9 &= 20 \\ \Rightarrow 20 + 4 \div 6 \div 4 \times 9 &= 20 \end{aligned}$$

$$\Rightarrow 20 + \frac{4}{6} \times \frac{1}{4} \times 9 = 20$$

$$\Rightarrow 20 + \frac{3}{2} \neq 20$$

101. (1)

$@ \Rightarrow +$	$\oplus \Rightarrow -$
$a \Rightarrow \div$	$\theta \Rightarrow \times$

$$\begin{aligned} 8900 \ a \ 100 \ \oplus \ 5 \ \theta \ 4 \ \oplus \ 121 \ a \ 11 \\ \Rightarrow 8900 \div 100 - 5 \times 4 - 121 \div 11 \\ \Rightarrow 89 - 20 - 11 \\ \Rightarrow 89 - 31 = 58 \end{aligned}$$

102. (1)

$-\Rightarrow \div$	$+\Rightarrow \times$
$\div \Rightarrow -$	$\times \Rightarrow +$

Option (1)

$$\begin{aligned} 100 + 5 - 10 \times 250 \div 200 &= 100 \\ \Rightarrow 100 \times 5 \div 10 + 250 - 200 &= 100 \\ \Rightarrow 50 + 250 - 200 &= 100 \end{aligned}$$

Option (2)

$$\begin{aligned} 200 + 10 - 20 \times 200 \div 100 &= 150 \\ \Rightarrow 200 \times 10 \div 20 + 200 - 100 &= 150 \\ \Rightarrow 100 + 200 - 100 &= 150 \\ \Rightarrow 300 - 100 &\neq 150 \end{aligned}$$

Option (3)

$$\begin{aligned} 50 \times 5 \div 10 + 100 - 75 &= 50 \\ \Rightarrow 50 + 5 - 10 \times 100 \div 75 &= 50 \end{aligned}$$

$$\Rightarrow 50 + 5 - \frac{10 \times 100}{75} = 50$$

$$\Rightarrow 50 + 5 - \frac{40}{3} = 50$$

$$\Rightarrow \frac{150 + 15 - 40}{3} = 50$$

$$\begin{aligned} \Rightarrow 165 - 40 &= 50 \times 3 \\ \Rightarrow 125 &\neq 150 \end{aligned}$$

Option (4)

$$\begin{aligned} 300 + 5 - 20 \times 200 \div 100 &= 200 \\ \Rightarrow 300 \times 5 \div 20 + 200 - 100 &= 200 \\ \Rightarrow 75 + 200 - 100 &= 200 \\ \Rightarrow 275 - 100 &\neq 200 \end{aligned}$$

103. (1)

$-\Rightarrow \times$	$\times \Rightarrow +$
$+\Rightarrow \div$	$\div \Rightarrow -$

$$\begin{aligned} 40 \times 12 + 3 - 6 \div 60 &= ? \\ \Rightarrow ? &= 40 + 12 \div 3 \times 6 - 60 \\ \Rightarrow ? &= 40 + 4 \times 6 - 60 \\ \Rightarrow ? &= 40 + 24 - 60 = \boxed{4} \end{aligned}$$

104. (4) $25 + 5 \div 2 = 40$

$$\begin{aligned} \Rightarrow (25 \times 2) - (5 \times 2) &= 40 \\ \Rightarrow 50 - 10 &= 40 \\ 35 + 5 \div 2 &= 60 \\ \Rightarrow (35 \times 2) - (5 \times 2) &= 60 \\ \Rightarrow 70 - 10 &= 60 \\ 45 + 5 \div 2 &= 90 \\ \Rightarrow (45 \times 2) - (5 \times 2) &= 90 \\ \Rightarrow 90 - 10 &= 80 \end{aligned}$$

105. (1)

$+\Rightarrow \div$	$\div \Rightarrow -$
$-\Rightarrow \times$	$\times \Rightarrow +$

$$\begin{aligned} 48 + 16 \times 4 - 2 \div 8 &= ? \\ \Rightarrow 48 \div 16 + 4 \times 2 - 8 &= ? \\ \Rightarrow 3 + 8 - 8 &= \boxed{3} \end{aligned}$$

106. (2)

$+\Rightarrow -$	$-\Rightarrow \times$
$\div \Rightarrow +$	$\times \Rightarrow \div$

$$\begin{aligned} 15 - 3 + 10 \times 5 \div 5 &= ? \\ \Rightarrow 15 \times 3 - 10 \div 5 + 5 &= ? \\ \Rightarrow 45 - 2 + 5 &= \boxed{48} \end{aligned}$$

107. (4)

$\div \Rightarrow -$	$-\Rightarrow +$
$\times \Rightarrow \div$	$+\Rightarrow \times$

$$\begin{aligned} 35 \div 4 - 25 \times 5 + 5 &= ? \\ \Rightarrow 35 - 4 + 25 \div 5 \times 5 &= ? \\ \Rightarrow 35 - 4 + 5 \times 5 &= ? \\ \Rightarrow 35 - 4 + 25 &= ? \\ \Rightarrow 35 + 25 - 4 &= ? \\ \Rightarrow 60 - 4 &= 56 \end{aligned}$$

108. (3)

$+\Rightarrow \times$	$-\Rightarrow \div$
$\times \Rightarrow -$	$\div \Rightarrow +$

$$\begin{aligned} 16 \div 4 \times 10 - 5 + 8 &= ? \\ \Rightarrow ? &= 16 + 4 - 10 \div 5 \times 8 \\ \Rightarrow ? &= 16 + 4 - 2 \times 8 \\ \Rightarrow ? &= 16 + 4 - 16 = \boxed{4} \end{aligned}$$

109. (4)

$+\Rightarrow \div$	$-\Rightarrow \times$
$\div \Rightarrow +$	$\times \Rightarrow -$

$$\begin{aligned} 36 \times 12 + 4 \div 6 + 2 - 3 &= ? \\ \Rightarrow ? &= 36 - 12 \div 4 + 6 \div 2 \times 3 \\ \Rightarrow ? &= 36 - 3 + 3 \times 3 \\ \Rightarrow ? &= 36 + 9 - 3 = \boxed{42} \end{aligned}$$

110. (2)

$+\Rightarrow /$	$/ \Rightarrow -$
$-\Rightarrow \times$	$\times \Rightarrow +$

$$\begin{aligned} 24 + 8/2 - 6 \times 6 &= ? \\ \Rightarrow ? &= 24 \div 8 - 2 \times 6 + 6 \\ \Rightarrow ? &= 3 - 2 \times 6 + 6 \\ \Rightarrow ? &= 3 - 12 + 6 = -3 \end{aligned}$$

111. (2)

$+$	$\Rightarrow$	$\div$	$-$	$\Rightarrow$	$\times$
$\times$	$\Rightarrow$	$-$	$\div$	$\Rightarrow$	$+$

$$16 \div 8 - 4 + 2 \times 4 = ?$$

$$\Rightarrow ? = 16 + 8 \times 4 \div 2 - 4$$

$$\Rightarrow ? = 16 + 8 \times 2 - 4$$

$$\Rightarrow ? = 16 + 16 - 4 = \boxed{28}$$

112. (2)

$\times$	$\Rightarrow$	$+$	$+$	$\Rightarrow$	$\div$
$-$	$\Rightarrow$	$\times$	$\div$	$\Rightarrow$	$-$

$$8 \times 7 - 8 + 40 \div 2 = ?$$

$$\Rightarrow ? = 8 + 7 \times 8 \div 40 - 2$$

$$\Rightarrow ? = 8 + 7 \times \frac{8}{40} - 2$$

$$\Rightarrow 8 + \frac{7}{5} - 2$$

$$\Rightarrow ? = \frac{40 + 7 - 10}{5}$$

$$\Rightarrow ? = \frac{37}{5}$$

113. (2) $8 \times 20 \div 3 + 9 - 5 = 38$
 $\Rightarrow 8 \times 20 \div 5 + 9 - 3 = 38$
 $\Rightarrow 8 \times 4 + 9 - 3 = 38$
 $\Rightarrow 32 + 9 - 3 = 38$

114. (1)

$+$	$\Rightarrow$	$\times$	$-$	$\Rightarrow$	$+$
$\times$	$\Rightarrow$	$\div$	$\div$	$\Rightarrow$	$-$

Option (1)

$$9 + 8 - 4 \times 2 \div 18 = 56$$

$$\Rightarrow 9 \times 8 + 4 \div 2 - 18 = 56$$

$$\Rightarrow 72 + 2 - 18 = 56$$

$$\Rightarrow 74 - 18 = 56$$

Option (2)

$$9 \times 8 + 4 \div 2 - 18 = 26$$

$$\Rightarrow 9 \div 8 \times 4 - 2 + 18 = 26$$

$$\Rightarrow \frac{9}{8} \times 4 - 2 + 18 = 26$$

$$\Rightarrow \frac{9 - 4 + 36}{2} = 26$$

$$\Rightarrow 45 - 4 \neq 26 \times 2$$

Option (3)

$$9 \times 8 \div 4 + 2 - 18 = 200$$

$$\Rightarrow 9 \div 8 - 4 \times 2 + 18 = 200$$

$$\Rightarrow \frac{9}{8} - 8 + 18 = 200$$

$$\Rightarrow \frac{9 - 64 + 144}{8} = 200$$

$$\Rightarrow 153 - 64 \neq 200 \times 8$$

Option (3)

$$9 - 8 \times 4 + 2 \div 18 = 203$$

$$\Rightarrow 9 + 8 \div 4 \times 2 - 18 = 203$$

$$\Rightarrow 9 + 2 \times 2 - 18 \neq 203$$

115. (*)

$-$	$\Rightarrow$	$\div$	$+$	$\Rightarrow$	$\times$
$\div$	$\Rightarrow$	$-$	$\times$	$\Rightarrow$	$+$

Option (1)

$$43 \times 7 \div 5 + 4 - 8 = 25$$

$$\Rightarrow 43 + 7 - 5 \times 4 \div 8 = 25$$

$$\Rightarrow 43 + 7 - \frac{5}{2} \neq 25$$

Option (2)

$$48 \div 5 + 8 \times 10 - 2 = 03$$

$$\Rightarrow 48 - 5 \times 8 + 10 \div 2 = 03$$

$$\Rightarrow 48 - 40 + 5 = 03$$

$$\Rightarrow 53 - 40 \neq 03$$

Option (3)

$$36 \times 4 - 12 + 5 \div 3 = 420$$

$$\Rightarrow 36 + 4 \div 12 \times 5 - 3 = 420$$

$$\Rightarrow 36 + \frac{5}{3} - 3 \neq 420$$

Option (4)

$$42 + 5 \div 6 \times 8 - 3 = 28$$

$$\Rightarrow 42 \times 5 - 6 + 8 \div 3 = 28$$

$$\Rightarrow 210 - 6 + \frac{8}{3} \neq 28$$

116. (1)

$P \Rightarrow +$	$Q \Rightarrow \times$
$R \Rightarrow \div$	$S \Rightarrow -$

$$44 Q 9 R 12 S 6 Q 4 P 16 = ?$$

$$\Rightarrow ? = 44 \times 9 \div 12 - 6 \times 4 + 16$$

$$\Rightarrow ? = 44 \times \frac{9}{12} - 6 \times 4 + 16$$

$$\Rightarrow ? = 11 \times 3 - 6 \times 4 + 16$$

$$\Rightarrow ? = 33 - 24 + 16$$

$$\Rightarrow ? = 49 - 24 = \boxed{25}$$

117. (2)

$+$	$\Rightarrow$	$-$	$-$	$\Rightarrow$	$\times$
$\div$	$\Rightarrow$	$+$	$\times$	$\Rightarrow$	$\div$

$$15 - 3 + 10 \times 5 \div 5$$

$$\Rightarrow 15 \times 3 - 10 \div 5 + 5$$

$$\Rightarrow 45 - 2 + 5 = \boxed{48}$$

118. (1)

$+$	$\Rightarrow$	$\times$	$-$	$\Rightarrow$	$+$
$\times$	$\Rightarrow$	$\div$	$\div$	$\Rightarrow$	$-$

$$12 \times 2 + 6 - 7 \div 5 = ?$$

$$\Rightarrow ? = 12 \div 2 \times 6 + 7 - 5$$

$$\Rightarrow ? = 6 \times 6 + 7 - 5$$

$$\Rightarrow ? = 36 + 7 - 5 = \boxed{38}$$

TYPE-I (ii)

1. (2)

$\rightarrow$	$\Rightarrow$	$+$	$\leftarrow$	$\Rightarrow$	$-$
$\uparrow$	$\Rightarrow$	$\div$	$\downarrow$	$\Rightarrow$	$\times$
$\nearrow$	$\Rightarrow$	$=$			

Option (1)

$$2 \downarrow 5 \leftarrow 6 \rightarrow 2 \nearrow 6$$

After putting the value of signs

$$2 \times 5 - 6 + 2 = 6$$

$$\text{or, } 10 - 6 + 2 = 6$$

TYPE-I (iii)

1. (2)

$a \rightarrow \div$	$b \rightarrow +$
$c \rightarrow -$	$d \rightarrow \times$

Given expression

$$24 a 6 d 4 b 9 c 8 = ?$$

After conversion

$$? = 24 \div 6 \times 4 + 9 - 8$$

$$\text{or, } ? = 4 \times 4 + 9 - 8$$

$$\text{or, } ? = 16 + 9 - 8$$

$$\text{or, } ? = 25 - 8 = 17$$

2. (4)

$A \Rightarrow +$	$B \Rightarrow -$	$C \Rightarrow \div$
$D \Rightarrow \times$	$E \Rightarrow <$	$F \Rightarrow >$
$G \Rightarrow =$		

Option (4)

$$9 A 7 B 4 C 2 G 14$$

$$\Rightarrow 9 + 7 - 4 \div 2 = 14$$

$$\Rightarrow 9 + 7 - 2 = 14$$

3. (3)

$P \Rightarrow +$	$Q \Rightarrow -$
$R \Rightarrow \div$	$S \Rightarrow \times$

$$18 S 36 R 12 Q 6 P 7 = ?$$

$$\text{or, } ? = 18 \times 36 \div 12 - 6 + 7$$

$$\text{or, } ? = 18 \times 3 - 6 + 7$$

$$\text{or, } ? = 54 - 6 + 7$$

$$\text{or, } ? = 61 - 6 = 55$$

4. (4)

$A \Rightarrow =$	$B \Rightarrow <$	$C \Rightarrow >$
$D \Rightarrow \neq$	$E \Rightarrow \neq$	$F \Rightarrow \neq$

Premises

$$3 X B \neq 2Y \text{ and } 2Y D Z$$

$$\text{or, } 3 X < 2Y \text{ and } 2Y \neq Z$$

It is clear that 2 Y is either equal to or less than Z.

Thus, $3 X < Z$

Option (1)

$$3 X A Z = 3 X = Z : \text{Not true}$$

Option (2)

$$3 X D Z = 3 X \neq Z$$

The equation implies that $3X$ is either equal to or less than Z . Therefore, the given equation is not true.

Option (3)

$$3X \times FZ = 3X \times Z$$

It is also not true.

Option (4)

$$3X \times BZ = 3X < Z$$

Clearly $3X$ is less than Z .

5. (2) Premises

$$7X < 3Y \text{ and}$$

$$6Y \geq 2Z$$

$$\text{or, } 6Y \leq 2Z$$

Therefore,

$$7X < 2Z \Rightarrow 7X \times B \times 2Z$$

6. (2)

$J \Rightarrow +$	$K \Rightarrow -$
$L \Rightarrow \div$	$M \Rightarrow \times$

Given expression

$$18M \times 36L \times 12K \times 6J \times 7 = ?$$

After conversion

$$? = 18 \times 36 \div 12 - 6 + 7$$

$$\text{or, } ? = 18 \times 3 - 6 + 7$$

$$\text{or, } ? = 54 - 6 + 7 = \boxed{55}$$

7. (4) Option (1)

$$15C \times 15B \times 8F \times 4B \times 6C \times 3$$

$$\Rightarrow 15 \div 15 + 8 < 4 + 6 \div 3$$

$$\Rightarrow 1 + 8 < 4 + 2$$

Option (2)

$$15B \times 5G \times 8B \times 4G \times 6F \times 3$$

$$\Rightarrow 15 + 5 - 8 + 4 - 6 < 3$$

$$\Rightarrow 24 - 14 < 3$$

Option (3)

$$15A \times 5E \times 8C \times 4B \times 6E \times 3$$

$$\Rightarrow 15 > 5 \times 8 \div 4 + 6 \times 3$$

$$\Rightarrow 15 \geq 10 + 18$$

Option (4)

$$15C \times 5F \times 8C \times 4B \times 6C \times 3$$

$$\Rightarrow 15 \div 5 < 8 \div 4 + 6 \div 3$$

$$\Rightarrow 3 < 2 + 2$$

8. (3)

$A \rightarrow +$	$B \rightarrow -$
$C \rightarrow \times$	$D \rightarrow \div$

Option (1)

$$8B \times 6D \times 2A \times 4C \times 3 = 15$$

$$\text{or, } 8 - 6 \div 2 + 4 \times 3 = 15$$

$$\text{or, } 8 - 3 + 4 \times 3 = 15$$

$$\text{or, } 8 - 3 + 12 = 15$$

$$\text{or, } 20 - 3 = 15$$

$$\text{or, } 17 \neq 15$$

Option (2)

$$9C \times 9B \times 9D \times 9A \times 9 = 17$$

$$\text{or, } 9 \times 9 - 9 \div 9 + 9 = 17$$

$$\text{or, } 9 \times 9 - 1 + 9 = 17$$

$$\text{or, } 81 - 1 + 9 = 17$$

$$\text{or, } 90 - 1 \neq 17$$

Option (3)

$$8A \times 8B \times 8C \times 8 = -48$$

$$\text{or, } 8 + 8 - 8 \times 8 = -48$$

$$\text{or, } 8 + 8 - 64 = -48$$

$$\text{or, } 16 - 64 = -48$$

9. (2)

$A \rightarrow \times$	$D \rightarrow +$
$G \rightarrow -$	

Given expression

$$7A \times 4D \times 4A \times 3G \times 2$$

After conversion

$$? = 7 \times 4 + 4 \times 3 - 2$$

$$\text{or, } ? = 28 + 12 - 2$$

$$\text{or, } ? = 40 - 2 = 38$$

10. (2)

$L \Rightarrow M$	$M \Rightarrow -$
$N \Rightarrow \times$	$P \Rightarrow \div$

Given expression

$$5N \times 5P \times 5L \times 5M \times 5 = ?$$

After changing the signs

$$? = 5 \times 5 \div 5 + 5 - 5$$

$$\text{or, } ? = 5 + 5 - 5 = 5$$

11. (1)

$L \Rightarrow +$	$M \Rightarrow -$
$N \Rightarrow \times$	$P \Rightarrow \div$

Given expression

$$14N \times 10L \times 42P \times 2M \times 8 = ?$$

After changing the signs

$$? = 14 \times 10 + 42 \div 2 - 8$$

$$\text{or, } ? = 14 \times 10 + 21 - 8$$

$$\text{or, } ? = 140 + 21 - 8 = \boxed{153}$$

12. (2) Option (1)

$$18F \times 3B \times 6E \times 8G \times 4E \times 12$$

$$\text{or, } 18 < 3 + 6 \times 8 - 4 \times 12$$

$$\text{or, } 18 < 3 + 48 - 48$$

$$\text{or, } 18 < 3$$

Option (2)

$$18C \times 3G \times 6B \times 8D \times 12$$

$$\text{or, } 18 \div 3 - 6 + 8 + 4 = 12$$

$$\text{or, } 6 - 6 + 12 = 12$$

$$\text{or, } 12 = 12$$

Option (3)

$$18A \times 3E \times 6B \times 8G \times 4B \times 12$$

$$\text{or, } 18 > 3 \times 6 + 8 - 4 + 12$$

$$\text{or, } 18 > 18 + 8 - 4 + 12$$

$$\text{or, } 18 > 34$$

Option (4)

$$18C \times 3D \times 6B \times 8C \times 4G \times 12$$

$$\text{or, } 18 \div 3 = 6 + 8 \div 4 - 12$$

$$\text{or, } 6 = 6 + 2 - 12$$

$$\text{or, } 6 \neq 4$$

13. (2)

$P \Rightarrow -$	$Q \Rightarrow \times$
$R \Rightarrow \div$	$S \Rightarrow +$

$$14 \times 3 - 12 + 4 \div 2$$

$$\Rightarrow 42 - 12 + 2 \Rightarrow 44 - 12 = \boxed{32}$$

14. (3)

$L \Rightarrow \times$	$M \Rightarrow \div$
$P \Rightarrow +$	$Q \Rightarrow -$

$$16P \times 24M \times 8Q \times 6M \times 2L \times 3 = ?$$

$$\Rightarrow ? = 16 + 24 \div 8 - 6 \div 2 \times 3$$

$$\Rightarrow ? = 16 + 3 - 3 \times 3$$

$$\Rightarrow ? = 16 + 3 - 9 = \boxed{10}$$

15. (3)

$A \Rightarrow +$	$M \Rightarrow \times$	$L \Rightarrow <$
$D \Rightarrow \div$	$G \Rightarrow >$	

$$20A \times 4D \times 4L \times 4A \times 6D \times 2$$

$$\Rightarrow 20 + 4 \div 4 < 4 + 6 \div 2$$

$$\Rightarrow 20 + 1 < 4 + 3$$

$$20D \times 5G \times 8D \times 4A \times 6M \times 3$$

$$\Rightarrow 20 \div 5 > 8 \div 4 + 6 \times 3$$

$$\Rightarrow 4 \not> 2 + 18$$

$$20D \times 4A \times 4L \times 4A \times 2M \times 3$$

$$\Rightarrow 20 \div 4 + 4 < 4 + 2 \times 3$$

$$\Rightarrow 5 + 4 < 4 + 6$$

$$20A \times 2G \times 10M \times 3A \times 12D \times 2$$

$$\Rightarrow 20 + 2 > 10 \times 3 + 12 \div 2$$

$$\Rightarrow 22 \not> 30 + 6$$

16. (3)

$X \Rightarrow +$	$Z \Rightarrow \div$
$Y \Rightarrow -$	$P \Rightarrow \times$

$$10P \times 2X \times 5Y \times 5 = ?$$

$$\Rightarrow ? = 10 \times 2 + 5 - 5$$

$$\Rightarrow ? = 20 + 5 - 5 = \boxed{20}$$

17. (1)

$P \Rightarrow +$	$Q \Rightarrow \times$
$R \Rightarrow \div$	$S \Rightarrow -$

$$44Q \times 9R \times 12S \times 6Q \times 4P \times 16 = ?$$

$$\Rightarrow ? = 44 \times 9 - 12 - 6 \times 4 + 16$$

$$\Rightarrow ? = 44 \times \frac{3}{4} - 24 + 16$$

$$\Rightarrow ? = 33 - 24 + 16 = \boxed{25}$$

18. (1)

$R \Rightarrow -$	$A \Rightarrow +$
$B \Rightarrow \div$	$C \Rightarrow \times$

$$25A \times 37C \times 2B \times 4R \times 1 = ?$$

$$\Rightarrow ? = 25 + 37 \times 2 \div 4 - 1$$

| SYMBOLS & NOTATIONS |

$$? = 62 \times 2 \div 4 + 1$$

$$? = 124 \div 4 + 1$$

$$? = 31 + 1 = \boxed{32}$$

19. (3)

$P \Rightarrow \div$	$Q \Rightarrow \times$
$R \Rightarrow +$	$S \Rightarrow -$

$$12 \text{ Q } 15 \text{ P } 3 \text{ R } 4 \text{ S } 6 = ?$$

$$\Rightarrow 12 \times 15 \div 3 + 4 - 6$$

$$\Rightarrow 12 \times 5 + 4 - 6$$

$$\Rightarrow 60 + 4 - 6 = \boxed{58}$$

20. (*)

$+ \Rightarrow >$	$\times \Rightarrow =$	$- \Rightarrow \geq$
$L \Rightarrow \neq$	$I \Rightarrow <$	$\phi \Rightarrow \leq$

$$A - B \phi C$$

$$\Rightarrow A \geq B \leq C$$

$$\Rightarrow A > B < C \text{ or, } A > B = C$$

$$\text{or, } A = B < C \text{ or, } A = B < C$$

Option (1)

$$A | B + C$$

$$\Rightarrow A < B > C : \text{Not True}$$

Option (2)

$$A | B | C$$

$$\Rightarrow A < B < C : \text{Not True}$$

Option (3)

$$A + B - C$$

$$\Rightarrow A > B \geq C$$

$$\Rightarrow A > B = C$$

$$\text{or, } A > B > C$$

Option (4)

$$A \phi B | C$$

$$\Rightarrow A \leq B < C$$

$$\Rightarrow A < B < C$$

$$\text{or, } A = B < C$$

Both the options (3) and (4) may be true.

21. (*)

$A \Rightarrow +$	$Q \Rightarrow -$
$V \Rightarrow \times$	$R \Rightarrow \div$

$$225 \text{ R } 5 \text{ A } 64 \text{ Q } 13 \text{ V } 6 = ?$$

$$\Rightarrow ? = 225 \div 5 + 64 - 13 \times 6$$

$$\Rightarrow ? = 45 + 64 - 78 = \boxed{31}$$

22. (3)

$P \Rightarrow \times$	$T \Rightarrow -$
$M \Rightarrow +$	$B \Rightarrow \div$

$$12 \text{ P } 6 \text{ M } 15 \text{ T } 16 \text{ B } 4 = ?$$

$$\Rightarrow ? = 12 \times 6 + 15 - 16 \div 4$$

$$\Rightarrow ? = 72 + 15 - 4 = \boxed{83}$$

23. (2)

$+ \Rightarrow >$	$\phi \Rightarrow \leq$	$- \Rightarrow \geq$
$\times \Rightarrow =$	$I \Rightarrow <$	$L \Rightarrow \neq$

$$A | B \times C$$

$$\Rightarrow A < B = C$$

Option (1)

$$B + C | A$$

$$\Rightarrow B > C < A$$

Option (2)

$$C - B + A$$

$$\Rightarrow C \geq B > A$$

Option (3)

$$B | A | C$$

$$\Rightarrow B < A < C$$

Option (4)

$$A \phi B | C$$

$$\Rightarrow A \leq B < C$$

24. (3)

$A \Rightarrow \leq$	$B \Rightarrow =$	$C \Rightarrow <$
$D \Rightarrow \geq$	$E \Rightarrow \neq$	$F \Rightarrow >$

$$2 \text{ M } B \text{ N}$$

$$\Rightarrow 2 \text{ M} = \text{N} \Rightarrow \text{M} = \frac{\text{N}}{2}$$

$$2 \text{ N } A \text{ 3K}$$

$$\Rightarrow 2 \text{ N} \leq 3\text{K} \Rightarrow 4\text{M} \leq 3\text{K}$$

Option (1)

$$2 \text{ M } D \text{ 3K}$$

$$\Rightarrow 2 \text{ M} \geq 3\text{K} : \text{Not True}$$

Option (2)

$$2 \text{ M } B \text{ 3 K}$$

$$\Rightarrow 2 \text{ M} = 3 \text{ K} : \text{Not True}$$

Option (3)

$$2 \text{ M } C \text{ 3 K}$$

$$\Rightarrow 2 \text{ M} < 3 \text{ K} : \text{True}$$

Option (4)

$$2 \text{ K } B \text{ 3 N}$$

$$\Rightarrow 2 \text{ K} = 3 \text{ N} : \text{Not True}$$

25. (*)

$B \Rightarrow +$	$G \Rightarrow -$	$E \Rightarrow \times$	$C \Rightarrow \div$
$D \Rightarrow =$	$A \Rightarrow >$	$F \Rightarrow <$	

Option (1)

$$15 \text{ C } 3 \text{ B } 2 \text{ A } 6 \text{ E } 2$$

$$\Rightarrow 15 \div 3 + 2 > 6 \times 2$$

$$\Rightarrow 5 + 2 \ngtr 12$$

Option (2)

$$15 \text{ B } 2 \text{ G } 5 \text{ A } 4 \text{ G } 4$$

$$\Rightarrow 15 + 2 - 5 > 4 - 4$$

$$\Rightarrow 12 > 0$$

Option (3)

$$15 \text{ C } 3 \text{ B } 2 \text{ D } 6 \text{ B } 1$$

$$\Rightarrow 15 \div 3 + 2 = 6 + 1$$

$$\Rightarrow 5 + 2 = 7$$

Option (4)

$$15 \text{ B } 3 \text{ D } 4 \text{ E } 6$$

$$\Rightarrow 15 + 3 = 4 \times 6$$

$$\Rightarrow 18 \neq 24$$

Both options (2) and (3) are correct.

26. (2)

$A \Rightarrow +$	$B \Rightarrow -$	$C \Rightarrow \times$
-------------------	-------------------	------------------------

$$(10 \text{ C } 4) \text{ A } (4 \text{ C } 4) \text{ B } 6 = ?$$

$$\Rightarrow ? = (10 \times 4) + (4 \times 4) - 6$$

$$\Rightarrow ? = 40 + 16 - 6 = \boxed{50}$$

27. (3)

$P \Rightarrow +$	$Q \Rightarrow -$
$R \Rightarrow \div$	$S \Rightarrow \times$

$$18 \text{ S } 36 \text{ R } 12 \text{ Q } 6 \text{ P } 7 = ?$$

$$\Rightarrow ? = 18 \times 36 \div 12 - 6 + 7$$

$$\Rightarrow ? = 18 \times 3 - 6 + 7$$

$$\Rightarrow ? = 54 - 6 + 7$$

$$\Rightarrow ? = 61 - 6 = \boxed{55}$$

28. (2)

$P \Rightarrow \div$	$Q \Rightarrow \times$
$R \Rightarrow +$	$S \Rightarrow -$

$$18 \text{ Q } 12 \text{ P } 4 \text{ R } 5 \text{ S } 6$$

$$\Rightarrow 18 \times 12 \div 4 + 5 - 6$$

$$\Rightarrow 18 \times 3 + 5 - 6$$

$$\Rightarrow 54 + 5 - 6 = 53$$

29. (2)

$A \Rightarrow +$	$B \Rightarrow \times$
$C \Rightarrow \div$	$D \Rightarrow -$

$$9 \text{ A } 2 \text{ B } 6 \text{ D } 4 \text{ C } 2 = ?$$

$$\Rightarrow ? = 9 + 2 \times 6 - 4 \div 2$$

$$\Rightarrow ? = 9 + 12 - 2 = 19$$

30. (3)

$D \Rightarrow \times$	$S \Rightarrow +$
$A \Rightarrow -$	$M \Rightarrow \div$

$$28 \text{ D } 6 \text{ S } 34 \text{ M } 2 \text{ A } 8 \text{ D } 6$$

$$\Rightarrow 28 \times 6 + 34 \div 2 - 8 \times 6$$

$$\Rightarrow 168 + 17 - 48$$

$$\Rightarrow 185 - 48 = 137$$

31. (2)

$A \Rightarrow +$	$B \Rightarrow -$
$C \Rightarrow \times$	

$$(10 \text{ C } 4) \text{ A } (4 \text{ C } 4) \text{ B } 6 = ?$$

$$\Rightarrow ? = (10 \times 4) + (4 \times 4) - 6$$

$$\Rightarrow ? = 40 + 16 - 6 = 50$$

32. (2)

$a \Rightarrow \times$	$b \Rightarrow \div$
$c \Rightarrow +$	$d \Rightarrow -$

$$8 \text{ a } 3 \text{ c } 24 \text{ b } 12 \text{ d } 19 = ?$$

$$\Rightarrow ? = 8 \times 3 + 24 \div 12 - 19$$

$$\Rightarrow ? = 24 + 2 - 19 = 7$$

33. (4)

$A \Rightarrow -$	$C \Rightarrow \times$
$D \Rightarrow \div$	$E \Rightarrow +$

$$14 \text{ C } 3 \text{ A } 12 \text{ E } 4 \text{ D } 2 = ?$$

$$\Rightarrow ? = 14 \times 3 - 12 + 4 \div 2$$

$$\Rightarrow ? = 42 - 12 + 2$$

$$\Rightarrow ? = 44 - 12 = \boxed{32}$$

34. (2)

$A \Rightarrow +$	$B \Rightarrow -$
$C \Rightarrow \times$	$D \Rightarrow \div$

Option (1)

$$8B6D2A4C3 = 15$$

$$\Rightarrow 8 - 6 \div 2 + 4 \times 3 = 15$$

$$\Rightarrow 8 - 3 + 4 \times 3 = 15$$

$$\Rightarrow 8 - 3 + 12 = 15$$

$$\Rightarrow 20 - 3 \neq 15$$

Option (2)

$$8A8B8C8 = -48$$

$$\Rightarrow 8 + 8 - 8 \times 8 = -48$$

$$\Rightarrow 16 - 64 = -48$$

Option (3)

$$9C9B9D9A9 = 17$$

$$\Rightarrow 9 \times 9 - 9 \div 9 + 9 = 17$$

$$\Rightarrow 81 - 1 + 9 = 17$$

$$\Rightarrow 90 - 1 \neq 17$$

Option (4)

$$3A3B3C3A3D3 = 4$$

$$\Rightarrow 3 + 3 - 3 \times 3 + 3 \div 3 = 4$$

$$\Rightarrow 3 + 3 - 9 + 1 = 4$$

$$\Rightarrow 7 - 9 \neq 4$$

35. (2)

$P \Rightarrow \times$	$T \Rightarrow -$
$M \Rightarrow +$	$B \Rightarrow \div$

$$28B7P8T6M4 = ?$$

$$\Rightarrow ? = 28 \div 7 \times 8 - 6 + 4$$

$$\Rightarrow ? = 4 \times 8 - 6 + 4$$

$$\Rightarrow ? = 32 - 6 + 4 = 30$$

36. (4)

$L \Rightarrow \times$	$M \Rightarrow \div$
$P \Rightarrow +$	$Q \Rightarrow -$

$$16P24M8Q6M2L3$$

$$\Rightarrow 16 + 24 \div 8 - 6 \div 2 \times 3$$

$$\Rightarrow 16 + 3 - 9 = 10$$

37. (1)

$L \Rightarrow +$	$M \Rightarrow -$
$N \Rightarrow \times$	$P \Rightarrow \div$

$$14N10L42P2M8 = ? \Rightarrow$$

$$? = 14 \times 10 + 42 \div 2 - 8$$

$$\Rightarrow ? = 140 + 21 - 8$$

$$\Rightarrow ? = 161 - 8 = 153$$

38. (1)

$P \Rightarrow \div$	$Q \Rightarrow \times$
$R \Rightarrow +$	$S \Rightarrow -$

$$18Q12P4R5S6 = ?$$

$$\Rightarrow ? = 18 \times 12 \div 4 + 5 - 6$$

$$\Rightarrow ? = 18 \times 3 + 5 - 6$$

$$\Rightarrow ? = 54 + 5 - 6 = 53$$

39. (1)

$K \Rightarrow -$	$L \Rightarrow \div$
$M \Rightarrow +$	$D \Rightarrow \times$

$$96L4K6M11D9 = ?$$

$$\Rightarrow ? = 96 \div 4 - 6 + 11 \times 9$$

$$\Rightarrow ? = 24 - 6 + 99$$

$$\Rightarrow ? = 123 - 6 = 117$$

TYPE-I (iv)

1. (2)

$X \Rightarrow +$	$V \Rightarrow -$	$U \Rightarrow =$	$\sigma \Rightarrow <$
$\wedge \Rightarrow \div$	$\Sigma \Rightarrow \times$	$\varnothing \Rightarrow >$	

Option (1)

$$3X8V2U12 \wedge 3$$

$$\Rightarrow 13 + 8 - 2 = 12 \div 3$$

$$\Rightarrow 9 \neq 4$$

Option (2)

$$13V12X9V2\varnothing 5\Sigma 1$$

$$\Rightarrow 13 - 12 + 9 - 2 > 5 \times 1$$

$$\Rightarrow 8 > 5$$

Option (3)

$$2\Sigma 3\Sigma 4\sigma 51 \wedge 3$$

$$\Rightarrow 2 \times 3 \times 4 < 51 \div 3$$

$$\Rightarrow 24 < 17$$

Option (4)

$$3\Sigma 2\Sigma 4U2X7V3$$

$$\Rightarrow 3 \times 2 \times 4 = 2 + 7 - 3$$

$$\Rightarrow 24 \neq 6$$

2. (4)

$+ \Rightarrow \delta$	$- \Rightarrow \bullet$	$\times \Rightarrow \gamma$	$\div \Rightarrow \eta$
$= \Rightarrow \omega$	$> \Rightarrow \beta$	$< \Rightarrow \alpha$	

Option (1)

$$3\gamma 6\eta 2\delta 84\omega 5$$

$$\Rightarrow 3 \times 6 \div 2 + 8 - 4 \times 5$$

$$\Rightarrow 3 \times 3 + 8 - 4 = 5$$

$$\Rightarrow 17 - 4 \neq 5$$

Option (2)

$$3\eta 6\gamma 2\delta 84\beta 5$$

$$\Rightarrow 3 \div 6 \times 2 + 8 - 4 > 5$$

$$\Rightarrow \frac{3}{6} \times 2 + 8 - 4 > 5$$

$$\Rightarrow 1 + 8 - 4 \neq 5$$

Option (3)

$$3\gamma 62\delta 8\eta 4\alpha 5$$

$$\Rightarrow 3 \times 6 - 2 + 8 \div 4 < 5$$

$$\Rightarrow 3 \times 6 - 2 + 2 < 5$$

$$\Rightarrow 18 - 2 + 2 \neq 5$$

Option (4)

$$3\delta 62\gamma 8\eta 4\omega 5$$

$$\Rightarrow 3 + 6 - 2 \times 8 \div 4 = 5$$

$$\Rightarrow 3 + 6 - 2 \times 2 = 5$$

$$\Rightarrow 9 - 4 = 5$$

3. (1) M A S T E R

$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
#	∇	H	\$	%	@

4. (4) & is the brother of $\neq$.
 $\neq$ is the daughter of @.
 # is the brother of @.
 Therefore, # is the uncle of &.

5. (*) $-\Rightarrow$ not less than

It means $\geq$

$$X - Y - Z$$

$$\Rightarrow X \geq Y \geq Z$$

$$\Rightarrow X > Y > Z$$

$$\text{or, } X > Y = Z$$

$$\text{or, } X = Y > Z$$

$$\text{or, } X = Y = Z$$

Option (1)

$$X \otimes Y \Delta Z$$

$$\Rightarrow X > Y \text{ or, } X < Y; Y = Z$$

Option (2)

$$X \circ Y + Z$$

$$\Rightarrow X \leq Y > Z$$

Option (3)

$$X \oslash Y - Z$$

$$\Rightarrow X < Y \geq Z$$

Option (4)

$$X \oslash Y + Z$$

$$\Rightarrow X < Y > Z$$

6. (4) $\{(13\%5)\$6\}\#15$

$$\Rightarrow \{(5^2 - 13^2) \div 6^2\} \times 2 \times 15$$

$$\Rightarrow \{(25 - 169) \div 36\} \times 30$$

$$\Rightarrow \{-144 \div 36\} \times 30$$

$$\Rightarrow -4 \times 30 = -120$$

7. (1)

$@ \Rightarrow +$	$\oplus \Rightarrow -$
$\alpha \Rightarrow \div$	$\ominus \Rightarrow \times$

$$107 \ominus 3 \oplus 64 \alpha 8 \oplus 2 \ominus 9 = ?$$

$$\Rightarrow ? = 107 \times 3 - 64 \div 8 - 2 \times 9$$

$$\Rightarrow ? = 321 - 8 - 18$$

$$\Rightarrow ? = 321 - 26 = 295$$

8. (*) $324 \oplus 289$

$$\Rightarrow \sqrt{324} + \sqrt{289}$$

$$\Rightarrow 18 + 17 = 35$$

$$441 \oplus 484$$

$$\Rightarrow \sqrt{441} + \sqrt{484}$$

$$\Rightarrow 21 + 22 = 43$$

Therefore, $625 \oplus 400$

$$\Rightarrow \sqrt{625} + \sqrt{400}$$

$$\Rightarrow 25 + 20 = 45$$

9. (4)

$\# \Rightarrow -$	$\& \Rightarrow \div$
$@ \Rightarrow +$	$\% \Rightarrow \times$

$$217 \& 7 \# 3 @ 2 \% 7 = ?$$

$$\Rightarrow ? = 217 \div 7 - 3 + 2 \times 7$$

$$\Rightarrow ? = 31 - 3 + 14$$

$$\Rightarrow ? = 45 - 3 = 42$$

TYPE-I (v)

1. (1) $(\square \times \triangle) \div \diamond = ?$
 $\Rightarrow ? = (8 \times 3) \div 4$
 $\Rightarrow ? = 24 \div 4 = 6 \Rightarrow \diamond 017$
2. (2) $(\triangle + \square) \div \bigcirc = ?$
 $= (15 + 12) \div 3 = 27 \div 3 = 9$
3. (1) $(12 \times 4) \div 6 = ?$
 or, $? = \frac{48}{6} = 8$

4. (3) $\triangle + \bigcirc - \square = ?$
 $? = 15 + 3 - 12$
 or, $? = 18 - 12 = 6 = \square$

5. (1) $\square \div \triangle = ?$
 or, $12 \div 4 = 3 \Rightarrow \bigcirc$

6. (3) $\frac{\text{Rectangle} + \text{Square}}{\text{Triangle}}$

$$\Rightarrow \frac{12+6}{15} = \frac{18}{15} = \frac{6}{5}$$

7. (3) $\begin{matrix} [&] & \cap & \cup & \downarrow \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 4 & 5 & 9 & 0 & 6 \end{matrix}$

8. (2) $24 \triangle 4 \triangle 5 \triangle 4$
 $\Rightarrow 24 = 4 \times 5 + 4$

9. (4) $\begin{matrix} \square & \bigcirc & W & \triangle & | \\ \square & \bigcirc & W & \triangle & | \\ \square & \bigcirc & W & & \end{matrix}$

10. (4)

# $\Rightarrow$ <	$\bigcirc \Rightarrow$ >	$\square \Rightarrow$ =
-------------------	--------------------------	-------------------------

$$a \bigcirc b \# c \square d$$

$$\Rightarrow a > b < c = d$$

Option (1)

$$a \square c \Rightarrow a = c : \text{Not True}$$

Option (2)

$$b \bigcirc d \Rightarrow b > d : \text{Not True}$$

Option (3)

$$b \square d \Rightarrow b = d : \text{Not True}$$

Option (4)

$$b \# d \Rightarrow b < d : \text{True}$$

11. (1)

* $\Rightarrow$ -	- $\Rightarrow$ $\div$
$\square \Rightarrow$ +	% $\Rightarrow$ $\times$

$$13 \square 3 * 6 \% 8 - 4 \square 14 = ?$$

$$\Rightarrow ? = 13 + 3 - 6 \times 8 \div 4 + 14$$

$$\Rightarrow ? = 13 + 3 - 6 \times 2 + 14$$

$$\Rightarrow ? = 13 + 3 - 12 + 14$$

$$\Rightarrow ? = 30 - 12 = \boxed{18}$$

TYPE-I (vi)

1. (4) Option (1)

$$6 > 3 < 2 \wedge 4 \vee 8 - 13$$

$$\text{or, } 6 \div 3 + 2 - 4 \times 8 > 13$$

$$\text{or, } 2 + 2 - 32 > 13$$

$$\text{or, } -28 \nless 13$$

- Option (2)

$$6 \wedge 3 < 2 > 4 \vee 8 + 13$$

$$\text{or, } 6 - 3 + 2 \div 4 \times 8 = 13$$

$$\text{or, } 6 - 3 + \frac{2}{4} \times 8 = 13$$

$$\text{or, } 6 - 3 + 4 \neq 13$$

- Option (3)

$$6 \vee 3 \wedge 2 > 4 < 8 \times 13$$

$$\text{or, } 6 \times 3 - 2 \div 4 + 8 < 13$$

$$\text{or, } 6 \times 3 - \frac{2}{4} + 8 < 13$$

$$\text{or, } 18 - \frac{2}{4} + 8 < 13$$

$$\text{or, } \frac{72 - 2 + 32}{4} \nless 13$$

- Option (4)

$$6 \vee 3 > 2 < 4 \wedge 8 \times 13$$

$$\text{or, } 6 \times 3 \div 2 + 4 - 8 < 13$$

$$\text{or, } \frac{6 \times 3}{2} + 4 - 8 < 13$$

$$\text{or, } 9 + 4 - 8 < 13$$

2. (4) Option (1)

$$6 > 2 > 3 \wedge 8 \vee 4 + 13$$

$$\Rightarrow 6 + 2 \div 3 - 8 \times 4 = 13$$

$$\Rightarrow 6 + \frac{2}{3} - 32 \neq 13$$

- Option (2)

$$6 \wedge 2 < 3 > 8 < 4 - 13$$

$$\Rightarrow 6 - 2 + 3 \div 8 + 4 > 13$$

$$\Rightarrow 6 - 2 + \frac{3}{8} + 4 > 13$$

$$\Rightarrow \frac{48 - 16 + 3 + 32}{8} > 13$$

$$\Rightarrow \frac{67}{8} \nless 13$$

- Option (3)

$$6 \vee 2 < 3 \wedge 8 > 4 \times 13$$

$$\Rightarrow 6 \times 2 + 3 - 8 \div 4 < 13$$

$$\Rightarrow 12 + 3 - 2 < 13$$

$$\Rightarrow 13 \nless 13$$

- Option (4)

$$6 > 2 \vee 3 < 8 \wedge 4 + 13$$

$$\Rightarrow 6 \div 2 \times 3 + 8 - 4 = 13$$

$$\Rightarrow 3 \times 3 + 8 - 4 = 13$$

$$\Rightarrow 9 + 8 - 4 = 13$$

3. (3)

$\div \Rightarrow$ >	$\times \Rightarrow$ +
$+$ $\Rightarrow$ $\div$	$- \Rightarrow$ =
$> \Rightarrow$ $\times$	$= \Rightarrow$ <
$< \Rightarrow$ -	

- Option (1)

$$5 > 2 < 1 - 3 \times 4 \times 1$$

$$\text{or, } 5 \times 2 - 1 = 3 + 4 + 1$$

$$\text{or, } 9 \neq 8$$

- Option (2)

$$5 < 2 \times 1 \div 3 > 4 \times 1$$

$$\text{or, } 5 - 2 + 1 > 3 \times 4 + 1$$

$$\text{or, } 4 \nless 13$$

- Option (3)

$$5 > 2 \times 1 - 3 > 4 < 1$$

$$\text{or, } 5 \times 2 + 1 = 3 \times 4 - 1$$

$$\text{or, } 11 = 11$$

- Option (4)

$$5 + 2 \times 1 = 3 + 4 > 1$$

$$\text{or, } 5 \div 2 + 1 < 3 \div 4 \times 1$$

$$\text{or, } \frac{5}{2} + 1 < \frac{3}{4}$$

$$\text{or, } \frac{7}{2} \nless \frac{3}{4}$$

4. (*)

$\times \Rightarrow$ +	$< \Rightarrow$ -	$+$ $\Rightarrow$ $\div$	$> \Rightarrow$ $\times$
$- \Rightarrow$ =	$\div \Rightarrow$ >	$= \Rightarrow$ <	

- Option (1)

$$5 > 8 + 4 = 10 < 4 \times 8$$

$$\Rightarrow 5 \times 8 \div 4 < 10 - 4 + 8$$

$$\Rightarrow 5 \times 2 < 18 - 4 \Rightarrow 10 < 14$$

- Option (2)

$$3 \times 4 > 2 - 9 + 3 < 3$$

$$\Rightarrow 3 + 4 \times 2 = 9 \div 3 - 3$$

$$\Rightarrow 3 + 8 \neq 3 - 3$$

- Option (3)

$$5 \times 3 < 3 \div 8 + 4 \times 1$$

$$\Rightarrow 5 + 3 - 3 > 8 \div 4 + 1$$

$$\Rightarrow 8 - 3 > 2 + 1$$

$$\Rightarrow 5 > 3$$

- Option (4)

$$3 \times 2 < 4 \div 16 > 2 \times 4$$

$$\Rightarrow 3 + 2 - 4 > 16 \times 2 \div 4$$

$$\Rightarrow 5 - 4 > \frac{16 \times 2}{4} \Rightarrow 1 \nless 8$$

Both options (1) and (3) are correct.

5. (3)

$+$ $\Rightarrow$ $\times$	$< \Rightarrow$ $\div$	$\div \Rightarrow$ -
$- \Rightarrow$ +	$\times \Rightarrow$ >	

- Option (1)

$$20 - 4 \div 4 + 8 < 2 \times 26$$

$$\Rightarrow 20 + 4 - 4 \times 8 \div 2 > 26$$

$$\Rightarrow 20 + 4 - 4 \times 4 > 26$$

$$\Rightarrow 24 - 16 \nless 26$$

Option (2)

$$20 \times 8 + 15 < 5 \div 9 - 8$$

$$\Rightarrow 20 > 8 \times 15 \div 5 - 9 + 8$$

$$\Rightarrow 20 > 8 \times 3 - 9 + 8$$

$$\Rightarrow 20 > 24 - 9 + 8$$

$$\Rightarrow 20 \nless 23$$

Option (3)

$$20 < 2 + 10 \div 4 - 6 \times 100$$

$$\Rightarrow 20 \div 2 \times 10 - 4 + 6 > 100$$

$$\Rightarrow 10 \times 10 - 4 + 6 > 100$$

$$\Rightarrow 100 - 4 + 6 > 100$$

$$\Rightarrow 106 - 4 > 100$$

Option (4)

$$20 < 5 + 25 \div 10 - 2 \times 96$$

$$\Rightarrow 20 \div 5 \times 25 - 10 + 2 > 96$$

$$\Rightarrow 4 \times 25 - 10 + 2 > 96$$

$$\Rightarrow 100 - 10 + 2 > 96$$

$$\Rightarrow 102 - 10 \nless 96$$

6. (1)

$= \Rightarrow \div$	$+ \Rightarrow -$	$\times \Rightarrow =$	$- \Rightarrow >$
$> \Rightarrow +$	$< \Rightarrow \times$	$\div \Rightarrow <$	

Option (1)

$$4 < 2 + 5 + 8 \times 5$$

$$\Rightarrow 4 \times 2 - 5 - 8 = 5$$

$$\Rightarrow 8 - 5 - 8 \neq 5$$

Option (2)

$$4 = 2 + 5 > 8 \times 5$$

$$\Rightarrow 4 \div 2 - 5 + 8 = 5$$

$$\Rightarrow 2 - 5 + 8 = 5$$

Option (3)

$$4 < 2 > 5 + 8 \times 5$$

$$\Rightarrow 4 \times 2 + 5 - 8 = 5$$

$$\Rightarrow 8 + 5 - 8 = 5$$

Option (4)

$$4 > 2 < 5 + 8 - 5$$

$$\Rightarrow 4 + 2 \times 5 - 8 > 5$$

$$\Rightarrow 4 + 10 - 8 > 5$$

$$\Rightarrow 14 - 8 > 5$$

$$\Rightarrow 6 > 5$$

7. (2)

$> \Rightarrow +$	$< \Rightarrow -$	$+ \Rightarrow \div$	$\wedge \Rightarrow \times$
$- \Rightarrow =$	$\times \Rightarrow >$	$= \Rightarrow <$	

Option (1)

$$13 > 7 < 6 + 2 = 3 \wedge 4$$

$$\Rightarrow 13 + 7 - 6 \div 2 < 3 \times 4$$

$$\Rightarrow 13 + 7 - 3 < 12$$

$$\Rightarrow 20 - 3 \nless 12$$

Option (2)

$$9 > 5 > 4 - 18 + 9 > 16$$

$$\Rightarrow 9 + 5 + 4 = 18 \div 9 + 16$$

$$\Rightarrow 18 = 2 + 16$$

Option (3)

$$9 < 3 < 2 > 1 \times 8 \wedge 2$$

$$\Rightarrow 9 - 3 - 2 + 1 > 8 \times 2$$

$$\Rightarrow 5 \nless 16$$

Option (4)

$$28 + 4 \wedge 2 = 6 \wedge 4 + 2$$

$$\Rightarrow 28 \div 4 \times 2 < 6 \times 4 \div 2$$

$$\Rightarrow 7 \times 2 \nless 6 \times 2$$

8. (4) $x \angle y + z$

$$x \neq y > z$$

$$\Rightarrow x > y > z$$

$$\text{or } x < y > z$$

Option (1)

$$x \times y \mid z$$

$$\Rightarrow x = y < z : \text{Not true}$$

Option (2)

$$x - y \times z$$

$$\Rightarrow x \nless y = z : \text{Not true}$$

Option (3)

$$x \angle y \phi z$$

$$\Rightarrow x \neq y \nless z : \text{Not true}$$

Option (4)

$$x - y \angle z$$

$$\Rightarrow x \nless y \neq z$$

$$\Rightarrow x > y > z$$

$$\text{or, } x > y < z$$

$$\text{or, } x = y > z$$

$$\text{or, } x = y < z$$

$$\text{Option (4) may be true.}$$

9. (4) $A \Rightarrow \nless \Rightarrow \geq$

$$B \Rightarrow \neq \Rightarrow >, <$$

$$C \Rightarrow \nless \Rightarrow \leq$$

$$D \Rightarrow >$$

$$E \Rightarrow <$$

$$F \Rightarrow =$$

$$4Y \text{ F } 3X \Rightarrow 4Y = 3X$$

$$3X \text{ F } 6Z \Rightarrow 3X = 6Z$$

$$X = \frac{4Y}{3} = 2Z$$

$$Y = \frac{3}{4} X$$

$$\frac{4}{3} Y = 2Z$$

$$\Rightarrow Z = \frac{2}{3} Y$$

Option (1)

$$2Y \text{ D } 3Z$$

$$\Rightarrow 2Y > 3Z$$

$$\Rightarrow 2Y > 3 \times \frac{2}{3} Y$$

$$\Rightarrow 2Y \nless 2Y$$

Option (2)

$$2Y < 3Z$$

$$\Rightarrow 2Y < 3Z$$

$$\Rightarrow 2Y < 3 \times \frac{2}{3} Y$$

$$\Rightarrow 2Y \nless 2Y$$

Option (3)

$$4Y \text{ B } 5Z$$

$$\Rightarrow 4Y > 5Z \text{ or } 4Y < 5Z$$

$$\Rightarrow 4Y > 5 \times \frac{2}{3} Y$$

$$\text{or } 4Y < 5 \times \frac{2}{3} Y$$

$$\Rightarrow 4Y > \frac{10}{3} Y \text{ or } 4Y \nless \frac{10}{3} Y$$

Option (4)

$$2Y \text{ F } 3Z$$

$$\Rightarrow 2Y = 3Z$$

$$\Rightarrow 2Y = 3 \times \frac{2}{3} Y$$

$$\Rightarrow 2Y = 2Y$$

10. (2)

$+ \Rightarrow \div$	$- \Rightarrow =$	$\times \Rightarrow +$
$\div \Rightarrow >$	$= \Rightarrow <$	$> \Rightarrow \times$
$< \Rightarrow -$		

Option (1)

$$5 \div 2 \times 1 = 3 + 4 > 1$$

$$\Rightarrow 5 > 2 + 1 < 3 \div 4 \times 1$$

$$\Rightarrow 5 > 3 < \frac{3}{4}$$

Option (2)

$$5 > 2 \times 1 - 3 > 4 < 1$$

$$\Rightarrow 5 \times 2 + 1 = 3 \times 4 - 1$$

$$\Rightarrow 11 = 11$$

11. (1)

$+ \Rightarrow -$	$\div \Rightarrow +$
$< \Rightarrow \times$	$> \Rightarrow \div$

$$9 \div 7 < 8 > (4 > 2) + 5$$

$$\Rightarrow 9 + 7 \times 8 \div (4 \div 2) - 5$$

$$\Rightarrow 9 + 7 \times 8 \div 2 - 5$$

$$\Rightarrow 9 + 7 \times 4 - 5$$

$$\Rightarrow 9 + 28 - 5 = \boxed{32}$$

12. (3)

$+ \Rightarrow \div$	$- \Rightarrow \times$	$\div \Rightarrow -$
$\times \Rightarrow +$	$< \Rightarrow <$	

Option (1)

$$(10 + 2) \div 7 < (10 \div 7) + 2$$

$$\Rightarrow (10 \div 2) - 7 < (10 - 7) \div 2$$

$$\Rightarrow 5 - 7 < 3 \div 2$$

$$\Rightarrow -2 < \frac{3}{2}$$

Option (2)

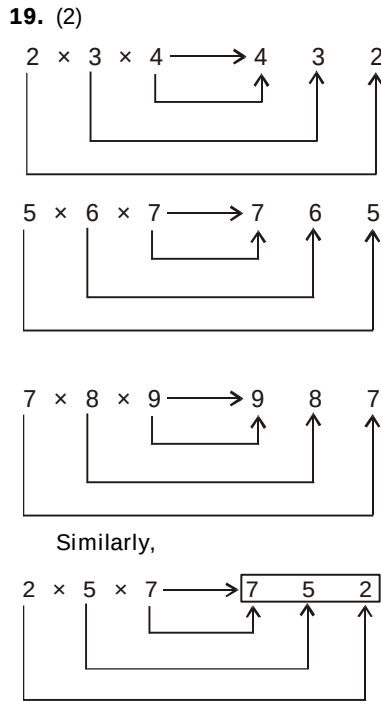
$$(10 - 7) \times 2 < (10 \times 2) - 7$$

$$\Rightarrow (10 \times 7) + 2 < (10 + 2) \times 7$$

$$\Rightarrow 70 + 2 < 12 \times 7$$

$$\Rightarrow 72 < 84$$

- 14. (4)** $5 \times 8 = 28$
 $\Rightarrow 5 \times 8 = 40$
 $\Rightarrow 5 + 8 = 13; 13 - 1 = 12$
 $\Rightarrow 40 - 12 = 28$
 $3 \times 7 = 12$
 $\Rightarrow 3 \times 7 = 21$
 $\Rightarrow 3 + 7 = 10, 10 - 1 = 9$
 $\Rightarrow 21 - 9 = 12$
 $8 \times 6 = 35$
 $\Rightarrow 8 \times 6 = 48$
 $\Rightarrow 8 + 6 = 14; 14 - 1 = 13$
 $\Rightarrow 48 - 13 = 35$
 $13 \times 13 = ?$
 $\Rightarrow 13 \times 13 = 169$
 $\Rightarrow 13 + 13 = 26; 26 - 1 = 25$
 $\Rightarrow 169 - 25 = 144$
- 15. (1)** $2 \times 4 \times 6 = 4$
 $\Rightarrow 6 - 4 = 2; 2 + 2 = 4$
 Third Number - Second Number
 + First Number = Result
 $9 \times 3 \times 7 = 13$
 $\Rightarrow 7 - 3 + 9 = 16 - 3 = 13$
 $4 \times 7 \times 6 = 3$
 $\Rightarrow 6 - 7 + 4 = 3$
 $\Rightarrow 10 - 7 = 3$
 $9 \times 7 \times 8$
 $\Rightarrow 8 - 7 + 9$
 $\Rightarrow 17 - 7 = 10$
- 16. (1)** $3 \times 5 \times 7 \times 2 = 24$
 $\Rightarrow 3 \times 5 + 7 + 2 = 24$
 $\Rightarrow 15 + 9 = 24$
 $2 \times 4 \times 6 \times 8 = 22$
 $\Rightarrow 2 \times 4 + 6 + 8 = 22$
 $8 + 14 = 22$
 $4 \times 4 \times 8 \times 9 = ?$
 $\Rightarrow ? = 4 \times 4 + 8 + 9$
 $\Rightarrow ? = 16 + 17 = 33$
- 17. (3)** $7 \times 8 = 49$
 $\Rightarrow 7 \times 8 - 7 = 56 - 7 = 49$
 $4 \times 4 = 12$
 $\Rightarrow 4 \times 4 - 4 = 16 - 4 = 12$
 $6 \times 4 = 18$
 $\Rightarrow 6 \times 4 - 6 = 24 - 6 = 18$
 Similarly,
 $9 \times 6 = ?$
 $\Rightarrow ? = 9 \times 6 - 9$
 $= 54 - 9 = 45$
- 18. (1)** $5 \odot 3 = -7$
 $\Rightarrow (5 \times 3) - (5 + 3)$
 $\Rightarrow 15 - 8 = 7$
 $7 \Rightarrow -7$
 $3 \odot 7 = -1$
 $\Rightarrow (3 \times 7) - (3 + 7)$
 $\Rightarrow 21 - 10 = 11$
 $11 \Rightarrow -11$
 $7 \odot 11$
 $\Rightarrow (7 \times 11) - (7 + 11)$
 $\Rightarrow 77 - 18 = 59$
 $59 \Rightarrow -59$



- 20. (2)** $2 \times 16 = 8$
 $\Rightarrow \frac{16}{2} = 8$
 $8 \times 8 = 1$
 $\Rightarrow \frac{8}{8} = 1$
 $6 \times 12 = 2$
 $\Rightarrow \frac{12}{6} = 2$
 Therefore,
 $12 \times 144 = ?$
 $\Rightarrow ? = \frac{144}{12} = 12$
- 21. (2)** $10 - 3 = 12$
 $\Rightarrow 10 - 3 = 7$ and $7 + 5 = 12$
 $12 - 4 = 13$
 $\Rightarrow 12 - 4 = 8$ and $8 + 5 = 13$
 $14 - 5 = 14$
 $\Rightarrow 14 - 5 = 9$ and $9 + 5 = 14$
 Similarly,
 $16 - 6 = ?$
 $\Rightarrow 16 - 6 = 10$ and $10 + 5 = 15$
- 22. (4)** Box - 1 $\Rightarrow \boxed{5}$
 Box - 2 $\Rightarrow \boxed{7}$
 $5 - 1 = 4$ and $7 + 1 = 8$
 $5 + 1 = 6$ and $7 - 1 = 6$

- 23. (4)** $0 \Rightarrow a$
 $1 \Rightarrow b$
 $2 \Rightarrow c$
 $3 \Rightarrow d$
 $9 \Rightarrow j$
 $dc \times f - (bf - d) \times d$
 $\Rightarrow 32 \times 5 - (15 - 3) \times 3$
 $\Rightarrow 160 - (12) \times 3$
 $\Rightarrow 160 - 36 = 124 \Rightarrow bce$
- 24. (3)** $29 \times 13 = 14$
 $\Rightarrow 29 + 13 = 42$
 $\Rightarrow \frac{42}{3} = 14$
 $76 \times 26 = 34$
 $\Rightarrow 76 + 26 = 102$
 $\Rightarrow \frac{102}{3} = 34$
 Therefore,
 64×14
 $\Rightarrow 64 + 14 = 78$
 $\Rightarrow \frac{78}{3} = 26$
- 25. (2)**
-
- Therefore,
-
- 26. (1)** $98 - 39 - 27 = 31$
 $\Rightarrow (9 \times 8) - (3 \times 9 + 2 \times 7) = 31$
 $\Rightarrow 72 - (27 + 14) = 31$
 $\Rightarrow 72 - 41 = 31$
 $87 - 38 - 34 = 20$
 $\Rightarrow (8 \times 7) - (3 \times 8 + 3 \times 4) = 20$
 $\Rightarrow 56 - (24 + 12) = 20$
 $\Rightarrow 56 - 36 = 20$
 $79 - 25 - 12 = ?$
 $\Rightarrow ? = (7 \times 9) - (2 \times 5 + 1 \times 2)$
 $\Rightarrow ? = 63 - (10 + 2)$
 $\Rightarrow ? = 63 - 12 = 51$

27. (4) $\begin{array}{ccccc} 5 & 3 & - & 3 & 4 \\ \downarrow & \swarrow & & \searrow & \downarrow \\ 5 & 3 & & 3 & 4 \\ \hline 6 & 5 & - & 4 & 6 \\ \downarrow & \swarrow & & \searrow & \downarrow \\ 6 & 4 & & 5 & 6 \end{array}$
Therefore,
 $\begin{array}{ccccc} 7 & 5 & - & 2 & 4 \\ \downarrow & \swarrow & & \searrow & \downarrow \\ 7 & 2 & & 5 & 4 \end{array}$

28. (4) $\begin{array}{ccccccc} & & & & & & \downarrow \\ 6 & \times & 2 & \times & 9 & \longrightarrow & 2 & 6 & 9 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 8 & \times & 7 & \times & 1 & \longrightarrow & 7 & 8 & 1 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 4 & \times & 1 & \times & 3 & \longrightarrow & 1 & 4 & 3 \end{array}$
Therefore,
 $\begin{array}{ccccccc} & & & & & & \downarrow \\ 4 & \times & 1 & \times & 3 & \longrightarrow & 1 & 4 & 3 \end{array}$

29. (1) $\begin{array}{ccccccc} & & & & & & \downarrow \\ 3 & \times & 4 & \times & 5 & \longrightarrow & 4 & 3 & 5 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 4 & \times & 3 & \times & 2 & \longrightarrow & 3 & 4 & 2 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 2 & \times & 3 & \times & 4 & \longrightarrow & 3 & 2 & 4 \end{array}$
Therefore,
 $\begin{array}{ccccccc} & & & & & & \downarrow \\ 2 & \times & 3 & \times & 4 & \longrightarrow & 3 & 2 & 4 \end{array}$

30. (2) $17 + 17 \Rightarrow 2895$
 $\Rightarrow 17 \times 17 = 289$
 $\Rightarrow 289 \Rightarrow 2895$
 $18 + 18 \Rightarrow 3245$
 $\Rightarrow 18 \times 18 = 324$
 $\Rightarrow 324 \Rightarrow 3245$
 $19 + 19 \Rightarrow 3615$
 $\Rightarrow 19 \times 19 = 361$
 $\Rightarrow 361 \Rightarrow 3615$
Similarly,
 $23 + 23 = ?$
 $\Rightarrow ? = 23 \times 23 = 529$
 $\Rightarrow ? = 529 \Rightarrow 5295$

31. (2) $24 \times 2 = 48 \Rightarrow 84$
 $32 \times 3 = 96 \Rightarrow 69$
 $13 \times 3 = 39 \Rightarrow 93$

32. (3) $4 + 3 = 25$
 $\Rightarrow (4)^2 + (3)^2 = 25$
 $\Rightarrow 16 + 9 = 25$
 $8 + 4 = 80$
 $\Rightarrow (8)^2 + (4)^2 = 80$
 $\Rightarrow 64 + 16 = 80$
Therefore,
 $3 + 2$
 $\Rightarrow (3)^2 + (2)^2$
 $\Rightarrow 9 + 4 = 13$

33. (3) $12 \times 16 = 188$
 $\Rightarrow 12 \times 16 = 192$
and, $192 - 4 = 188$
 $14 \times 18 = 248$
 $\Rightarrow 14 \times 18 = 252$
and, $252 - 4 = 248$
Therefore,
 $16 \times 20 = 320$
and, $320 - 4 = 316$

34. (4) $64 + 7 = 460$
 $\Rightarrow 64 \times 7 + 12 = 460$
 $25 + 8 = 212$
 $\Rightarrow 25 \times 8 + 12 = 212$
Therefore,
 $43 + 8 = ?$
 $\Rightarrow 43 \times 8 + 12$
 $= 344 + 12 = 356$

35. (3) $\begin{array}{ccccccc} & & & & & & \downarrow \\ 6 & \times & 9 & \times & 3 & \Rightarrow & 9 & 6 & 3 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 4 & \times & 8 & \times & 5 & \Rightarrow & 8 & 4 & 5 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 9 & \times & 4 & \times & 7 & \Rightarrow & 4 & 9 & 7 \end{array}$
Therefore,
 $\begin{array}{ccccccc} & & & & & & \downarrow \\ 9 & \times & 4 & \times & 7 & \Rightarrow & 4 & 9 & 7 \end{array}$

36. (2) (i) $4 \times 3 = 14 \Rightarrow (4 + 3) \times 2$
 $= 7 \times 2 = 14$
(ii) $5 \times 4 = 18 \Rightarrow (5 + 4) \times 2$
 $= 9 \times 2 = 18$
(iii) $6 \times 5 = 22 \Rightarrow (6 + 5) \times 2$
 $= 11 \times 2 = 22$
Similarly,
(iv) $7 \times 6 \Rightarrow (7 + 6) \times 2$
 $= 13 \times 2 = 26$

37. (2) $\begin{array}{ccccccc} & & & & & & \downarrow \\ 8 & \times & 5 & \times & 0 & \longrightarrow & 8 & 0 & 5 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ 7 & \times & 4 & \times & 6 & \longrightarrow & 7 & 6 & 4 \\ & \uparrow & & \uparrow & & & \uparrow & & \end{array}$

Therefore,
 $\begin{array}{ccccccc} & & & & & & \downarrow \\ 6 & \times & 8 & \times & 9 & \longrightarrow & 6 & 9 & 8 \\ & \uparrow & & \uparrow & & & \uparrow & & \end{array}$

38. (1) $34 \times 34 \times 34 = 39304$
 $27 \times 27 \times 27 = 19683$
 $13 \times 13 \times 13 = 2197$

39. (4) $\begin{array}{ccccccc} 1 & \times & 3 & \times & 5 & = & 1 & 9 & 25 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ & & \times 1 & & & & & & \\ & & \times 3 & & & & & & \\ & & \times 5 & & & & & & \end{array}$

$\begin{array}{ccccccc} 7 & \times & 9 & \times & 11 & \longrightarrow & 49 & 81 & 121 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ & & \times 7 & & & & & & \\ & & \times 9 & & & & & & \\ & & \times 11 & & & & & & \end{array}$

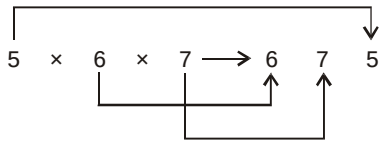
Therefore,

$\begin{array}{ccccccc} 19 & \times & 21 & \times & 23 & \longrightarrow & 361 & 441 & 529 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ & & \times 19 & & & & & & \\ & & \times 21 & & & & & & \\ & & \times 23 & & & & & & \end{array}$

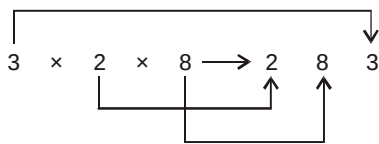
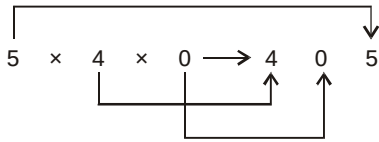
40. (2) $2 + 4 + 6 = 48$
 $\Rightarrow 2 \times 4 \times 6 = 48$
 $3 + 2 + 8 = 48$
 $\Rightarrow 3 \times 2 \times 8 = 48$
 $2 + 5 + 7 = ?$
 $\Rightarrow ? = 2 \times 5 \times 7 = 70$

41. (2) $\begin{array}{ccccccc} & & & & & & \downarrow \\ 1 & \times & 2 & \times & 3 & \longrightarrow & 2 & 3 & 1 \\ & \uparrow & & \uparrow & & & \uparrow & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ 3 & \times & 4 & \times & 5 & \longrightarrow & 4 & 5 & 3 \\ & \uparrow & & \uparrow & & & \uparrow & & \end{array}$

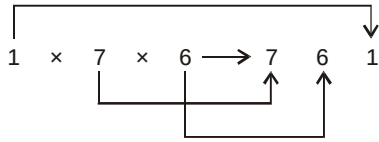
Therefore,



42. (4)



Therefore,



43. (1) $56 \times 11 = 9$

$$\Rightarrow 56 - 11 = 9$$

$$\Rightarrow 45 \Rightarrow 4 + 5 = 9$$

$$37 \times 13 = 6$$

$$\Rightarrow 37 - 13 = 6$$

$$\Rightarrow 24 \Rightarrow 2 + 4 = 6$$

$$42 \times 12 = 3$$

$$\Rightarrow 42 - 12 = 3$$

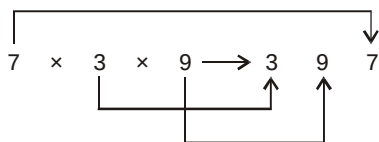
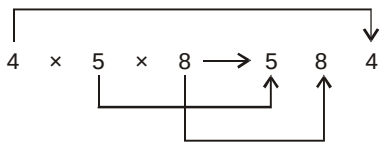
$$\Rightarrow 30 \Rightarrow 3 + 0 = 3$$

Therefore,

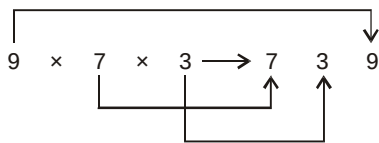
$$87 \times 77$$

$$\Rightarrow 87 - 77 = 10 \Rightarrow 1 + 0 = 1$$

44. (4)



Therefore,



45. (1) $879 = 8$

$$\Rightarrow 7 + 9 - 8 = 8$$

$$625 = 1$$

$$\Rightarrow 2 + 5 - 6 = 1$$

$$586 = 9$$

$$\Rightarrow 8 + 6 - 5 = 9$$

$$785 = ?$$

$$\Rightarrow 8 + 5 - 7 = 6$$

46. (1) $84 + 96 = 4842$

$$\Rightarrow \frac{84}{2}; \frac{96}{2} \Rightarrow 42; 48 \Rightarrow 4842$$

Therefore,

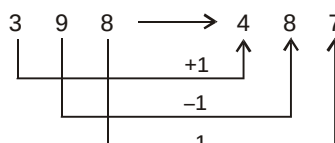
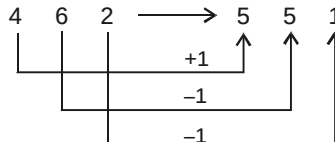
$$36 + 78$$

$$\Rightarrow \frac{36}{2}; \frac{78}{2}$$

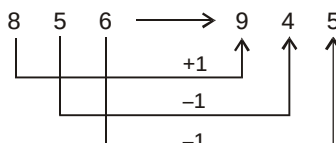
$$\Rightarrow 18; 39$$

$$\Rightarrow 3918$$

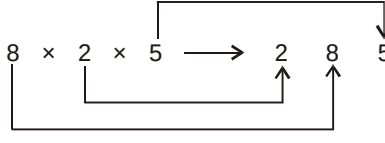
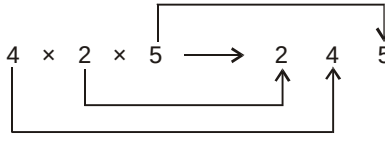
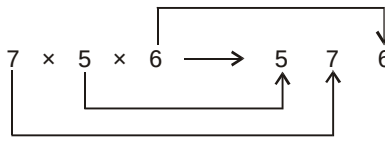
47. (3)



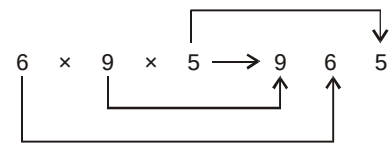
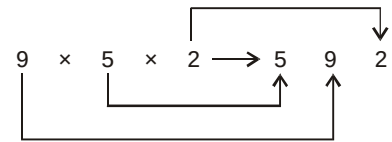
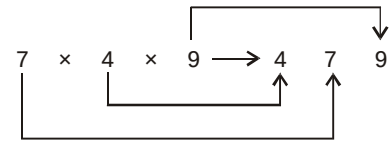
Therefore,



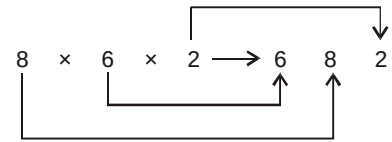
48. (2)



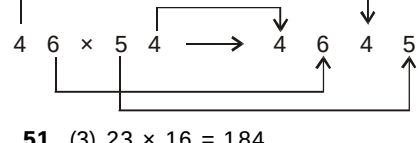
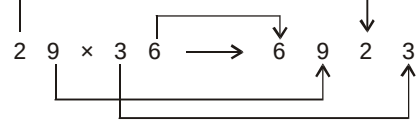
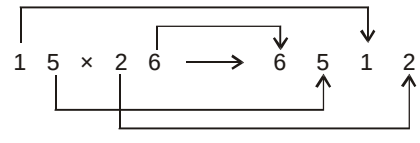
49. (2)



Therefore,



50. (2)



51. (3) $23 \times 16 = 184$

$$\Rightarrow \frac{23 \times 16}{2} = 184$$

$$37 \times 10 = 185$$

$$\Rightarrow \frac{37 \times 10}{2} = 185$$

Therefore,

$$? = 85 \times 12$$

$$\Rightarrow \frac{85 \times 12}{2} = \boxed{510}$$

52. (4) $5 \times 4 \times 3 = 70$

$$\Rightarrow 5 \times 4 \times 3 + 10 = 70$$

$$6 \times 5 \times 4 = 140$$

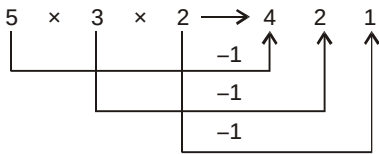
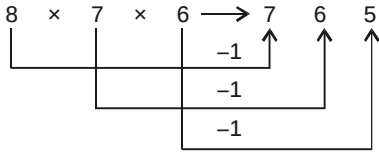
$$\Rightarrow 6 \times 5 \times 4 + 20 = 140$$

Therefore,

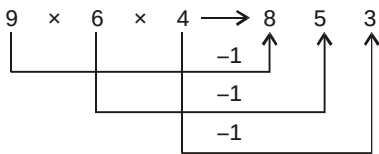
$$? = 7 \times 6 \times 5 + 30$$

$$\Rightarrow ? = 210 + 30 = 240$$

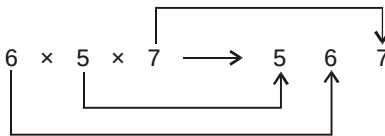
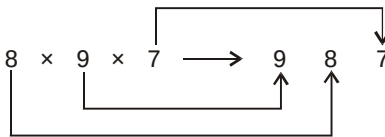
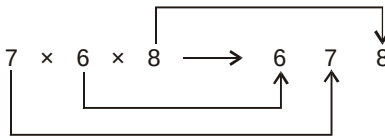
53. (2)



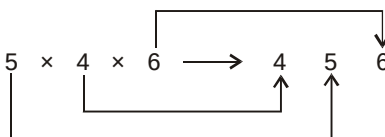
Therefore,



54. (1)



Therefore,

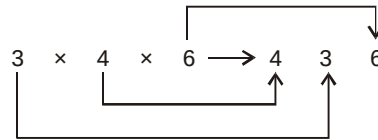
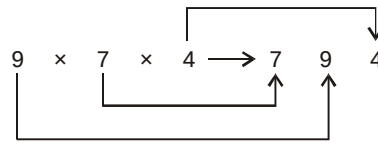


55. (3) $\frac{6 \times 4}{2} = 12$

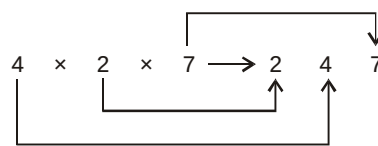
$$\frac{4 \times 12}{2} = 24$$

$$\frac{6 \times 9}{2} = 27$$

56. (2)



Therefore,



57. (2)

$$8 + 5 - 5 = 45$$

$$\Rightarrow 8 \times 5 + 5 = 45$$

$$12 + 6 - 5 = 77$$

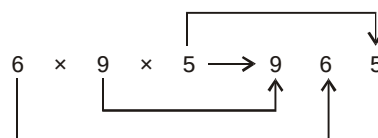
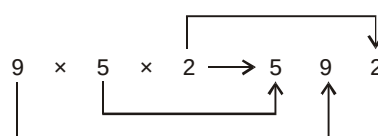
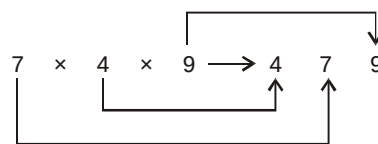
$$\Rightarrow 12 \times 6 + 5 = 77$$

$$14 + 5 - 10$$

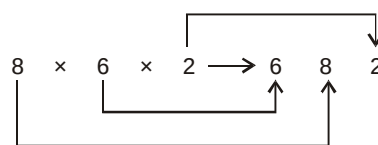
$$\Rightarrow 14 \times 5 + 10$$

$$\Rightarrow 70 + 10 = 80$$

58. (2)



Therefore,



TYPE-II

1. (1) $6 \div 3 + 4 > 45$

$$\Rightarrow 2 + 4 > 45$$

2. (*) Option (2)

$$24 = 4 \times 5 + 4$$

$$\Rightarrow 24 = 20 + 4$$

Option (4)

$$24 = 4 + 5 \times 4$$

$$\Rightarrow 24 = 4 + 20$$

Both options (2) and (4) are correct.

3. (1) $25 \times 2 - 6 = 4 \times 11 + 0$

$$\Rightarrow 50 - 6 = 44 + 0 \Rightarrow 44 = 44$$

4. (1) $8 \times 5 = 27 - 3 + 16$

$$\Rightarrow 40 = 24 + 16$$

$$\Rightarrow 40 = 40$$

5. (1) $96 \div 6 = 8 \times 2$

$$\Rightarrow 16 = 16$$

6. (4) $17 * 3 * 6 * 45$

$$\Rightarrow 17 \times 3 - 6 = 45$$

$$\Rightarrow 51 - 6 = 45$$

$$\Rightarrow 45 = 45$$

7. (1) $3 * 2 * 1 * 7$

$$\Rightarrow 3 \times 2 + 1 = 7$$

$$\Rightarrow 6 + 1 = 7$$

$$\Rightarrow 7 = 7$$

8. (4) $12 * 3 * 4 * 8 * 0$

$$\text{or, } 4 + 4 - 8 = 0$$

$$\text{or, } 8 - 8 = 0$$

9. (3) $7 * 4 * 8 * 2$

$$\Rightarrow 28 - 4 = 24$$

10. (3) $5 * 0 * 3 * 5$

$$\text{or, } 5 + 15 = 20$$

11. (3) $23 + 3 = 26$ and $26 + 1 = 27$

12. (1) $65 - 40 + 11 = 25 + 11 = 36$

13. (4) $31 - 1 \div 2 + 1 = 16$

$$\Rightarrow 30 \div 2 + 1 = 16$$

$$\Rightarrow 16 = 16$$

14. (4) $20 \div 5 + 8 \times 2 - 17$

$$\text{or, } 4 + 16 - 17 = 3$$

15. (2) $16 \div 4 + 5 = 9 \times 1$

$$\Rightarrow 4 + 5 = 9$$

16. (3) $7 \times 5 + 5 = 4 \times 10$

$$\Rightarrow 40 = 40$$

17. (4) $34 + 12 = 46$; $46 \div 2 = 23$

$$28 + 76 = 104$$
; $104 \div 2 = 52$

$$97 + 39 = 136$$
; $136 \div 2 = 68$

Therefore,

$$37 + 73 = 110$$
; $\frac{110}{2} = 55$

18. (4) $8 - 8 + 1 = 11 \div 11$

$$\Rightarrow 1 = 1$$

19. (3) $16 \div 4 \times 5 = 14 + 6$
 $\Rightarrow 4 \times 5 = 20$
20. (2) $7 + 7 - 2 \times 1 = 12$
 $\Rightarrow 7 + 7 - 2 = 12$
21. (4) $9 + 7 = 16$; $9 - 7 = 2$
 $16 \times 2 = 32$
 $13 + 7 = 20$; $13 - 7 = 6$
 $20 \times 6 = 120$
 $17 + 9 = 26$; $17 - 9 = 8$
 $26 \times 8 = 208$
 $19 + 11 = 30$; $19 - 11 = 8$
 $30 \times 8 = 240$
22. (4) $8 \times 5 - 9 = 31$
 $\Rightarrow 40 - 9 = 31$
23. (2) $4 \times 6 - 6 + 2 = 20$
 $\Rightarrow 24 - 6 + 2 = 20$
 $\Rightarrow 26 - 6 = 20$
24. (4) $8 \quad 5 \quad 2 \quad 72 \quad 4$
 $\Rightarrow 8 + 5 \times 2 = 72 \div 4$
 $\Rightarrow 8 + 10 = 18$
25. (2) $15 \times 3 - 5 = 20 \times 2$
 $\Rightarrow 45 - 5 = 40$
26. (2) $2 \times 3 - 2 + 4 = 8$
 $\Rightarrow 6 - 2 + 4 = 8$
27. (4) $16 - 2 - 24 \div 3 = 6$
 $\Rightarrow 16 - 2 - 8 = 6$
 $\Rightarrow 16 - 10 = 6$
28. (4) $16 - 4 \times 3 \div 4 = 13$
 $\Rightarrow 16 - 3 = 13$
29. (3) $6 * 15 * 10 * 3 * 12$
 $\Rightarrow 6 \times 15 \div 10 + 3 = 12$
 $\Rightarrow 9 + 3 = 12$
30. (2) $18 \times 6 \div 3 - 12 = 24$
 $\Rightarrow 18 \times 2 - 12 = 24$
 $\Rightarrow 36 - 12 = 24$
31. (2) $16 \times 4 > 64 \div 4 \Rightarrow 64 > 16$
32. (1) $28 \div 4 + 9 = 16$
 $\Rightarrow 7 + 9 = 16$
33. (4) $16 \times 6 \div 4 = 24$
 $\frac{96}{4} = 24$
34. (2) $16 - 8 \div 1 = 8$
 $\Rightarrow 16 - 8 = 8$
35. (1) $9 \div 3 \times 3 - 3 = 6$
 $\Rightarrow 3 \times 3 - 3 = 6$
 $\Rightarrow 9 - 3 = 6$
36. (2) $8 * 6 * 96 * 2 = 0$
 $\Rightarrow 8 \times 6 - 96 \div 2 = 0$
 $\Rightarrow 48 - 48 = 0$
37. (2) $\frac{264}{2} = 132$
 $1 + 3 + 2 = 6$

- $\frac{870}{3} = 290$
 $2 + 9 + 0 = 11$
 Similarly,
 $\frac{735}{5} = 147$
 $1 + 4 + 7 = 12$
38. (1) $24 * 16 * 8 * 32$
 $\Rightarrow 24 + 16 - 8 = 32$
 $\Rightarrow 40 - 8 = 32$
39. (2) $15 * 24 * 3 * 6 * 17$
 $\Rightarrow 15 + 24 \div 3 - 6 = 17$
 $\Rightarrow 15 + 8 - 6 = 17$
40. (1) $5 * 5 * 5 * 3 * 10$
 $\Rightarrow 5 \times 5 + 5 = 3 \times 10$
 $\Rightarrow 30 = 30$
41. (2) $21 * 7 * 6 * 9$
 $\Rightarrow 21 \div 7 + 6 = 9$
 $\Rightarrow 3 + 6 = 9$
42. (4) $6 * 4 * 12 * 12$
 $\Rightarrow 6 \times 4 - 12 = 12$
 $\Rightarrow 24 - 12 = 12$
43. (3) $8 * 8 * 1 * 7 = 8$
 $\Rightarrow 8 \div 8 \times 1 + 7 = 8$
 $\Rightarrow 1 \times 1 + 7 = 8$
44. (2) $13 * 3 * 4 * 3 = 4$
 $\Rightarrow 13 - 3 \times 4 + 3 = 4$
 $\Rightarrow 13 - 12 + 3 = 4$
 $\Rightarrow 1 + 3 = 4$
45. (3) $2 * 4 * 3 * 4 * 9$
 $\Rightarrow 2 \times 4 - 3 + 4 = 9$
 $\Rightarrow 8 - 3 + 4 = 9$
 $\Rightarrow 5 + 4 = 9$
46. (1) $5 * 6 = 35$
 $\Rightarrow 6 \times 5 + 5 = 35$
 $8 * 4 = 28$
 $4 \times 5 + 8 = 28$
 Similarly,
 $6 * 8$
 $8 \times 5 + 6 = 46$
47. (3) $12 * 3 * 4 = 6 * 8 * 8$
 $\Rightarrow 12 \times 3 + 4 = 6 \times 8 - 8$
 $\Rightarrow 36 + 4 = 48 - 8$
 $\Rightarrow 40 = 40$
48. (1) $5 * 9 * 3 * 6 * 8$
 $\Rightarrow 5 \times 9 + 3 = 6 \times 8$
 $\Rightarrow 45 + 3 = 48$
49. (2) $33 * 11 * 3 * 6 = 115$
 $\Rightarrow 33 \times 11 \div 3 - 6 = 115$
 $\Rightarrow \left(\frac{33 \times 11}{3} \right) - 6 = 115$
 $\Rightarrow 11 \times 11 - 6 = 115$
 $\Rightarrow 121 - 6 = 115$

50. (2) $13 * 12 * 5 * 4$
 $\Rightarrow 13 = 12 + 5 - 4$
 $\Rightarrow 13 = 17 - 4$
51. (1) $73 * 17 = 45$
 $\Rightarrow \frac{73+17}{2} = \frac{90}{2} = 45$
 $68 * 40 = 54$
 $\Rightarrow \frac{68+40}{2} = \frac{108}{2} = 54$
 Similarly,
 $83 * 15 = ?$
 $\frac{83+15}{2} = \frac{98}{2} = \boxed{49}$
52. (*) **Option (1)**
 $(\sqrt{121} - 9) + 5 \times 4 = 1$
 $\Rightarrow (11 - 9) + 5 \times 4 = 1$
 $\Rightarrow 2 + 5 \times 4 = 1$
 $\Rightarrow 2 + 20 \neq 1$
- Option (2)**
 $(\sqrt{121} + 9) \div 5 \times 4 = 1$
 $\Rightarrow (11 + 9) \div 5 \times 4 = 1$
 $\Rightarrow \frac{20}{5} \times 4 \neq 1$
- Option (3)**
 $(\sqrt{121} = 9) + 5 \times 4 \div 1$
 $\Rightarrow 11 \neq 9 + 20$
- Option (4)**
 $(\sqrt{121} - 9) \times 5 + 4 = 1$
 $\Rightarrow (11 - 9) \times 5 + 4 = 1$
 $\Rightarrow 2 \times 5 + 4 \neq 1$
53. (2) $45 * 3 * 6 * 2 * 16$
 $\Rightarrow (45 + 3) \div 6 \times 2 = 16$
 $\Rightarrow 48 \div 6 \times 2 = 16$
 $\Rightarrow 8 \times 2 = 16$
54. (3) $8 * 5 * 10 * 2 * 25$
 $\Rightarrow 8 \times 5 + 10 = 2 \times 25$
 $\Rightarrow 40 + 10 = 50$
55. (1) $5 * 3 * 3 * 5 * 0$
 $\Rightarrow 5 \times 3 \div 3 - 5 = 0$
 $\Rightarrow 5 - 5 = 0$
56. (*)
- | | | | | |
|----------|---------------|--------|---------------|----------|
| $+$ | $\Rightarrow$ | $-$ | $\Rightarrow$ | $\times$ |
| $\times$ | $\Rightarrow$ | $\div$ | $\Rightarrow$ | $+$ |
- Option (1)**
 $42 * 4 * 12 * 20 * 9$
 $\Rightarrow 42 - 4 \div 12 \times 20 + 9$
 After changing the signs
 $42 \times 4 + 12 \div 20 - 9$

$$\Rightarrow 42 \times 4 + \frac{12}{20} - 9$$

$$\Rightarrow 168 + \frac{3}{5} - 9 \neq 0$$

Option (2)

$$42 * 4 * 12 * 20 * 9$$

$$\Rightarrow 42 \div 4 + 12 - 20 \times 9$$

After changing the signs

$$42 + 4 - 12 \times 20 \div 9$$

$$\Rightarrow 42 + 4 - \frac{12 \times 20}{9}$$

$$\Rightarrow 42 + 4 - \frac{80}{3}$$

$$\Rightarrow 46 - \frac{80}{3} \neq 0$$

Option (3)

$$42 * 4 * 12 * 20 * 9$$

$$\Rightarrow 42 + 4 - 12 \div 20 \times 9$$

After changing the signs

$$\Rightarrow 42 - 4 \times 12 + 20 \div 9$$

$$\Rightarrow 42 - 4 \times 12 + \frac{20}{9}$$

$$\Rightarrow 42 - 48 + \frac{20}{9} \neq 0$$

Option (4)

$$42 * 4 * 12 * 20 * 9$$

$$\Rightarrow 42 \times 4 - 12 \div 20 + 9$$

After changing the signs

$$42 \div 4 \times 12 + 20 - 9$$

$$\Rightarrow \frac{21}{2} \times 12 + 20 - 9$$

$$\Rightarrow 21 \times 6 + 20 - 9 \neq 0$$

57. (2) $(16 + 18) * (21 - 11) * 32 * 8$

$$\Rightarrow 34 - 10 = 32 - 8$$

$$= 24 = 24$$

58. (4)

$*$	$\Rightarrow \times$	$@$	$\Rightarrow \div$
$?$	$\Rightarrow +$	$=$	$\Rightarrow -$

$$10 * 10 = 5 * 10 ? 50 @ 10$$

$$\Rightarrow 10 \times 10 - 5 \times 10 + 50 \div 10$$

$$\Rightarrow 10 \times 10 - 5 \times 10 + 5$$

$$\Rightarrow 100 - 50 + 5 = \boxed{55}$$

59. (4) $37 * 14 = 17$

$$\Rightarrow \frac{37 + 14}{3} = 17$$

$$\Rightarrow \frac{51}{3} = 17$$

$$69 * 33 = 34$$

$$\Rightarrow \frac{69 + 33}{3} = 34$$

$$\Rightarrow \frac{102}{3} = 34$$

$$91 * 125 = 72$$

$$\Rightarrow \frac{91 + 125}{3} = 72$$

$$\Rightarrow \frac{216}{3} = 72$$

Therefore,
 $28 * 56$

$$\Rightarrow \frac{28 + 56}{3} \Rightarrow \frac{84}{3} = 28$$

60. (2) $5 * 3 = 19$

$$\Rightarrow (5 \times 2) + (3 \times 3) = 19$$

$$[2 = 5 - 3]$$

$$\Rightarrow 10 + 9 = 19$$

$$8 * 5 = 49$$

$$\Rightarrow (8 \times 3) + (5 \times 5) = 49$$

$$[3 = 8 - 5]$$

$$\Rightarrow 24 + 25 = 49$$

Therefore,

$$6 * 4$$

$$\Rightarrow (6 \times 2) + (4 \times 4)$$

$$[2 = 6 - 4]$$

$$\Rightarrow 12 + 16 = 28$$

61. (4)

$$5 * 4 * 2 * 1 \rightarrow 1 \quad 4 \quad 2 \quad 5$$

$$7 * 8 * 1 * 6 \rightarrow 6 \quad 8 \quad 1 \quad 7$$

Therefore,

$$9 * 3 * 7 * 5 \rightarrow 5 \quad 3 \quad 7 \quad 9$$

62. (1) $4 * 2 = 3 \Rightarrow \frac{4}{2} + 1 = 3$

$$8 * 4 = 3 \Rightarrow \frac{8}{4} + 1 = 3$$

Therefore,

$$21 * 7 = ? \Rightarrow \frac{21}{7} + 1 = 4$$

63. (2) $3 * 4 = 10$

$$\Rightarrow 3 \times 2 + 4 = 10$$

$$5 * 8$$

$$\Rightarrow 5 \times 2 + 8 = 18$$

$$7 * 7$$

$$\Rightarrow 7 \times 2 + 7 = 21$$

64. (3) $48 * 4 * 6 * 3 * 30$

$$\Rightarrow 48 \div 4 + 6 \times 3 = 30$$

$$\Rightarrow 12 + 18 = 30$$

65. (1) $1 * 2 = 1$

$$\Rightarrow (1 + 2) - (1 \times 2)$$

$$\Rightarrow 3 - 2 = 1$$

$$2 * 3 = -1$$

$$\Rightarrow (2 + 3) - (2 \times 3) = -1$$

$$\Rightarrow 5 - 6 = -1$$

$$3 * 4 = -5$$

$$\Rightarrow (3 + 4) - (3 \times 4) = -5$$

$$\Rightarrow 7 - 12 = -5$$

Therefore,

$$7 * 9$$

$$\Rightarrow (7 + 9) - (7 \times 9)$$

$$\Rightarrow 16 - 63 = -47$$

66. (1) $\frac{3 * 2}{\downarrow} \quad \frac{* 8 * 4}{\downarrow}$

$$3 \times 2$$

$$8 \times 4$$

$$\downarrow$$

$$\downarrow$$

$$6$$

$$32$$

$$\frac{2 * 4}{\downarrow}$$

$$\frac{* 4 * 4}{\downarrow}$$

$$\downarrow$$

$$\downarrow$$

$$2 \times 4$$

$$4 \times 4$$

$$\downarrow$$

$$\downarrow$$

$$8$$

$$16$$

Therefore,

$$\frac{3 * 3}{\downarrow}$$

$$\frac{* 5 * 1}{\downarrow}$$

$$\downarrow$$

$$\downarrow$$

$$3 \times 3$$

$$5 \times 1$$

$$\downarrow$$

$$\downarrow$$

$$9$$

$$5$$

67. (2)

$$7 * 4 * 3 \rightarrow 4 \quad 3 \quad 7$$

$$8 * 6 * 4 \rightarrow 6 \quad 4 \quad 8$$

Therefore,

$$4 * 3 * 6 \rightarrow 3 \quad 6 \quad 4$$

68. (2) $13 * 45 = 29$

$$\Rightarrow \frac{13 + 45}{2} = 29$$

$$24 * 58 = 41$$

$$\Rightarrow \frac{24 + 58}{2} = 41$$

$$74 * 32 = 53$$

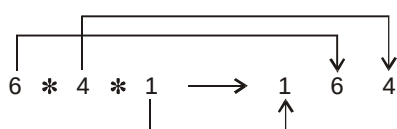
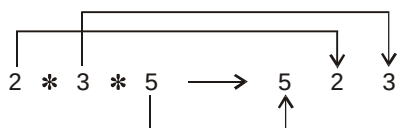
$$\Rightarrow \frac{74 + 32}{2} = 53$$

Therefore,

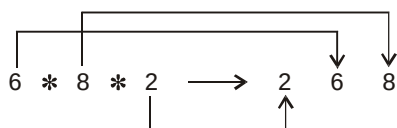
$$97 * 47 \\ \Rightarrow 97 + 47 = 144$$

$$\Rightarrow \frac{144}{2} = 72$$

69. (2)



Therefore,



70. (3) $9 * 8 = 63$

$$\Rightarrow 9 \times 7 = 63$$

$$7 * 8 = 49$$

$$\Rightarrow 7 \times 7 = 49$$

$$5 * 6 = 25$$

$$\Rightarrow 5 \times 5 = 25$$

$$11 * 7 = ?$$

$$\Rightarrow 11 \times 6 = \boxed{66}$$

TYPE-III

1. (4) $(12 + 6) \times 18 = 36$

$$\Rightarrow (18 \div 6) \times 12 = 36$$

$$\Rightarrow 3 \times 12 = 36$$

2. (2) $5 = 15 \div 3$

3. (3) $(12 \div 6) + 3 \times 7 = 42$

$$\Rightarrow (12 \div 6) \div 3 \times 7 = 42$$

$$\Rightarrow 18 \div 3 \times 7 = 42$$

4. (3) $35 + 7 \times 5 \div 5 - 6 = 24$

$$\Rightarrow 35 \div 7 \times 5 + 5 - 6 = 24$$

$$\Rightarrow 5 \times 5 + 5 - 6 = 24$$

$$\Rightarrow 25 + 5 - 6 = 24$$

5. (3) $24 \div 6 \times 3 + 3 - 1 = 14$

$$\Rightarrow 4 \times 3 + 3 - 1 = 14$$

$$\Rightarrow 12 + 3 - 1 = 14$$

$$\Rightarrow 15 - 1 = 14$$

6. (3) $(5 + 2) \times 2 - 10 = 16$

$$\Rightarrow (5 - 2) \times 2 + 10 = 16$$

$$\Rightarrow 3 \times 2 + 10 = 16$$

$$\Rightarrow 6 + 10 = 16$$

7. (3) $30 \div 6 - 4 + 2 \times 3 = 7$

$$\Rightarrow 5 - 4 + 6 = 7$$

8. (1) $5 + 6 \times 3 - 12 \div 2 = 17$

$$\Rightarrow 5 + 18 - 6 = 17$$

$$\Rightarrow 23 - 6 = 17$$

9. (4) $(7 + 2) \times 3 \times 4 - 1 = 20$

$$\Rightarrow (7 \times 2) + 3 + 4 - 1 = 20$$

$$\Rightarrow 14 + 3 + 4 - 1 = 20$$

10. (1) $(16 - 4) \times 6 \div 2 + 8 = 30$

$$\Rightarrow (16 \div 4) \times 6 - 2 + 8 = 30$$

$$\Rightarrow 4 \times 6 - 2 + 8 = 30$$

$$\Rightarrow 24 - 2 + 8 = 30$$

$$\Rightarrow 32 - 2 = 30$$

11. (3) $6 \times 4 + 2 = 16$

$$\Rightarrow 4 + 6 \times 2 = 16$$

$$\Rightarrow 4 + 12 = 16$$

12. (4) $(20 - 4) \times 4 + 16 = 36$

$$\Rightarrow (20 \div 4) \times 4 + 16 = 36$$

$$\Rightarrow 5 \times 4 + 16 = 36$$

13. (1) $2 \times 3 + 6 - 12 \div 4 = 17$

$$\Rightarrow 2 + 3 \times 6 - 12 \div 4 = 17$$

$$\Rightarrow 2 + 18 - 3 = 17$$

14. (3) $10 + 10 \div 10 - 10 \times 10 = 10$

$$\Rightarrow 10 \times 10 \div 10 - 10 + 10 = 10$$

$$\Rightarrow 10 - 10 + 10 = 10$$

15. (4) $(8 - 8) + 8 \times 32 = 64$

$$\Rightarrow (8 + 8) \div 8 \times 32 = 64$$

$$\Rightarrow 16 \div 8 \times 32 = 64$$

$$\Rightarrow 2 \times 32 = 64$$

16. (4) $8 \times 20 \div 3 + 9 - 5 = 38$

$$\Rightarrow 8 \times 20 \div 5 + 9 - 3 = 38$$

$$\Rightarrow 8 \times 4 + 9 - 3 = 38$$

$$\Rightarrow 32 + 9 - 3 = 38$$

17. (2) $(18 \div 9) + 3 \times 5 = 45$

$$\Rightarrow (18 + 9) \div 3 \times 5 = 45$$

$$\Rightarrow 27 \div 3 \times 5 = 45$$

18. (3) $8 \times 6 + 2 = 22$

$$\Rightarrow 6 + 8 \times 2 = 22$$

$$\Rightarrow 6 + 16 = 22$$

19. (3) Given expression

$$64 - 8 \times 9 \times 8 = 64$$

After interchange

$$(64 + 8) \div 9 \times 8 = 64$$

$$72 \div 9 \times 8 = 64$$

$$\Rightarrow 8 \times 8 = 64$$

20. (2) Option (1)

$$(30 \div 5) \times 10 = 24$$

$$\Rightarrow (30 \times 10) \div 5 = 24$$

$$\Rightarrow 300 \div 5 = 24$$

$$\Rightarrow 60 \neq 24$$

Option (2)

$$(30 \times 10) \div 5 = 60$$

$$\Rightarrow (30 \div 5) \times 10 = 60$$

$$\Rightarrow 6 \times 10 = 60$$

Option (3)

$$(30 \div 10) \times 5 = 18$$

$$\Rightarrow (30 \times 5) \div 10 = 18$$

$$\Rightarrow 150 \div 10 = 18$$

$$\Rightarrow 15 \neq 18$$

Option (4)

$$(10 \div 30) \times 5 = 70$$

$$\Rightarrow (5 \times 30) \div 10 = 70$$

$$\Rightarrow 150 \div 10 = 70$$

$$\Rightarrow 15 \neq 70$$

21. (3) $(6 + 3) + (4 \times 7) = 29$

$$\Rightarrow (6 \times 3) + (4 + 7) = 29$$

$$\Rightarrow 18 + 11 = 29$$

22. (1) $28 - 7 + 2 \times 2 = 0$

$$\Rightarrow 28 - 7 \times 2 \times 2 = 0$$

$$\Rightarrow 28 - 28 = 0$$

23. (*) There are two equations and it is not possible to correlate the two equations as per the information given in the question.

24. (2) $6 \times 4 + 2 = 16$

$$\Rightarrow 4 + 6 \times 2 = 16$$

$$\Rightarrow 4 + 12 = 16$$

25. (4) $6 + 2 - 3 = 16$

$$\Rightarrow 6 \times 3 - 2 = 16$$

$$\Rightarrow 18 - 2 = 16$$

26. (3) Option (1)

$$8 - 7 + 3 \times 5 = 35$$

$$\Rightarrow 7 + 8 - 3 \times 5 = 35$$

$$\Rightarrow 7 + 8 - 15 \neq 35$$

Option (2)

$$7 \times 8 + 6 - 9 = 25$$

$$\Rightarrow 8 \times 7 - 6 + 9 = 25$$

$$\Rightarrow 56 - 6 + 9 \neq 25$$

Option (3)

$$6 + 8 \times 2 - 7 = 0$$

$$\Rightarrow 6 - 7 \times 2 + 8 = 0$$

$$\Rightarrow 6 - 14 + 8 = 0$$

$$\Rightarrow 14 - 14 = 0$$

Option (4)

$$8 \times 2 + 7 - 6 = 9$$

$$\Rightarrow 7 \times 2 - 8 + 6 = 9$$

$$\Rightarrow 14 - 14 \neq 9$$

27. (1) $5 + 3 \times 8 - 12 \div 4 = 3$

$$\Rightarrow 5 + 3 \times 8 \div 12 - 4 = 3$$

$$\Rightarrow 5 + \frac{3 \times 8}{12} - 4 = 3$$

$$\Rightarrow 5 + 2 - 4 = 3$$